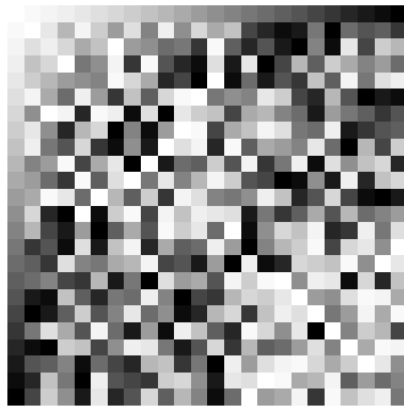


Lecture notes on sequences, series, differential calculus and groups



by Dr Killian O'Brien

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5.9 Going
fur-
ther:
Sub-
groups
and
La-
grange's
the-
o-
rem542

Ap-
pen-
dices547

A A
(very)
brief
look
at
num-
ber
sys-
tems549

Chapter 1

Sequences

Some
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ti-
va-
tion
and
con-
text

In
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unit
we
are
work-
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to-
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CHAPTER

1.

SEQUENCES

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smaller
changes,
or
in-
creases
with-
out
bound
etc.

In
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in
study-
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the
con-
ver-
gence
of
in-
fi-
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quences
of
num-
bers.
Se-
quences
con-
verge
when
their

CHAPTER

1.

SEQUENCES

terms

get

closer

and

closer

to

some

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ing

value.

An

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and

why

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quences

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con-

verge

is

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for

un-

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stand-

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hind

many

tech-

niques

of

math-

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mat-

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cal

mod-

elling.

When we model real physical processes our models are only ever approximate to the real situation. The techniques for solving these approximate models themselves only offer approximate solutions. This is fine as long

CHAPTER

1.

SEQUENCES

as

in

prin-

ci-

ple

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can

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and

closer

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weather)

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1.1.
BA-
SIC
will
DEFINITIONS
prob-

a-
bly
de-
part
fur-
ther
and
fur-
ther
from
the
real
so-
lu-
tion
when
ex-
tended
fur-
ther
into
the
fu-
ture,
i.e.
our
ap-
prox-
i-
ma-
tion
so-
lu-
tion
di-
verges
from
the
real
so-
lu-
tion.

1.1 Basic def- i- ni- tions

Definition

1.1
(Se-
quence).

CHAPTER
1.
SEQUENCES

A
sequence
is
a
list
of
el-
e-
ments
(usu-
ally
num-
bers)
in-
dexed
by
the
pos-
i-
tive
in-
te-
gers.

Sequences

can
con-
sist
of
a
fi-
nite
or
an
in-
fi-
nite
num-
ber
of
el-
e-
ments.
The
el-
e-
ments
of
a
se-
quence
could
be
any
type

1.1.
BA-
SIC
of
DEFINITIONS

math-
e-
mat-
i-
cal
ob-
ject
how-
ever
we
shall
be
con-
sid-
er-
ing
in-
fi-
nite
se-
quences
of
num-
bers,
un-
less
oth-
er-
wise
stated,
and
these
se-
quences
will
of-
ten
be
de-
fined
by
a
sin-
gle
rule
as
in
the
ex-
am-
ple
above.
We
use
a

CHAPTER
1.
SEQUENCES

sin-
gle
let-
ter
to-
gether
with
a
sub-
script
to
de-
note
the
el-
e-
ments
of
a
se-
quence,
as
in

$$x_1, x_2, x_3, x_4, \dots$$

We
can
rep-
re-
sent
the
se-
quence

$$x_1, x_2, x_3, x_4, \dots$$

us-
ing
the
com-
pact
no-
ta-
tion
 $\{x_n\}_{n=1}^{\infty}$

or
just
sim-
ply
 $\{x_n\}$
if
the
 n
in-
dex-
ing
is
un-

1.1.
BA-
SIC
for
DEFINITIONS
stood.

**Adivce
on
no-
ta-
tion**

Be
aware
of
the
dif-
fer-
ences
be-
tween
a
se-
quence
and
a
set.
Un-
for-
tu-
nately
we
use
the
same
brack-
ets
{ }
to
de-
note
each
but
re-
mem-
ber
that
the
el-
e-
ments
of
a
set
are
un-
ordered
whereas
the

CHAPTER

1.

SEQUENCES

el-

e-

ments

of

a

se-

quence

are

or-

dered,

i.e.

there

is

a

first

el-

e-

ment

,

a

sec-

ond

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e-

ment

and

so

on.

You

will

need

to

be

aware

from

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con-

text

what

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ob-

ject

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a

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or

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se-

quence

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ferred

1.1.
BA-
SIC
to
DEFINITIONS
Also

be
aware
that
the
in-
dex-
ing
of
a
se-
quence
need
not
nec-
es-
sar-
ily
be-
gin
at
1,
but
could
be-
gin
at
some
other
ini-
tial
value
such
as
0
etc.
When
deal-
ing
with
a
par-
tic-
u-
lar
se-
quence
you
should
al-
ways
be
aware
of
what

CHAPTER

1.

SEQUENCES

the
start-
ing
value
of
the
in-
dex
is.

Example

1.1.

The
fol-
low-
ing
ex-
am-
ples
from
here
to
Ex-
am-
ple

1.5

show
some
ba-
sic
se-
quences
and
dif-
fer-
ent
ways
to
em-
ploy
the
no-
ta-
tion
for
the
se-
quence
terms
and
in-
dex-
ing
vari-
able
to

1.1.
BA-
SIC
DEFINITIONS

One
pha-
sise
that
one
should
not
get
wed-
ded
to
al-
ways
us-
ing
 x
and
 n .

- The
se-
quence
of
pos-
i-
tive
in-
te-
gers
 $1, 2, 3, \dots$,
or
 $\{z_m\}$,
where
the
rule
is

$$z_m = m.$$

- The
Fi-
bonacci
se-
quence:
 $1, 1, 2, 3, 5, 8, 13, \dots$
or
 $\{f_n\}$
,
where
the
rule
is

$$f_1 = 1, f_2 = 1 \text{ and } f_m = f_{m-1} + f_{m-2} \text{ for all } m \geq 2,$$

CHAPTER

1.

SEQUENCES

i.e.
each
el-
e-
ment
is
the
sum
of
the
pre-
vi-
ous
two
el-
e-
ments.
This
type
of
rule
for
a
se-
quence,
where
each
term
is
de-
fined
us-
ing
the
val-
ues
of
some
pre-
vi-
ous
terms,
is
called
a
re-
cur-
rence
re-
la-
tion.

- The
se-
quence

1.1.

BA-

SIC

DEFINITIONS

odd

pos-

i-

tive

in-

te-

gers:

1, 3, 5, 7, ...or

$\{a_n\}$,

where

$$a_n = 2n - 1.$$

•

The

se-

quence

of

prime

num-

bers:

2, 3, 5, 7, 11, 13, ...

or

$\{p_m\}$

,

where

$$p_m = ???.$$

Two

im-

por-

tant

types

of

se-

quences

are

arith-

metic

and

ge-

o-

met-

ric

se-

quences.

Definition

1.2

(Arith-

metic

and

Ge-

o-

met-

ric

CHAPTER

1.

SEQUENCES

se-
quences).

An

*arith-
metic*

*se-
quence*

has

a

com-
mon

dif-
fer-
ence

be-
tween

suc-
ces-
sive

el-
e-

ments

and

a

*ge-
o-*

*met-
ric*

*se-
quence*

has

a

com-
mon

ra-
tio

of

suc-
ces-
sive

el-
e-

ments.

So

we

can

write

an

arith-
metic

se-
quence

as

$a, a+d, a+2d, a+3d, \dots$

or

1.1.
 BA-
 SIC
 DEFINITIONS

,
 where

$$a_n = a + (n-1)d$$

.
 Here

a
 is
 the
 ini-
 tial
 el-
 e-
 ment

and
 d
 is
 the
 the
 com-
 mon
 dif-
 fer-
 ence.

A
 ge-
 o-
 met-
 ric
 se-
 quence
 can
 be
 writ-
 ten
 as

$$a, ar, ar^2, ar^3, \dots$$

or
 $\{a_n\}$

,
 where

$$a_n = ar^{n-1}.$$

Here

a
 is
 the
 ini-
 tial
 el-
 e-
 ment
 and

CHAPTER

1.

SEQUENCES

r

is

the

com-

mon

ra-

tio.

Example

1.2

(Se-

quence

of

odd

in-

te-

gers).

The

se-

quences

of

pos-

i-

tive

in-

te-

gers

and

odd

pos-

i-

tive

in-

te-

gers

above

are

ac-

tu-

ally

arith-

metic

se-

quences.

The

se-

quence

of

pos-

i-

tive

in-

te-

gers

can

be

1.1.
 BA-
 SIC
 with
 DEFINITIONS

ten
 as
 $\{a_n\}$
 ,
 where

$$a_n = n = 1+(n-1),$$

so
 1
 is
 the
 ini-
 tial
 el-
 e-
 ment
 and
 1
 is
 the
 com-
 mon
 dif-
 fer-
 ence.
 The

se-
 quence
 of
 odd
 pos-
 i-
 tive
 in-
 te-
 gers
 can
 be
 writ-
 ten
 as
 $\{a_n\}$
 ,
 where

$$a_n = 1+(n-1)2,$$

so
 1
 is
 the
 ini-
 tial
 el-
 e-
 ment

CHAPTER
1.
SEQUENCES

and
2
is
the
com-
mon
dif-
fer-
ence.

Example

1.3
(Se-
quence
of
pow-
ers
of
 $\frac{1}{2}$).
The
se-
quence,

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$$

of
pow-
ers
of
 $\frac{1}{2}$,
is
ge-
o-
met-
ric
as
it
has
the
form
 $\{a_n\}$
,
where

$$a_n = \left(\frac{1}{2}\right)^{n-1},$$

so
1
is
the
ini-
tial
el-
e-
ment
and

1.1.
BA-
SIC
2
DEFINITIONS
the
com-
mon
ra-
tio.

Example

1.4
(Com-
pound
in-
ter-
est
rates).
Com-
pound
in-
ter-
est
awarded
on
in-
vest-
ments
or
charged
on
debts
pro-
vide
ex-
am-
ples
of
ge-
o-
met-
ric
se-
quences.
Sup-
pose
a
bank
of-
fers
an
an-
nual
com-
pound
in-
ter-
est
rate

CHAPTER
1.
SEQUENCES

of
 $r\%$
on
an
ini-
tial
de-
posit
of
 P
units.
Then
the
value
of
the
de-
posit
af-
ter
 n
years
is
given
by

$$v_n = P \left(1 + \frac{r}{100} \right)^n.$$

Example

1.5
(Nei-
ther
arith-
metic
nor
ge-
o-
met-
ric).
The
se-
quence
 $\{x_n\}$,
where

$$x_n = \frac{1}{n^2},$$

is
nei-
ther
arith-
metic
nor
ge-
o-
met-
ric,

1.2.
MONO-
TONIC
AND
BOUNDED
SEQUENCES
is
not
pos-
si-
ble
to
ex-
press
it
in
the
re-
quired
form.

1.2 Monotonic and bounded se- quences

The
fol-
low-
ing
def-
i-
ni-
tions
de-
scribe
the
be-
haviour
of
cer-
tain
se-
quences.

Definition

1.3.

A
se-
quence
 $\{x_n\}$
is
*in-
creas-
ing*
if
its
el-

CHAPTER

1.

SEQUENCES

e-

ments

sat-

isfy

$$x_1 \leq x_2 \leq x_3 \leq \dots$$

and

de-

creas-

ing

if

they

sat-

isfy

$$x_1 \geq x_2 \geq x_3 \geq \dots$$

If

the

in-

equal-

i-

ties

are

all

strict

then

the

se-

quence

can

be

called

strictly

in-

creas-

ing

or

strictly

de-

creas-

ing

as

ap-

pro-

pri-

ate.

If

the

in-

equal-

i-

ties

only

hold

from

1.2. MONO- TONIC ~~SOME~~ AND ~~THE~~ BOUNDED SE- QUENCES

lar
term
 x_k
on-
wards
then
we
can
speak
of
the
se-
quence
as
in-
creas-
ing/de-
creas-
ing
from
 k
as
ap-
pro-
pri-
ate.

The
term
*mono-
tonic*
refers
to
a
se-
quence
that
is
ei-
ther
in-
creas-
ing
or
de-
creas-
ing.
Sim-
i-
larly
for
strictly
mono-

CHAPTER
1.
SEQUENCES
tonic.

Proving
in-
creas-
ing
or
de-
creas-
ing
be-
haviour

If
one
has
a
com-
putable
for-
mula
for
defin-
ing
a
se-
quence
then
the
quick-
est
way
to
see
if
the
func-
tion
is
in-
creas-
ing
or
de-
creas-
ing
might
be
to
plot
some
terms
of
the
se-

1.2.
MONO-
TONIC
AND
BOUNDED
SEQUENCES

a
com-
puter.
There
are
many
ways
to
do
this.
Here
is
a
quick
ex-
am-
ple
of
how
to
do
this
with
the
Mat-
lab
soft-
ware
and
the
Sage-
Math
soft-
ware.

The
fol-
low-
ing
Mat-
lab
com-
mands
show
how
to
plot
the
first
hun-
dred
el-
e-
ments

CHAPTER 31

SEQUENCES

of
the
se-
quence
 $\{x_n\}$
de-
fined
by
 $x_n =$
 $f(n) =$
 $(n^2 +$
 $2)/2^n$.

From
the
re-
sult-
ing
plot
we
can
clearly
see
that
the
se-
quence
is
de-
creas-
ing
af-
ter
the
sec-
ond
el-
e-
ment.

```
f = @(n) (n^2+2)/2^n ;
N=1:100;
S=arrayfun(f,N);
plot(S, 'r')
```

Here
is
some
Sage-
Math¹

¹You
can
ex-

1.2.
MONO-
TONIC
AND
BOUNDED
SEQUENCES
code that achieves the same thing.

```
S=[(n^2+2)/2^n for n in range(1,101)]  
  
list_plot(S)
```

Investigating
the
first
few
terms
of
a
sequence
like
this
may
provide
evidence

per-
i-
ment
with
Sage
and
Python
code
us-
ing
the
on-
line
open-
source
com-
put-
ing
en-
vi-
ron-
ment
at

[www.
cocalc.
com](http://www.cocalc.com)

CHAPTER

1.

SEQUENCES

that

a

se-

quence

is

in-

creas-

ing

or

de-

creas-

ing.

How-

ever

a

more

rig-

or-

ous

ap-

proach

is

needed

if

we

are

to

for-

mally

prove

that

a

given

se-

quence

is

in-

creas-

ing

or

de-

creas-

ing.

There

are

(at

least)

two

ap-

proaches

that

we

can

take

to

this

1.2.
MONO-
TONIC
AND
BOUNDED
SEQUENCES

Suppose
we
are
given
a
se-
quence
 $\{a_n\}$
where
the
term
 a_n
is
de-
fined
by
some
for-
mula
de-
pend-
ing
on
 n .
Sup-
pose
we
want
to
in-
ves-
ti-
gate
and
prove
it
is
an
in-
creas-
ing
se-
quence.
We
can
con-
sider
the
dif-
fer-
ence
be-
tween

CHAPTER

1.

SEQUENCES

con-

sec-

u-

tive

terms,

i.e.

$a_{n+1} -$

a_n

and

then

try

to

es-

tab-

lish

that

the

in-

equal-

ity

$$a_{n+1} - a_n \geq 0$$

holds

for

all

rel-

e-

vant

val-

ues

of

the

in-

dex-

ing

vari-

able

n .

If

we

can

show

that

the

strict

in-

equal-

ity

$$a_{n+1} - a_n > 0,$$

holds

then

that

will

es-

1.2. MONO- TONIC AND BOUNDED SEQUENCES

strictly
in-
creas-
ing
be-
haviour
of
the
se-
quence.

If
it
is
de-
creas-
ing
be-
haviour
that
has
to
be
proved
then
the
rel-
e-
vant
in-
equal-
i-
ties
to
tackle
are

$$a_{n+1}-a_n \leq 0, \text{ or } a_{n+1}-a_n < 0.$$

Depending
on
the
for-
mula
defin-
ing
the
se-
quence
terms
it
may
be
more
ap-

CHAPTER

1.

SEQUENCES

pro-
pri-
ate
to
con-
sider
the
quo-
tient
of
con-
sec-
u-
tive
terms,

$$\frac{a_{n+1}}{a_n},$$

and
then
prove
that
this
is
ei-
ther
greater
than
or
less
than
1,
which
will
es-
tab-
lish
in-
creas-
ing
or
de-
creas-
ing
be-
haviour.
Clearly
though,
this
ap-
proach
can
only
be
used
if

1.2.
MONO-
TONIC
AND
BOUNDED
SEQUENCES

terms
are
never
equal
to
zero
as
this
would
lead
to
the
ra-
tio
be-
ing
un-
de-
fined.

Example

1.6

(In-
creas-
ing
se-
quence).

In-
ves-
ti-
gate
the
se-
quence
 $\{a_n\}$
de-
fined
by

$$a_n = \frac{5^n}{n!},$$

and
de-
scribe
its
in-
creas-
ing
/
de-
creas-
ing
be-
haviour.

CHAPTER

1.

SEQUENCES

Solution.

We
will
con-
sider
the
ra-
tio
of
con-
sec-
u-
tive
terms
which
is
given
by

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{5^{n+1}}{(n+1)!} \frac{n!}{5^n} \\ &= \frac{5}{n+1}.\end{aligned}$$

Now
the
quan-
tity
 $\frac{5}{n+1}$
can
eas-
ily
be
com-
pared
to
1.
It
is
greater
than
1
for
 n
less
than
4
and
less
than
1
for
 n
greater
than
5.
So

1.2. MONO- TONIC AND BOUNDED SEQUENCES

We shall include that the sequence $\{a_n\}$ is increasing for n up to 4 and then decreases for all $n \geq 5$.

□

Of primary interest in our study of sequences is how an infinite sequence behaves as it develops. Do

CHAPTER

1.

SEQUENCES

the

terms

vary

with

any

ap-

par-

ent

pat-

tern?

Do

they

stay

within

up-

per

and/or

lower

bounds?

Do

they

get

closer

and

closer

to

some

lim-

it-

ing

value?

To

be-

gin

with

we

in-

tro-

duce

the

con-

cept

of

a

bounded

se-

quence.

Definition

1.4.

A

se-

quence

$\{x_n\}$

is

said

1.2.
MONO-
TONIC
AND
BOUNDED
SEQUENCES

if
there
ex-
ists
some
pos-
i-
tive
num-
ber
 M
such
that
for
all
 n ,
the
el-
e-
ments
of
the
se-
quence
sat-
isfy
the
con-
di-
tion

$$|x_n| \leq M.$$

A
se-
quence
 $\{x_n\}$
is
said
to
be
bounded
above
if
there
ex-
ists
some
num-
ber
 M
such
that
for

CHAPTER

1.

SEQUENCES

all

n ,

the

el-

e-

ments

of

the

se-

quence

sat-

isfy

the

con-

di-

tion

$$x_n \leq M.$$

A

se-

quence

$\{x_n\}$

is

said

to

be

bounded

be-

low

if

there

ex-

ists

some

num-

ber

M

such

that

for

all

n ,

the

el-

e-

ments

of

the

se-

quence

sat-

isfy

the

con-

di-

1.3. CONVERGENCE

tion

$x_n \geq M$.

The
num-
ber
 M
is
called
the
(*up-
per
/
lower*)
bound,
as
ap-
pro-
pri-
ate.

1.3 Convergence

Now
we
in-
tro-
duce
the
very
im-
por-
tant
con-
cept
of
con-
ver-
gence.

Definition

1.5

(In-
for-
mal
def-
i-
ni-
tion
of
con-
ver-
gence).

Let
 L
be

CHAPTER

1.

SEQUENCES

a

num-

ber.

A

se-

quence

$\{x_n\}$

con-

verges

to

the

limit

L

if

as

n

in-

creases,

the

el-

e-

ments

of

the

se-

quence

get

closer

and

closer

to

the

num-

ber

L ,

or

equiv-

a-

lently,

the

dis-

tance

$|x_n - L|$

gets

closer

and

closer

to

0.

Another

way

of

say-

ing

that

1.3. CONVERGENCE

a
se-
quence
 $\{x_n\}$
con-
verges
to
the
limit
 L
is
to
say
that
 x_n
tends
to
 L
as
 n
tends
to
in-
fin-
ity,
or
in
sym-
bols

$$x_n \rightarrow L \text{ as } n \rightarrow \infty,$$

or

$$\lim_{n \rightarrow \infty} x_n = L.$$

Such
se-
quences
are
called
*con-
ver-
gent*
and
if
a
se-
quence
is
not
con-
ver-
gent
then
we
call

CHAPTER

1.

SEQUENCES

it

di-

ver-

gent.

Some

se-

quences

are

di-

ver-

gent

be-

cause

their

terms

keep

in-

creas-

ing

in

ab-

so-

lute

value

with-

out

any

up-

per

bound,

in

which

case

we

can

write

$y_n \rightarrow \infty$ as $n \rightarrow \infty$,

or

$y_n \rightarrow -\infty$ as $n \rightarrow \infty$.

The

first

of

these

says

that

the

term

y_n

in-

crease

in

pos-

i-

tive

1.3. CONVERGENCE

value
with-
out
any
up-
per
bound,
and
the
sec-
ond
says
that
the
terms
 y_n
get
large
and
neg-
a-
tive
with-
out
any
lower
bound.
How-
ever
al-
though
this
no-
ta-
tion
and
phras-
ing
is
used
quite
com-
monly
we
should
be
care-
ful.
Such
se-
quences
are
still
di-
ver-
gent,

CHAPTER
1.
SEQUENCES

and
we
should
cer-
tainly
not
treat
 ∞
(what-
ever
it
is
re-
ally)
as
a
num-
ber
like
any
other.

Example
1.7.

It
is
not
too
hard
to
ap-
pre-
ci-
ate
the
con-
ver-
gence
of
some
sim-
ple
se-
quences,
for
in-
stance

$$\lim_{m \rightarrow \infty} \frac{1}{m} = 0.$$

The
num-
bers
 $\frac{1}{m}$
are
get-
ting

1.3. CONVERGENCE

smaller
and
smaller
and
more-
over
they
are
get-
ting
ar-
bi-
trar-
ily
small,
i.e.
given
any
small
num-
ber
 ϵ ,
even-
tu-
ally
the
frac-
tions
 $\frac{1}{m}$
will
get
smaller
than
 ϵ .

While
the
def-
i-
ni-
tion
given
above
al-
lows
us
to
un-
der-
stand
the
idea
of
con-
ver-
gence,

CHAPTER

1.

SEQUENCES

a

more

rig-

or-

ous

for-

mal

def-

i-

ni-

tion

is

needed

to

prove

re-

sults

about

the

con-

cept

of

con-

ver-

gence.

This

def-

i-

ni-

tion

is:

Definition

1.6

(For-

mal

def-

i-

ni-

tion

of

con-

ver-

gence).

The

se-

quence

$\{x_n\}$

con-

verges

to

the

limit

L

if

given

1.3. CONVERGENCE

any
 $\epsilon >$
0
we
can
find
a
pos-
i-
tive
in-
te-
ger
 N_ϵ
such
that
for
all
 $n \geq$
 N_ϵ
the
in-
equal-
ity
 $|x_n - L| <$
 ϵ
holds.

This
def-
i-
ni-
tion
is
say-
ing
that
 $\{x_n\}$
con-
verges
to
 L
if
for
all
small
pos-
i-
tive
dis-
tances
 ϵ ,
the
se-
quence
even-

CHAPTER

1.

SEQUENCES

tu-

ally

(for

all

$n \geq$

N_ϵ

)

lies

within

a

dis-

tance

ϵ

of

the

limit

L .

Properties

of

con-

ver-

gence

Proofs

of

most

of

these

re-

sults

will

be

dis-

cussed

in

lec-

tures.

The

ideas

and

tech-

niques

you

see

in

these

proofs

will

oc-

cur

again

when

prov-

ing

re-

1.3. CONVERGENCE

lated
prop-
er-
ties
later
in
the
unit.
The
first
prop-
erty
of
con-
ver-
gence
we
will
prove
is
that
a
se-
quence
could
not
con-
verge
to
more
than
one
limit.

Theorem

1.1

(Unique-
ness
of
lim-
its).

If

a

se-

quence

con-

verges

then

its

limit

is

unique,

i. e.

if

$x_n \rightarrow$

L

CHAPTER
1.
SEQUENCES
and
 $x_n \rightarrow$
 M
as
 $n \rightarrow$
 ∞
then
 $L =$
 M .

Proof.
This
ar-
gu-
ment
uses
the
prin-
ci-
ple
of
proof
by
con-
tra-
dic-
tion.

adiction
nique of proof-by- Suppose
make the assumption that
e are trying to prove a
From this assumption a
clear contradiction of se-
easily know to be true. quence
resolve this then is to
assumption we made a_n
that the result we are con-
in fact true! verges

to
lim-
its
 $L, M \in$
 $\mathbb{R}$.
If
we
as-
sume
that
 L
and
 M
are
dis-
tinct,
i.e.
that
 $|L -$
 $M| >$

1.3. CONVERGENCE

0,
then
we
can
let
 $\epsilon =$
 $|L -$
 $M|/2 >$
0.
Since
 $a_n \rightarrow$
 L
there
ex-
ists
an
 N_1
such
that
for
ev-
ery
 $n \geq$
 N_1
we
have
 $|a_n -$
 $L| <$
 ϵ .
Sim-
i-
larly,
since
 $a_n \rightarrow$
 M
there
ex-
ists
an
 N_2
such
that
for
ev-
ery
 $n \geq$
 N_2
we
have
 $|a_n -$
 $M| <$
 ϵ .

We
wish
to
make

CHAPTER 1. SEQUENCES

use
of
both
of
these
in-
equal-
i-
ties
si-
mul-
ta-
ne-
ously.
So
we
set
 $N =$
 $\max(N_1, N_2)$.

Now
for
ev-
ery
 $n \geq$
 N
we
can
say

$$\begin{aligned} |L - M| &= |L - a_n + a_n - M| \\ &\leq |L - a_n| + |a_n - M|, \text{ by the triangle inequality} \\ &< \epsilon + \epsilon \\ &= |L - M|/2 + |L - M|/2 \\ &= |L - M|. \end{aligned}$$

ality

ue, $|x|$, of real num-
the following property:
ers x and y we have
 $|x + y|$. This can be shown
e four possibilities for
gativity of x and y .
f the triangle inequal-
other geometric and
(e.g. vector spaces).

This
chain
of
re-
la-
tions
has
es-
tab-
lished
that
for
all
 $n \geq$
 N

$$|L - M| < |L - M|.$$

But
this

1.3.
CONVERGENCE

last
state-
ment
is
a
clear
con-
tra-
dic-
tion
as
a
real
num-
ber
can-
not
be
strictly
less
than
it-
self.

So
we
con-
clude
that
the
as-
sump-
tion
that
led
to
this
con-
clu-
sion
must
it-
self
be
false,
i.e.
that
 L
and
 M
are
in-
deed
the
same.
This
proves

CHAPTER
1.
SEQUENCES
the
re-
sult.
□

The
con-
ver-
gence
and
bound-
ed-
ness
prop-
er-
ties
are
re-
lated,
as
de-
scribed
by
the
fol-
low-
ing
two
the-
o-
rems.

Theorem
1.2.
*Ev-
ery
con-
ver-
gent
se-
quence
is
bounded.*

Proof.
Consider
a
con-
ver-
gent
se-
quence
 $\{x_n\}$.
Sup-
pose
that
 $x_n \rightarrow$

1.3. CONVERGENCE

x
as
 $n \rightarrow$

∞ .
We
know
there
ex-
ists
a

pos-
i-
tive
in-
te-
ger
 N

such
that
for
all
 $n \geq$
 N

we
have
 $|x_n -$
 $x| <$

1,
this
im-
plies
that

for
all
 $n \geq$
 N

we
have
 $|x_n| <$
1 +
 $|x|$.

Why $\epsilon = 1$?

Nothing special about the choice of $\epsilon =$
1. Here, any choice of $\epsilon > 0$ would do,
that just have to make a choice.

the
se-
quence
is
even-
tu-
ally
bounded,
i.e.
af-
ter
el-

CHAPTER 1. SEQUENCES

ment
 N .

But
this
just
leaves

a
fi-
nite
num-
ber
of
terms
to
in-
clude.

So
let
 $M =$

$\max(x_1, x_2, \dots, x_{N-1}, 1 +$
 $|x|).$

nd minimums of
s

imum (or minimum)
ion of numbers is un-
s simply the largest
ne numbers. However
en when trying to talk
(or minimum) of an
of numbers, as their
imum) might not ex-

Then
we
can
say
that
for
all
 $n \geq$
1

we
have
 $|x_n| \leq$

$M,$
i.e.

that
the
se-

quence
is

bounded.

□

Knowing
that

a
se-
quence

is
bounded

and
ei-

ther
in-

creas-

1.3.
CONVERGENCE

ing
or
de-
creas-
ing
is
enough
to
guar-
an-
tee
that
it
con-
verges.

Theorem

1.3

(Mono-
tone
con-
ver-
gence
the-
o-
rem).

*Ev-
ery
bounded
mono-
tonic
se-
quence
is
con-
ver-
gent.*

Proof.

We
shall
not
dis-
cuss
the
proof
of
this
is
in
de-
tail
in
this
course.
It

CHAPTER

1.

SEQUENCES

es-
sen-
tially
fol-
lows
from
the
least-
upper
bound
prin-
ci-
ple
of
the
real
num-
bers.

□

Theorem

1.4

(Al-
ge-
bra
of
lim-
its
the-
o-
rem).

Let

$\{a_n\}$

and

$\{b_n\}$

be

two

con-

ver-

gent

se-

quences

with

$$\lim_{n \rightarrow \infty} a_n = a \text{ and } \lim_{n \rightarrow \infty} b_n = b.$$

New

se-

quences

can

be

formed

from

the

el-

e-

ments

1.3.
CONVERGENCE

a_n
and
 b_n
in
many
ways.
The
fol-
low-
ing
new
se-
quences
are
all
con-
ver-
gent
and
have
the
ex-
pected
lim-
its:

- **Linear
Com-
bi-
na-
tions:**

$$\lim_{n \rightarrow \infty} (\alpha a_n + \beta b_n) = \alpha a + \beta b,$$

where
 $\alpha, \beta \in$
 $\mathbb{R}$
are
any
pair
of
co-
ef-
fi-
cients,

- **Products:**

$$\lim_{n \rightarrow \infty} (a_n b_n) = ab,$$

- **Quotients:**
(as-
sum-
ing

CHAPTER

1.

SEQUENCES

that

$$b \neq$$

$$0$$

and

for

all

n ,

$$b_n \neq$$

0)

$$\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{a}{b}.$$

Proof.

First

we

prove

the

lin-

ear

com-

bi-

na-

tions

part.

Assume

that

we

have

con-

ver-

gent

se-

quences

$$\{a_n\}$$

and

$$\{b_n\}$$

with

$$a_n \rightarrow$$

$$a$$

and

$$b_n \rightarrow$$

$$b.$$

Let

$$\alpha, \beta \in$$

$$\mathbb{R}.$$

Let

$$\epsilon >$$

$$0$$

be

given.

Since

both

se-

quences

are

1.3.
CONVERGENCE

con-
ver-
gent

we
can
say
that
there

ex-
ists

N_1
and
 N_2

such
that
for
all

$n \geq$
 N_1

we
have

$|a_n -$
 $a| <$

$\frac{\epsilon}{2|\alpha|},$
and

that
for

all
 $n \geq$

N_2
we

have
 $|b_n -$

$b| <$
 $\frac{\epsilon}{2|\beta|}.$

We
wish

to
make

use
of

both
in-

equal-
i-

ties
si-

mul-
ta-

ne-
ously

so
we

let
 $N =$

$\max(N_1, N_2).$

CHAPTER
1.
SEQUENCES

Then
for
all
 $n \geq$
 N
we
have

$$\begin{aligned} |(\alpha a_n + \beta b_n) - (\alpha a + \beta b)| &= |\alpha(a_n - a) + \beta(b_n - b)| \\ &\leq |\alpha||a_n - a| + |\beta||b_n - b|, \text{ triangle inequality and factorising} \\ &< |\alpha|\frac{\epsilon}{2|\alpha|} + |\beta|\frac{\epsilon}{2|\beta|}, \text{ from above} \\ &= \epsilon. \end{aligned}$$

From
the
above
re-
la-
tions
we
can
ex-
tract
the
re-
quired
cri-
te-
rion
for
con-
ver-
gence,
namely
for
all
 $n \geq$
 N
we
have
 $|(\alpha a_n + \beta b_n) - (\alpha a + \beta b)| <$
 ϵ .
This
proves
the
lin-
ear-
ity
part
of
the

1.3. CONVERGENCE

the-
o-
rem.

Next
we
prove
the
prod-
ucts
part.

Suppose
that
 $a_n \rightarrow$
 a
and
 $b_n \rightarrow$
 b
as
be-
fore
and
that
 $\epsilon >$
 0
is
given.

Understanding this part

There is much to unpick here in order
to gain a proper understanding of how
this proof operates. Spend some time
on it to see how rewriting the absolute
value allows us to use our control over
the sizes of $|a_n - a|$ and $|b_n - b|$ to
achieve the desired inequality for the
convergence of $a_n b_n$.

ist
pos-
i-
tive
in-
te-
gers
 N_1
and
 N_1
such
that
for
all
 $n \geq$
 N_1
we
have
 $|a_n -$
 $a| <$
 $\min\left(\sqrt{\frac{\epsilon}{3}}, \frac{\epsilon}{3|b|}\right)$
and

CHAPTER
1.
SEQUENCES

for
all
 $n \geq$
 N_2
we
have
 $|b_n -$
 $b| <$
 $\min\left(\sqrt{\frac{\epsilon}{3}}, \frac{\epsilon}{3|a|}\right).$

We
wish
to
use
both
in-
equal-
ties
so
we
let
 $N =$
 $\max(N_1, N_2)$
and
now
for
all
 $n \geq$
 N
we
have

$$\begin{aligned} |a_n b_n - ab| &= |(a_n - a)(b_n - b) + b(a_n - a) + a(b_n - b)| \\ &\leq |(a_n - a)(b_n - b)| + |b|(a_n - a)| + |a|(b_n - b)| \\ &< \sqrt{\frac{\epsilon}{3}} \sqrt{\frac{\epsilon}{3}} + |b| \frac{\epsilon}{3|b|} + |a| \frac{\epsilon}{3|a|} \\ &= \epsilon \end{aligned}$$

From
the
above
re-
la-
tions
we
can
ex-
tract
the
re-
quired
cri-
te-
rion
for
con-

1.3.
CONVERGENCE

ver-
gence,
namely
for
all
 $n \geq$
 N
we
have
 $|a_n b_n -$
 $ab| <$
 $\epsilon.$

This
proves
the
prod-
uct
part
of
the
the-
o-
rem.

Next
we
prove
the
quo-
tients
part.
We
will
sim-
plify
mat-
ters
by
first
prov-
ing
a
spe-
cial
case
that
if
 $b_n \rightarrow$
 b
(with
 $b_n, b \neq$
 0)
then
 $1/b_n \rightarrow$
 $1/b.$
Then
the

CHAPTER
1.
SEQUENCES

gen-
eral
quo-
tient
case
can
be
de-
rived
from
this
spe-
cial
case
and
the
prod-
uct
case,
since
we
can
re-
express
 a_n/b_n
as
 $a_n(1/b_n)$.

The
rel-
e-
vant
ab-
so-
lute
value
to
ex-
am-
ine
is

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \left| \frac{b - b_n}{b_n b} \right| = \frac{1}{|b||b_n|} |b_n - b|.$$

To
en-
sure
that
this
ab-
so-
lute
value
is
less
than
 ϵ

1.3. CONVERGENCE

we
will
have
to
ar-
range
a
lower
bound
for
the
de-
nom-
i-
na-
tor
and
a
suit-
able
up-
per
bound
for
the
nu-
mer-
a-
tor.

Understanding this part

Again, you should spend time com-
ing to a good understanding of the ar-
gument here. You need to appreci-
ate how we are controlling the size of
the $\frac{1}{|b||b_n|} |b_n - b|$ with a combination of
lower and upper-bounds.

ists
a
pos-
i-
tive
in-
te-
ger
 N_1
such
that
for
all
 $n \geq$
 N_1
we
have
 $|b_n -$
 $b| <$
 $|b|/2,$
which

CHAPTER

1.

SEQUENCES

im-

plies

that

$|b_n| >$

$|b|/2$.

Also

by

con-

ver-

gence,

there

ex-

ists

a

pos-

i-

tive

in-

te-

ger

N_2

such

that

for

all

$n \geq$

N_2

we

have

$|b_n -$

$b| <$

$\epsilon|b|^2/2$.

So

let-

ting

$N =$

$\max(N_1, N_2)$

then

for

all

$n \geq$

N

we

have

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{1}{|b||b_n|} |b_n - b| < \frac{2}{|b||b|} \frac{|b|^2}{2} \epsilon = \epsilon,$$

which

gives

us

the

re-

quired

in-

equal-

ity

1.3.
CONVERGENCE

to
con-
clude
that
 $1/b_n \rightarrow$
 $1/b$
as
claimed.
To-
gether
with
the
re-
marks
above
this
proves
the
quo-
tients
part
of
the
the-
o-
rem.
 $\square$

The
al-
ge-
bra
of
lim-
its
the-
o-
rem
can
al-
low
us
to
find
the
lim-
its
of
se-
quences
that
are
formed
from
se-
quences

CHAPTER

1.

SEQUENCES

with

known

con-

ver-

gence

prop-

er-

ties.

By

re-

fer-

ring

to

the

the-

o-

rem

we

can

pro-

vide

rig-

or-

ous

ar-

gu-

ments

to

jus-

tify

the

limit

of

the

se-

quence

in

ques-

tion,

based

on

lim-

its

of

sim-

pler

se-

quences.

Example

1.8

(Ap-

ply-

ing

the

al-

1.3.
CONVERGENCE

ge-
bra
of
lim-
its
the-
o-
rem).
Con-
sider
the
se-
quence
 $\{x_n\}_{n=1}^\infty$
where

$$x_n = \frac{2}{3} \frac{2n^2 - n}{(-3n^2 + 2)}.$$

By
ap-
ply-
ing
the
var-
i-
ous
parts
of
the
al-
ge-
bra
of
lim-
its
the-
o-
rem
we
can
see
how
to
find
the
limit
of
this
con-
ver-
gent
the-
o-

CHAPTER
1.
~~SE~~QUENCES
rem.

$$\begin{aligned}
 \lim_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} \frac{2}{3} \frac{2n^2 - n}{(-3n^2 + 2)} \\
 &= \frac{2}{3} \lim_{n \rightarrow \infty} \frac{2 - \frac{1}{n}}{\left(-3 + \frac{2}{n^2}\right)}, \quad \text{linearity} \\
 &= \frac{2}{3} \frac{\lim_{n \rightarrow \infty} \left(2 - \frac{1}{n}\right)}{\lim_{n \rightarrow \infty} \left(-3 + \frac{2}{n^2}\right)}, \quad \text{quotient rule} \\
 &= \frac{2}{3} \left(\frac{2}{-3} \right) = \frac{-4}{9}, \quad \text{linearity and known convergence of } \frac{1}{n} \text{ and } \frac{1}{n^2}.
 \end{aligned}$$

1.4 L'Hôpital's rule

Another
use-
ful
tool
for
work-
ing
with
lim-
its
of
se-
quences
is
L'Hôpital's
rule.
It
can
be
used
to
find
the
lim-
its
of
quo-
tients
of
se-
quences,
where
the
de-
nom-
i-
na-
tor
and
nu-

1.4.

L'HÔPITAL'S

~~R~~*ULE*

mer-

a-

tor

se-

quences

do

not

have

lim-

its,

or

both

tend

to

the

limit

zero.

When

we

con-

sider

dif-

fer-

en-

ti-

a-

tion

next

term

we

shall

be

able

to

prove

some

of

the

cases

of

this

rule.

The

rule

ap-

plies

to

se-

quences

of

the

form

$\{a_n\}$,

where

$$a_n = \frac{f(n)}{g(n)},$$

CHAPTER
1.
SEQUENCES

where

f

and

g

are

a

pair

of

dif-

fer-

en-

tiable

func-

tions.

If

it

is

the

case

that

$f(n) \rightarrow \pm\infty$ and $g(n) \rightarrow \pm\infty$ as $n \rightarrow \infty$,

or

$f(n) \rightarrow 0$ and $g(n) \rightarrow 0$ as $n \rightarrow \infty$,

then

we

can

work

out

the

limit

us-

ing

$$\lim_{n \rightarrow \infty} a_n = \frac{\lim_{n \rightarrow \infty} f'(n)}{\lim_{n \rightarrow \infty} g'(n)},$$

where

f'

and

g'

are

the

deriva-

tives

of

f

and

g .

Exercises

1.1.

These

prob-

lems

1.4.

L'HÔPITAL'S

~~R~~*ULE*

re-
quire
the
ap-
pli-
ca-
tion
of
var-
i-
ous
con-
cepts
and
re-
sults
from
this
chap-
ter.
We
shall
use
these
as
a
ba-
sis
for
dis-
cus-
sion
in
the
tu-
to-
ri-
als.

1.

Write
down
the
first
five
terms
of
the
se-
quence
 $\{x_n\}_{n=1}^{\infty}$
where
 x_n
is
given
by

CHAPTER

1.

SEQUENCES

a)

$$x_n = 2^{n-1},$$

b)

$$x_n = \frac{2n}{2n-1},$$

c)

$$x_n = (-1)^n \frac{n^2}{n!},$$

d)

$$x_n = \cos \frac{n\pi}{4}.$$

2.

For each of the sequences in question 1 above write down a simplified expression for the term x_{2m+3} .

3.

Determine whether the sequence $\{x_n\}_{n=1}^{\infty}$ is increasing or decreasing,

1.4.

L'HÔPITAL'S

RULE where

x_n

is

given

by

a)

$$x_n = \frac{n}{2n+1},$$

b)

$$x_n = \frac{10^n}{(2n)!},$$

c)

$$x_n = \frac{e^n}{(n+1)!},$$

d)

$$x_n = \frac{n}{2^n}.$$

4.

Consider

the

se-

quence

$\{x_n\}_{n=1}^{\infty}$

de-

ter-

mined

by

the

fol-

low-

ing

rules

for

x_n .

Which

ones

are

con-

ver-

gent?

De-

ter-

mine

the

limit

if

one

ex-

ists.

Hint:

l'Hospital's

rule

CHAPTER
1.
SEQUENCES

can
be
used
for
the
first
two.

a)

$$x_n = \frac{\ln n}{2n},$$

b)

$$x_n = n^2 e^{-n},$$

c)

$$x_n = \frac{(n+1)(n+2)}{2n^2},$$

d)

$$x_n = \frac{2n^3}{n^2-1},$$

e)

$$x_n = \left(1 - \frac{1}{2^n}\right).$$

5.

Use
the
Mono-
tone
Con-
ver-
gence
The-
o-
rem
to
prove
that
the
se-
quence
 $\{r_n\}$
con-
verges
where

$$r_n = \frac{\prod_{m=0}^n (2m+1)}{\prod_{m=0}^n (2m+2)} = \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n+1)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot (2n+2)}.$$

6.

Prove
that

1.4.

L'HÔPITAL'S

~~83~~**RULE** the

se-
quence

$\{a_n\}$

is

di-

ver-

gent,

where

$$a_n = \frac{n!}{2^n}.$$

Hint:

Show

that

the

se-

quence

is

un-

bounded.

7.

Consider

the

gen-

eral

arith-

metic

se-

quence

$\{x_n\}_{n=1}^{\infty}$

where

$x_n =$

$a +$

$(n -$

$1)d.$

Let

S_m

de-

note

the

sum

of

the

first

m

el-

e-

ments

of

the

se-

quence

$\{x_n\}.$

Ob-

CHAPTER

1.

SEQUENCES

tain

a

for-

mula

for

S_m

in

terms

of

a, d, m .

Hint:

Con-

sider

the

quan-

tity

$2S_m$

as

the

sum

of

S_m

with

it-

self

in

the

re-

verse

or-

der.

8.

Consider

the

gen-

eral

ge-

o-

met-

ric

se-

quence

$\{x_n\}_{n=1}^{\infty}$

where

$x_n =$

ar^{n-1} .

Let

S_m

de-

note

the

sum

of

the

1.4.

L'HÔPITAL'S

~~83~~**RULE.** first
the
elements
of
the
sequence
 $\{x_n\}$.
Ob-
tain
a
for-
mula
for
 S_m
in
terms
of
 a, r, m .
Hint:
Con-
sider
the
quan-
tity
 $S_m -$
 rS_m .

9.

Using
the
al-
ge-
bra
of
se-
quences
the-
o-
rem
find
the
limit
of
the
fol-
low-
ing
con-
ver-
gent
se-
quences.

CHAPTER
1.
SEQUENCES

a)

$$\{a_n\}$$

where

$$a_n = \frac{(2n+1)(n-2)}{n^3+2},$$

b)

$$\{b_n\}$$

where

$$b_n = \frac{2}{7n^2} - \frac{3n}{n^2} + \frac{2n}{7n+1}.$$

Point
out
care-
fully
the
places
where
the
the-
o-
rem
is
be-
ing
used.

10.

Consider
the
func-
tion
 f
where

$$f(x) = x^3 - 2.$$

When
New-
ton's
method
for
root
find-
ing
is
ap-
plied
to
 f
the
se-
quence
 $\{c_n\}$

1.4.

L'HÔPITAL'S

~~R~~*ULE*

is
pro-
duced
where
 c_n
is
de-
fined
by
the
re-
cur-
rence
re-
la-
tion

$$c_1 = 1,$$
$$c_{n+1} = \frac{2}{3}c_n + \frac{2}{3c_n^2}.$$

Newton's
method
claims
that
 $\{c_n\}$
should
con-
verge
to
 $2^{\frac{1}{3}}$,
the
root
of
 f .
Use
the
al-
ge-
bra
of
se-
quences
the-
o-
rem
to
prove
that:
if
 $\{c_n\}$
has
a
non-
zero
limit
then

CHAPTER

1.

SEQUENCES

that
limit
equals
 $2^{\frac{1}{3}}$.

11.

Investigate
the
long
term
be-
hav-
ior
of
the
fol-
low-
ing
se-
quences,
i.e.
are
they
in-
creas-
ing
or
de-
creas-
ing,
do
they
con-
verge
or
di-
verge,
are
they
bounded
or
un-
bounded?

a)

$$\left\{ \frac{1}{\sqrt{n^2 + 1} - \sqrt{n}} \right\}_{n=1}^{\infty},$$

b)

$$\left\{ \frac{\cos n}{n} \right\}_{n=1}^{\infty},$$

c)

$$\left\{ \sqrt[n]{n} \right\}_{n=1}^{\infty},$$

1.4.

L'HÔPITAL'S

~~89~~ RULE d)

$$\left\{ n \cos \left(\frac{\pi}{2n} \right) \right\}_{n=1}^{\infty}.$$

Chapter 2

Series

In
this
chap-
ter
we
build
on
the
con-
cepts
we
de-
vel-
oped
in
the
se-
quences
chap-
ter
and
ap-
ply
them
to
se-
ries.
We
will
look
at
the
con-
cept
of
the
con-
ver-
gence
of

CHAPTER

2.

~~S~~ERIES

an

in-

fi-

nite

se-

ries

and

re-

sults

that

give

var-

i-

ous

con-

di-

tions

for

their

con-

ver-

gence

or

di-

ver-

gence.

Two

im-

por-

tant

uses

of

se-

ries

are

in

func-

tions

ex-

pan-

sions

(such

as

Tay-

lor

se-

ries)

and

the

def-

i-

ni-

tion

of

the

Rie-

2.1.
BA-
SIC
DEFINITIONS
in-
te-
gral.

2.1 Basic def- i- ni- tions

A
se-
ries
is
the
name
given
to
the
pro-
cess
where
we
take
a
se-
quence
of
terms
and
add
them
all
to-
gether
term
by
term.

Definition

2.1.

A
*se-
ries*
is
the
sum
of
the
terms
of
a
se-
quence.

CHAPTER

2.

SERIES

We

can

rep-

re-

sent

se-

ries

us-

ing

ad-

di-

tion

and

sub-

trac-

tion

signs

as

in

$$1-2+3-4+5-6+7-8+9-10,$$

or

us-

ing

the

sigma

no-

ta-

tion

(Σ

is

the

cap-

i-

tal

Greek

let-

ter

sigma)

as

in

$$\sum_{k=1}^{10} (-1)^{k-1} k.$$

A

se-

ries

can

con-

tain

a

fi-

nite

num-

ber

2.1.

BA-

SIC

DEFINITIONS

terms

as

in

$\sum_{n=1}^N x_n$ or equivalently $x_1+x_2+\cdots+x_N$,

or

an

in-

fi-

nite

num-

ber

of

terms

as

in,

$\sum_{n=1}^{\infty} x_n$ or equivalently $x_1+x_2+\dots$

If

a

se-

ries

con-

tains

only

a

fi-

nite

num-

ber

of

terms

then

the

se-

ries

will

sum

to

a

fi-

nite

value

(as-

sum-

ing,

as

we

are

here,

that

the

CHAPTER

2.

~~SERIES~~

terms

of

the

se-

ries

are

all

fi-

nite

quan-

ti-

ties).

If

a

se-

ries

has

an

in-

fi-

nite

num-

ber

of

terms

then

the

se-

ries

may

or

may

not

sum

to

a

fi-

nite

value.

The

pre-

cise

def-

i-

ni-

tion

of

sum-

ming

to

a

fi-

nite

value

is

given

2.1.
 BA-
 SIC
 DEFINITIONS

the
 fol-
 low-
 ing.
 First
 we
 de-
 fine
 what
 is
 meant
 by
 the
 par-
 tial
 sums
 of
 an
 in-
 fi-
 nite
 se-
 ries

Definition

2.2
 (Par-
 tial
 sums).
 Con-
 sider
 an
 in-
 fi-
 nite
 se-
 ries
 $\sum_{n=1}^{\infty} x_n$.

The
 k^{th} -
partial
sum,
 S_k ,
 of
 the
 se-
 ries
 is
 the
 fi-
 nite
 se-
 ries

$$S_k = \sum_{n=1}^k x_n = x_1 + x_2 + \cdots + x_k.$$

2.2 Convergence**Definition****2.3**

(Con-

ver-

gence

for

se-

ries).

The

se-

ries

 $\sum_{n=1}^{\infty} x_n$

con-

verges

to

the

limit

(or

sum)

 L

if

the

se-

quence

of

its

par-

tial

sums,

 $\{S_k\}_{k=1}^{\infty}$,

is

a

con-

ver-

gent

se-

quence

and

$$\lim_{k \rightarrow \infty} S_k = L.$$

If

this

is

the

case

then

we

can

sim-

ply

write

$$\sum_{n=1}^{\infty} x_n = L.$$

2.2. CONVERGENCE

A
se-
ries
is
di-
ver-
gent
if
the
se-
quence
of
its
par-
tial
sums
is
a
di-
ver-
gent
se-
quence.

It
is
not
hard
to
think
of
ex-
am-
ples
of
di-
ver-
gent
se-
ries.

Example

2.1
(A
di-
ver-
gent
se-
ries).
Con-
sider
the
se-
ries

$$\sum_{m=1}^{\infty} m.$$

CHAPTER

2.

SERIES

This
se-
ries
is
clearly
di-
ver-
gent
as
the
par-
tial
sums
are
in-
creas-
ing
with-
out
any
up-
per
bound,
and
so
can-
not
form
a
con-
ver-
gent
se-
quence
(see
the-
o-
rem
[1.2](#)).

However

it
is
not
so
ob-
vi-
ous
that
con-
ver-
gent
se-
ries
ex-
ist

2.2. CONVERGENCE

(be-
yond
the
triv-
ial
ex-
am-
ple
of
a
se-
ries
where
the
terms
are
all
zero
af-
ter
some
point).

Example

2.2

(A
con-
ver-
gent
se-
ries).
Con-
sider
the
se-
ries

$$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m.$$

The
con-
ver-
gence
of
this
se-
ries
can
be
ap-
pre-
ci-
ated
when
we
in-

CHAPTER

2.

SERIES

ter-

pret

the

terms

as

mea-

sur-

ing

the

size

of

step

taken

along

a

line.

We

see

that

the

par-

tial

sums

of

the

se-

ries

get

closer

and

closer

to

1.

In-

deed

the

se-

ries

does

con-

verge

to

1.

This

can

be

proved

us-

ing

the

par-

tial

sum

for-

mula

for

2.2. CONVERGENCE

a
ge-
o-
met-
ric
se-
ries
and
mak-
ing
use
of
the
al-
ge-
bra
of
se-
quences
the-
o-
rem.

This
ex-
am-
ple
shows
that
it
is
pos-
si-
ble
to
have
an
in-
fi-
nite
num-
ber
of
terms
and
yet
for
them
still
to
sum
to
a
fi-
nite
value.

CHAPTER

2.

SERIES

This happens because the terms are getting smaller and smaller, the terms themselves are converging to zero, when viewed as sequence. However this type of behaviour does not guarantee convergence for a series, as the next example shows.

Example

2.2. CONVERGENCE

2.3

(A
di-
ver-
gent
se-
ries:
The
Har-
monic
Se-
ries).
Con-
sider
the
se-
ries

$$\sum_{m=1}^{\infty} \frac{1}{m}.$$

The
terms
of
this
se-
ries
are
in-
deed
con-
verg-
ing
to
zero,
but
yet
the
se-
ries
does
not
con-
verge.
It
in-
creases
with-
out
any
up-
per
bound,
al-
beit
very
slowly.

CHAPTER

2.

SERIES

This
fact
can
be
proved
with
the
aid
of
a
group-
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ment.

Note

that
chang-
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of
terms
of
a
se-
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will
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whether
a
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ries
is
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ver-
gent
or
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ver-
gent
(al-
though
it
may
well
change
the
value
of

2.2. CONVERGENCE

its
sum)
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for
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ries
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pends
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its
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term'
be-
haviour.

For
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CHAPTER

2.

SERIES

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Definition

2.4.

A

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$$\sum_{m=1}^{\infty} a_m$$

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$$\sum_{m=1}^{\infty} |a_m|,$$

is

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ries.

If

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$$\sum_{m=1}^{\infty} a_m$$

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2.2. CONVERGENCE

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ues,
 $\sum_{m=1}^{\infty} |a_m|$,
is
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then
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The
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proof.

Theorem
2.1.
If

CHAPTER

2.

SERIES

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sum.

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Proof.

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2.3 A

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$$\sum_{m=1}^{\infty} a_m = a_1 + a_2 + a_3 + a_4 + \dots$$

we

note

CHAPTER

2.

SERIES

that

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a_1

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$a_1 +$

δ_1

and

a_2

to

$a_2 +$

δ_2 .

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$\delta_1 +$

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sums,

$S_k =$

$\sum_{m=1}^k a_m,$

fails

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The

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CHAPTER

2.

SERIES

sums,

S'_k ,

of

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$k \geq$

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$S'_k =$

$S_k +$

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δ_2 ,

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CHAPTER

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Theorem

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ries

$\sum_{m=1}^{\infty} a_m$

and

$\sum_{m=1}^{\infty} b_m$

such

that

$$\sum_{m=1}^{\infty} a_m = A, \text{ and } \sum_{m=1}^{\infty} b_m = B.$$

Then

for

any

pair

$\alpha, \beta \in$

$\mathbb{R}$

we

have

a

CHAPTER

2.

SERIES

~~new~~

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ries

$$\sum_{m=1}^{\infty} (\alpha a_m + \beta b_m),$$

which

is

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$$\sum_{m=1}^{\infty} (\alpha a_m + \beta b_m) = \alpha A + \beta B.$$

Proof.

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2.4. SOME SE- RIES CON- VER- GENCE PRO- PERTIES

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sums.

Letting

$A_k =$

and

$B_k =$

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sums,

i.e.

$A_k =$

$\sum_{n=1}^k a_n$

and

$B_k =$

$\sum_{n=1}^k b_n$

we

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tions.

Direction of presentation

Why was the presentation reversed

Here? $\beta B = \lim_{k \rightarrow \infty} \beta B_k$

claimed limit $\alpha A + \beta B$ and worked our

way towards $\lim_{k \rightarrow \infty} (\alpha A_k + \beta B_k)$, by linearity of limits for sequences

$$\begin{aligned} &= \lim_{k \rightarrow \infty} \left(\alpha \sum_{n=1}^k a_n + \beta \sum_{n=1}^k b_n \right) \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k (\alpha a_n + \beta b_n), \text{ manipulation of sum} \\ &= \sum_{n=1}^{\infty} (\alpha a_n + \beta b_n), \text{ by def. of series convergence.} \end{aligned}$$

CHAPTER

2.

~~SERIES~~



2.5 Convergence tests

We will introduce various tests that can be used to assess the convergence / divergence of a series.

re of a statement $A \Rightarrow$
nt $\neg B \Rightarrow \neg A$. These
e *logically equivalent*,
have the same truth
true then so is the
sult one is trying to
rm of an implication,
the contrapositive of
e easier to prove di-

Theorem

2.3

(General term test).

If a sequence has a non-zero limit (or no limit at all) then the

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CHAPTER

2.

SERIES

$$\lim_{m \rightarrow \infty} a_m \neq 0 \Rightarrow \sum_{m=1}^{\infty} a_m \text{ diverges,}$$

•

$$\sum_{m=1}^{\infty} a_m \text{ converges} \Rightarrow \lim_{m \rightarrow \infty} a_m = 0.$$

Proof.

Suppose

that

$$\sum_{n=1}^{\infty} a_n$$

con-

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and

let

$$A_k$$

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$$A_k \rightarrow$$

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$$K$$

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have

$$|A_k - a| <$$

$$\epsilon/2.$$

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 $A_{k+1}| =$
 $|a_{k+1}| <$
 $\epsilon,$
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 $a_n \rightarrow$
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 $\infty.$
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CHAPTER

2.

SERIES

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$$\sum_{m=1}^{\infty} \frac{1}{m},$$

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$$\frac{1}{m} \rightarrow$$

0

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$$m \rightarrow$$

∞ .

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2.5.
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$\sum_{n=1}^{\infty} ar^{n-1}$

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if

$a =$

0.

If

$a \neq$

0

then

it

con-

verges

if

and

only

if

$|r| <$

1,

2.5.
CON-
VER-
GENCE
TESTS
which
case

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}.$$

Proof.

The
con-
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case

$a =$
0
is
clear.

So
we
as-
sume
that
 $a \neq$
0.

If
 $r =$
1
then
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clear

as
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is
just

$a +$
 $a +$
 $a +$
 $a +$
....

So
as-
sume
 $r \neq$

1
and
let
 A_k
de-
note
the
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tial

CHAPTER

2.

SERIES

sum,

i.e.

$A_k =$

$\sum_{n=1}^k a_n.$

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rA_k

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$$A_k = \frac{a(1 - r^k)}{1 - r}.$$

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$\sum_{n=1}^{\infty} ar^{n-1} = \lim_{k \rightarrow \infty} A_k = \lim_{k \rightarrow \infty} \frac{a(1 - r^k)}{1 - r} = \frac{a(1 - \lim_{k \rightarrow \infty} r^k)}{1 - r}.$

If

$r =$

-1

or

$|r| >$

1

then

r^k

di-

verges,

and

hence

so

does

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If

$|r| <$

1

then

$r^k \rightarrow$

0

as

$k \rightarrow$

$\infty,$

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$a/(1 -$

$r),$

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□

Theorem

2.5

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CHAPTER

2.

SERIES

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$\sum_{m=1}^{\infty} a_m$

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that

$a_m =$

$f(m)$

for

all

$m \geq$

1.

If

the

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gral

$$\int_1^{\infty} f(t) dt = \lim_{T \rightarrow \infty} \int_1^T f(t) dt$$

exists

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2.5.
CON-
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TESTS
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Proof.
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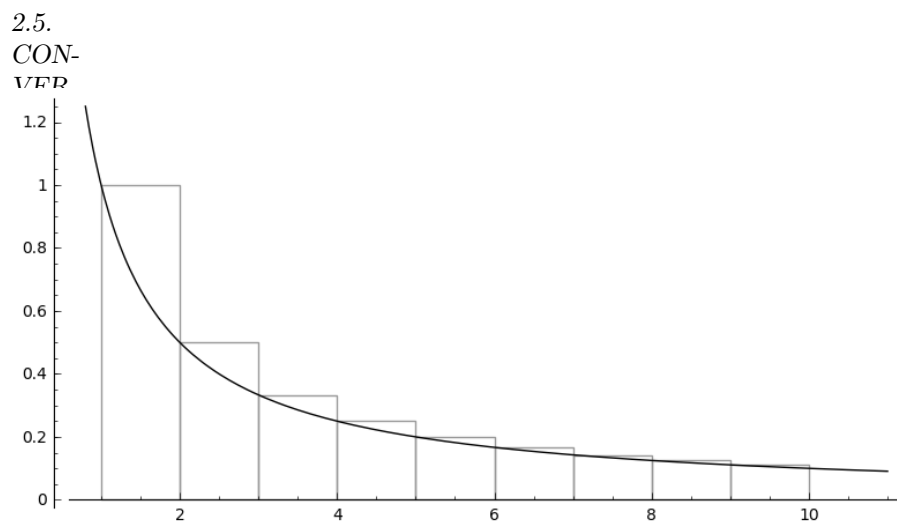


Figure 2.1:
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 $a_i =$
 $f(i)$.
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 $\int_1^\infty f(t) dt$
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CHAPTER

2.

SERIES

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2.5. CON- VER- GENCE TESTS

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Euler-Mascheroni constant

in the case, where the function f is
the reciprocal function $f(t) = 1/t$,
the quantity γ is known as the Euler-
Mascheroni constant, an important
quantity in analysis and number the-
ory where it occurs in many settings.
Note that it is still not known whether γ is irra-
tional or not, though most would bet
is that it is.

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CHAPTER

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$\sum_{m=1}^{\infty} a_m$

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$a_m \geq$

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$\sum_{m=1}^{\infty} c_m$

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terms.

Then

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all

$m \geq$

1,

$a_m \leq$

c_m ,

then

$\sum_{m=1}^{\infty} a_m$
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$m \geq$

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$a_m \geq$

d_m ,

then

$\sum_{m=1}^{\infty} a_m$
is

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Good and bad behaviour

~~Proof~~ A good way to think about/remem-
ber this result is in terms of the good
and bad behaviour shown by a series.
We sketch

A convergent series $\sum c_m$ of positive
terms converges because the terms c_m
behave well (i.e. are small enough). If
 $a_m \leq c_m$ then the terms a_m are be-
having better and so $\sum a_m$ also con-
verges.

Similarly a divergent series $\sum d_m$ di-
verges because its terms behave badly
(i.e. they're too large), so if $a_m \geq d_m$
then the terms a_m are behaving worse,
and so $\sum a_m$ also diverges.

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2.

SERIES

Let

c

de-

note

the

sum

of

the

con-

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com-

para-

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ries,

i.e.

$$\sum_{m=1}^{\infty} c_m =$$

c .

Then

the

par-

tial

sums

$$\sum_{m=1}^k c_m$$

of

the

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in-

creas-

ing

se-

quence

con-

verg-

ing

to

c .

Sup-

pose

that

for

all

$$m \geq$$

1

we

have

$$a_m \leq$$

c_m .

Since

each

2.5.
 CON-
 VER-
 GENCE
 OF
 TESTS
 sum
 $\sum_{m=1}^k a_m$
 is
 bounded
 above
 by
 c
 (as
 c
 is
 an
 upper-
 bound
 for
 the
 par-
 tial
 sum
 $\sum_{m=1}^k c_m$),
 these
 par-
 tial
 sums
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 which
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 bounded
 above.
 There-
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 tial
 sums
 $\sum_{m=1}^k a_m$
 is
 con-
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 gent
 as
 $k \rightarrow$
 ∞ ,
 and

CHAPTER

2.

SERIES

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to

a

limit

a

which

also

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$a \leq$

c .

The

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sums,

$\sum_{m=1}^k d_m,$

of

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quence.

Sup-

pose

that

for

all

$m \geq$

1

we

have

$a_m \geq$

d_m .

Then

the

par-

2.5.
CON-
VER-
GENCE
TESTS
sat-
isfy

$$\sum_{m=1}^k a_m \geq \sum_{m=1}^k d_m,$$

and
hence
the
par-
tial
sums
 $\sum_{m=1}^k a_m$
also
form
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creas-
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bounded
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quence,
hence
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ver-
gent.
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Theorem

2.7

(Limit
com-
par-
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son
test).

*Sup-
pose
that
 $\sum_{m=1}^{\infty} a_m$
is*

*a
se-
ries
con-
sist-
ing
of
non-
negative
terms,
i. e.
for*

CHAPTER

2.

SERIES

all

m ,

$a_m \geq$

0,

and

that

$\sum_{m=1}^{\infty} c_m$

is

a

known

con-

ver-

gent

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ries

of

strictly

pos-

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tive

terms

(i.e.

for

all

m ,

$c_m >$

0)

and

$\sum_{m=1}^{\infty} d_m$

is

a

known

di-

ver-

gent

se-

ries

of

strictly

pos-

i-

tive

terms.

Then

•

if
 $\lim_{m \rightarrow \infty} \frac{a_m}{c_m} =$

L

and

$0 \leq$

$L <$

∞

then

$\sum_{m=1}^{\infty} a_m$

is

2.5. CON- VER- GENCE^{con-} TESTS^{ver-} gent,

•

if
 $\lim_{m \rightarrow \infty} \frac{a_m}{d_m} =$

L

and

$0 <$

$L \leq$

∞

,

then

$\sum_{m=1}^{\infty} a_m$

is

di-

ver-

gent.

Good/bad behaviour

Again this result can be boiled
 down to good/bad behaviour. If
 this $\lim_{m \rightarrow \infty} a_m/x_m = L$ then *eventually*
 limit a_m acts like a scalar multiple of x_m .
 can if $\sum x_m$ behaves well (or badly),
 can does $\sum a_m$.

i-

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test

can

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First

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prove

CHAPTER

2.

~~SERIES~~

the
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gent
case.

By
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of

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tios

a_m/c_m

there
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ists

a
pos-
i-
tive
in-
te-
ger

M
such
that
for
all
 $m \geq$
 M

we
have
 $|a_m/c_m -$
 $L| <$

1,

hence

$a_m/c_m <$
 $L +$

1

and

hence

$a_m <$
 $(L +$
1) c_m .

There-

fore

the

stan-
dard

com-

par-

i-

son

test

2.5.
 CON-
 VER-
 GENCE
 says
 that
 the
 TESTS
 the
 tail
 se-
 ries
 $\sum_{m=M}^{\infty} a_m$
 con-
 verges
 and
 hence
 so
 does
 the
 orig-
 i-
 nal
 se-
 ries
 $\sum_{m=1}^{\infty} a_m$.

The
 di-
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 man-
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 By
 the
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 ver-
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 a_m/d_m
 there
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 in-
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 ger
 M

CHAPTER

2.

SERIES

such

that

for

all

$m \geq$

M

we

have

$|a_m/d_m -$

$L| <$

$L/2,$

hence

$a_m/d_m >$

$L/2$

and

hence

$a_m >$

$(L/2)d_m.$

There-

fore

the

stan-

dard

com-

par-

i-

son

test

says

that

the

tail

se-

ries

$\sum_{m=M}^{\infty} a_m$

di-

verges

and

hence

so

does

the

orig-

i-

nal

se-

ries

$\sum_{m=1}^{\infty} a_m.$

□

The

choice

for

the

com-

par-

2.5.
CON-
VER-
GENCE
TESTS

se-
ries
is
of-
ten
a
ge-
o-
met-
ric
se-
ries
or
a
*hyper-
harmonic*
se-
ries

$$\sum_{m=1}^{\infty} \frac{1}{m^p}$$

for
some
 p .

Us-
ing
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we
can
show
that
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se-
ries
con-
verges
when
 $p >$
1
and
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The
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CHAPTER

2.

SERIES

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ries

$$\sum_{m=1}^{\infty} \frac{(-1)^n}{m^p}$$

converges

ab-

so-

lutely

when

$p >$

1,

con-

verges

con-

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tion-

ally

when

$0 <$

$p \leq$

1

and

di-

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wise.

Theorem

2.8

(Al-

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ing

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ries

test).

Con-

sider

the

al-

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ries

$$\sum_{m=1}^{\infty} (-1)^m a_m,$$

where

the

2.5.
CON-
VER-
GENCE
TESTS

all
pos-
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tive
and
de-
creas-
ing,
i.e.
for
all
 $m \geq$
1,
 $a_m \geq$
 $a_{m+1} \geq$
0.
If
 $\lim_{m \rightarrow \infty} a_m =$
0
then
the
orig-
i-
nal
al-
ter-
nat-
ing
se-
ries
is
con-
ver-
gent.
But
if
 $\lim_{m \rightarrow \infty} a_m \neq$
0
then
the
al-
ter-
nat-
ing
se-
ries
is
di-
ver-
gent.

Proof.
We
give

CHAPTER

2.

SERIES

a

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of

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main

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The

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tial

sums,

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[2.2](#)

The

par-

tial

sums

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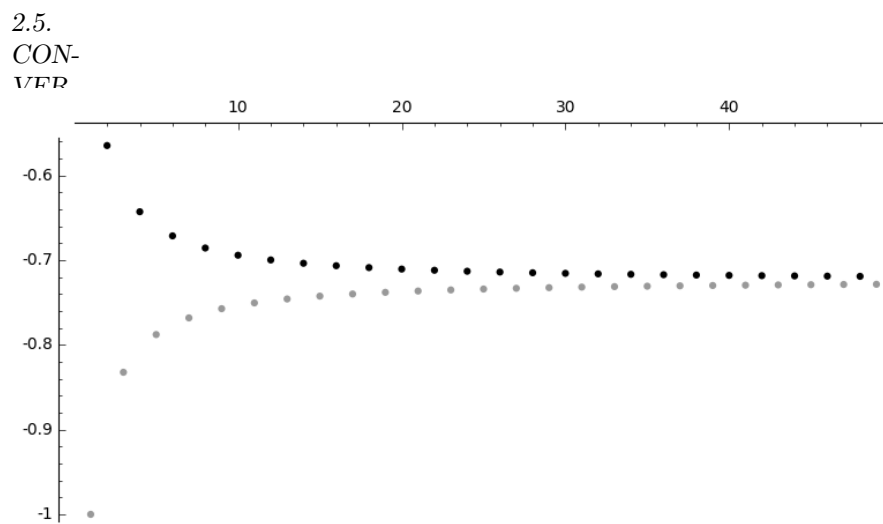


Figure 2.2:

The
par-
tial
sums,
 A_k ,
of
a
con-
ver-
gent
al-
ter-
nat-
ing
se-
ries

into
two
sub-
se-
quences,
those
of
odd
in-
dex,
 $A_1, A_3, A_5, \dots$,
shown
in
grey,
and
those
of
even
in-
dex,
 $A_2, A_4, A_6, \dots$,
shown

CHAPTER

2.

~~SERIES~~

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black.

Due

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2.5.
 CON-
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 TESTS
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 which
 is
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 low.
 By
 the
 Mono-
 tone
 Con-
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 gence
 The-
 o-
 rem
 each
 sub-
 se-
 quence
 is
 con-
 ver-
 gent,
 i.e.
 $A_{2k} \rightarrow$
 L_{even}
 and
 $A_{2k+1} \rightarrow$
 L_{odd}
 as
 $k \rightarrow$

CHAPTER

2.

SERIES

∞ .

But

since

the

step

size,

$|a_m|$,

con-

verges

to

zero,

the

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fer-

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be-

tween

L_{even}

and

L_{odd}

must

be

ar-

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trar-

ily

small,

i.e.

zero

and

hence

$L_{\text{even}} =$

L_{odd}

and

the

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se-

quence

of

par-

tial

sums

is

con-

ver-

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to

this

limit,

i.e.

the

se-

ries

is

con-

2.5.
CON-
VER-
GENCE
TESTS
□

Theorem

2.9

(Ra-
tio
test).

*Con-
sider
the*

*se-
ries*
 $\sum_{m=1}^{\infty} a_m$
of

*pos-
i-
tive
terms,*

i.e.

*for
all*

m,
 $a_m >$
 $0.$

*Sup-
pose
that
the
se-
quence*

*of
the
ra-
tios
of
con-
sec-
tu-
tive
terms*

*is
con-
ver-
gent
and
that*

$$\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = L.$$

*If
 $0 \leq$
 $L <$
 1
then
the*

CHAPTER

2.

SERIES

se-

ries

is

con-

ver-

gent.

If

$L >$

1

then

the

se-

ries

is

di-

ver-

gent.

For

the

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case

$L =$

1

we

can

not

draw

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about

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gence.

Proof.

The

proof

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2.5.
 CON-
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 and
 the
 use
 of
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 ge-
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 ric
 se-
 ries
 test.

Learning this result

This result boils down to the key idea:
 Suppose
 if $\lim_{m \rightarrow \infty} a_{m+1}/a_m = L$ then $\sum a_m$ is
 similar (compares to) to a geometric
 series with common ratio L . Hence
 $\sum a_m$ converges or diverges accord-
 ingly as $L < 1$ or $L > 1$.

the
 con-
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 o-
 rem
 and
 that

$$\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = L < 1.$$

Choose
 a

CHAPTER

2.

SERIES

suit-
able

ϵ ,

say

$\epsilon =$

$(1 -$
 $L)/2$.

Then

there

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M

such

that

for

all

$m \geq$

M

we

have

$|a_{m+1}/a_m -$

$L| <$

$(1 -$

$L)/2$,

i.e.

$a_{m+1}/a_m <$

$(L +$

$1)/2 <$

1.

Hence,

for

all

$m \geq$

M

we

have

$$a_{m+1} < \frac{L+1}{2} a_m.$$

Repeatedly

ap-

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lows

2.5.
CON-
VER-
GENCE
TESTS

We
say
that
for
all
 $n \geq$
1
we
have

$$a_{M+n} < \left(\frac{L+1}{2} \right)^n a_M.$$

Note
that
the
se-
ries
 $\sum_{n=1}^{\infty} \left(\frac{L+1}{2} \right)^n a_M$
is

a
con-
ver-
gent
ge-
o-
met-
ric
se-
ries,
since
 $\frac{L+1}{2} <$
1.

So
by
the
stan-
dard
com-
par-
i-
son
test
the
se-
ries
 $\sum_{n=1}^{\infty} a_{M+n}$
is
also
con-
ver-
gent.
There-
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CHAPTER

2.

SERIES

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ries

$\sum_{m=1}^{\infty} a_m$

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and

hence

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quired.

On

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other

hand

is

we

sup-

pose

that

$$\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = L > 1,$$

then

we

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that

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2.5.
 CON-
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 TESTS
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 se-
 ries
 $\sum_{n=1}^{\infty} \left(\frac{L+1}{2}\right)^n a_M,$
 for
 some
 M
 (re-
 mem-
 ber
 this
 time
 $(L+1)/2 >$
 $1)$.
 This
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 plies
 that
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 ries
 has
 a
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 ver-
 gent
 tail
 and
 hence
 is
 it-
 self
 di-
 ver-
 gent,
 as
 re-
 quired.
 In
 the
 re-
 main-
 ing
 case
 where
 $\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = 1,$
 then
 try-

CHAPTER

2.

~~SERIES~~

ing

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above

ideas

will

only

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say

that

$\sum a_m$

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2.5.
CON-
VER-
GENCE
TESTS



Theorem

2.10

(Root
test).

*Con-
sider
the*

*se-
ries*

$\sum_{m=1}^{\infty} a_m$
of

non-

*negative
terms,*

i.e.

for

all

m,

$a_m \geq$

$0.$

*Sup-
pose*

that

$\lim_{m \rightarrow \infty} (a_m)^{\frac{1}{m}} = L.$

If

$0 \leq$

$L <$

1

then

the

orig-

i-

nal

se-

ries

is

con-

ver-

gent,

if

$1 <$

$L \leq$

∞

then

the

se-

ries

di-

verges.

For

the

CHAPTER

2.

SERIES

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main-

ing

case

$L =$

1

we

can

not

draw

a

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about

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con-

ver-

gence.

Proof.

We

can

prove

this

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ing

$\sum a_m$

to

a

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L .

Hence

the

re-

appearance

of

the

$L <$

1

and

$L >$

2.6.
POWER
SERIES

The
de-
tails
are
left
as
an
ex-
er-
cise.
□

Example
2.4.

We
shall
con-
sider
ex-
am-
ples
of
the
ap-
pli-
ca-
tions
of
these
tests
in
the
lec-
tures.

2.6 Power
se-
ries

Using
se-
ries
we
can
de-
fine
a
class
of
func-
tions
called

CHAPTER

2.

~~SERIES~~

power

se-

ries.

We

shall

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some

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sic

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am-

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here

and

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turn

to

this

topic

later

when

we

are

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the

dif-

fer-

en-

tial

cal-

cu-

lus.

Definition

2.5.

A

se-

quence

of

terms

$\{a_m\}_{m=1}^{\infty}$

de-

fines

a

power

se-

ries

func-

tion

A

by

the

for-

2.6.

POWER

~~SERIES~~

mula

$$A(x) = \sum_{m=1}^{\infty} a_m x^m.$$

Of

course

this

for-

mula

for

$A(x)$

may

only

make

sense

for

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tain

val-

ues

of

the

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gu-

ment

x .

For

some

val-

ues

of

x

the

se-

ries

may

con-

verge

but

for

some

val-

ues

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may

di-

verge.

We

can

in-

ves-

ti-

gate

this

mat-

ter

CHAPTER

2.

SERIES

us-
ing
the
ra-
tio
test.

According

to
the
ra-
tio
test
the
se-
ries
for-
mula
for
 $A(x)$
will
con-
verge
when

$$\lim_{m \rightarrow \infty} \left| \frac{a_{m+1}x^{m+1}}{a_mx^m} \right| < 1,$$

which
is
equiv-
a-
lent,
af-
ter
sim-
pli-
fy-
ing,
to

$$|x| < \lim_{m \rightarrow \infty} \left| \frac{a_m}{a_{m+1}} \right|.$$

Definition

2.6.

The
quan-
tity
 R
de-
fined
by

$$R = \lim_{m \rightarrow \infty} \left| \frac{a_m}{a_{m+1}} \right|,$$

is
known

2.6.
 POWER
~~SERIES~~
 as
 the
ra-
dious
of
con-
ver-
gence
 of
 the
 se-
 ries
 $\sum_{m=1}^{\infty} a_m x^m$.
 The
 se-
 ries
 con-
 verges
 if
 $|x| <$
 R
 and
 di-
 verges
 if
 $|x| >$
 R .

Note
 that
 the
ra-
tio
 test
 does
 not
 tell
 us
 about
 the
 con-
 ver-
 gence
 of
 the
 se-
 ries
 when
 $|x| =$
 R .
 That
 will
 have
 to
 be
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CHAPTER

2.

SERIES

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by
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ods.

An
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tant
source
of
Power
se-
ries
comes
from
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ply-
ing
Tay-
lor's
the-
o-
rem.

Theorem

2.11

(Tay-
lor's
the-
o-
rem).

*Sup-
pose
that
on
an
in-
ter-
val
[a, x]
the
func-
tion
f
can
be
dif-
fer-
en-
ti-
ated
n+
1
times,*

2.6.
POWER
SERIES

for
some
 $n \geq$
0.

Then
there
ex-
ists

a
num-
ber
 $c \in$

(a, x)
such
that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n,$$

where
 R_n
is
the

La-
grange
re-
main-
der
and

is
given
by

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Taylor's

the-
o-
rem
pro-
vides

us
with
a
way
of
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tions

CHAPTER

2.

SERIES

to

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tion

val-

ues.

By

writ-

ing

$a +$

x

for

x

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above

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sion

of

f

about

the

point

a :

$$f(a+x) = f(a) + f'(a)x + \frac{f''(a)}{2!}x^2 + \cdots + \frac{f^{(n)}(a)}{n!}x^n + R_n.$$

It

is

of-

ten

the

case

that

$R_n(x) \rightarrow$

0

as

2.6.
~~POWER~~
~~SERIES~~
 $n \rightarrow$

∞
 and
 then
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 may
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lor
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ex-
pan-
sion
about
the
point
a:

$$f(a+x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} x^n.$$

Two
 im-
 por-
 tant
 things
 to
 note
 are
 that
 for
 all
 $x \in$
 $\mathbb{R}$
 we
 have

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

and
 thatn
 if
 $-1 <$
 $x <$
 1
 then

$$\lim_{n \rightarrow \infty} x^n = 0.$$

CHAPTER

2.

SERIES

The

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cial

case

where

we

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func-

tion

about

the

point

$a =$

0

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called

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Maclau-

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ries.

Definition

2.7.

The

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of

a

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tion

f

about

the

point

0

is

called

the

Maclau-

rin

se-

ries

of

2.6.
~~POWER~~
~~SERIES~~
~~f~~,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$$

We
 shall
 re-
 turn
 to
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 topic
 of
 Tay-
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In
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[2.1](#)
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 Maclau-
 rin
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 ries
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 dard
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 tions.

CHAPTER

2.

SERIES.

Maclarin series	Interval of convergence
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$	$(-\infty, \infty)$
$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$	$(-\infty, \infty)$
$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$	$(-\infty, \infty)$
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$	$(-1, 1)$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^k}{n}$	$(-1, 1)$

Table 2.1:

Some
stan-
dard
Maclau-
rin
se-
ries

Exercises

2.1.

These
prob-
lems
re-
quire
the
ap-
pli-
ca-
tion
of
var-
i-
ous
con-
cepts
and
re-
sults
from
this
chap-
ter.
We

2.6.
~~POWER~~
~~SERIES~~
 shall
 use
 these
 as
 a
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 sion
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 ri-
 als.

1. Investigate
 the
 con-
 ver-
 gence
 of
 the
 fol-
 low-
 ing
 se-
 ries,

a)

$$\sum_{n=1}^{\infty} (2n-1) = 1+3+5+7+9+\dots,$$

b)

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\dots,$$

c)

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\frac{1}{20}+\dots$$

2. Using
 a
 di-
 a-
 gram

CHAPTER

2.

SERIES

of
the
num-
ber
line
in-
ves-
ti-
gate
the
con-
ver-
gence
of
the
fol-
low-
ing
se-
ries,

$$(a) \sum_{n=1}^{\infty} (-1)^n, \quad (b) \sum_{n=1}^{\infty} (-1)^n \left(\frac{n+1}{n+2} \right).$$

3.

The
ge-
o-
met-
ric
se-
ries
 $\sum_{n=1}^{\infty} ar^{n-1}$
has
par-
tial
sums
 S_k
given
by

$$S_k = \sum_{n=1}^k ar^{n-1} = \frac{a(1-r^k)}{1-r}.$$

Use
this
re-
sult
to
in-
ves-
ti-
gate
the

2.6.

POWER

~~SERIES~~

con-

ver-

gence

of

the

se-

ries

$\sum_{n=1}^{\infty} nar^{n-1}$.

Hint:

Let

T_k

de-

note

k^{th}

par-

tial

sum

of

this

se-

ries.

Con-

sider

the

quan-

tity

$(1 -$

$r)T_k \dots$

4.

Recall

from

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ture

that

the

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met-

ric

se-

ries

$\sum_{n=1}^{\infty} ar^{n-1}$

has

par-

tial

sums

S_k

given

by

$$S_k = \sum_{n=1}^k ar^{n-1} = \frac{a(1 - r^k)}{1 - r}.$$

Therefore

CHAPTER

2.

SERIES

the
se-
ries
is
con-
ver-
gent
only
when
 $|r| <$
1,
in
which
case
the
in-
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nite
se-
ries
sums
to
 $\frac{a}{1-r}$.

For
the
fol-
low-
ing
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tions
of
 x_n
de-
ter-
mine
whether
the
se-
ries
 $\sum_{n=1}^{\infty} x_n$
con-
verges
and
if
so
find
the
sum
of
the
se-
ries.

a)

2.6.

POWER

SERIES

$$x_n = \frac{2}{3} \left(\frac{1}{2}\right)^n - \frac{1}{4} \left(\frac{2}{3}\right)^n.$$

b)

$$x_n = \left(\frac{3}{2}\right)^{2n} - \frac{1}{2} \left(\frac{2}{3}\right)^{n+1}$$

5.

Consider

the

se-

ries

$$\sum_{n=1}^{\infty} \frac{2}{n^2 + 2n}.$$

Show

that

this

is

a

con-

ver-

gent

se-

ries

by

find-

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the

par-

tial

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pan-

sion

of

$$\frac{2}{n^2 + 2n}$$

and

ex-

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ing

the

k^{th}

par-

tial

sum

S_k .

6.

Consider

the

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CHAPTER

2.

~~SE~~RIES

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sion

$2.104104104104\dots$

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sion.

7.

Determine
whether
the
fol-
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ing
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ries
is
con-
ver-
gent

2.6.

POWER

~~SERIES~~

or

di-

ver-

gent

$$\sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)}.$$

8.

Determine

whether

the

fol-

low-

ing

se-

ries

are

con-

ver-

gent

or

di-

ver-

gent

a)

$$\sum_{n=1}^{\infty} \frac{2}{n!},$$

b)

$$\sum_{n=1}^{\infty} \frac{n!}{n^n},$$

c)

$$\sum_{n=1}^{\infty} \left(\frac{2n-1}{3+4n} \right)^n,$$

d)

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}.$$

9.

Determine

whether

the

fol-

low-

ing

se-

ries

CHAPTER

2.

~~SERIES~~

con-
verges
or
di-
verges:

$$\sum_{n=1}^{\infty} a_n, \quad \text{where } a_n = \frac{2n^2 + 4}{3n^3 - 2n}.$$

10.

Determine
whether
the
fol-
low-
ing
se-
ries
con-
verges
or
di-
verges:

$$\sum_{n=1}^{\infty} b_n \text{ where } b_n = \frac{2n + 5}{\sqrt{n^6 + n}}.$$

11.

Determine
whether
the
fol-
low-
ing
se-
ries
con-
verges
or
di-
verges:

$$\sum_{n=1}^{\infty} c_n \text{ where } c_n = \frac{n!}{5^{n+2}}.$$

12.

Determine
whether
the
fol-
low-
ing
se-
ries

2.6.

POWER

~~SERIES~~

con-

verges

or

di-

verges:

$$\sum_{n=1}^{\infty} d_n \text{ where } d_n = \frac{n^2 2^{n+5}}{2^{2n}}.$$

13.

Determine

whether

the

fol-

low-

ing

se-

ries

con-

verges

or

di-

verges:

$$\sum_{n=1}^{\infty} e_n \text{ where } e_n = \frac{\sin^2(n)}{n(n+3)}.$$

14.

Determine

whether

the

fol-

low-

ing

se-

ries

con-

verges

or

di-

verges:

$$\sum_{n=1}^{\infty} f_n \text{ where } f_n = \frac{\sqrt{n+1}}{n\sqrt{n+4}}.$$

15.

Determine

whether

the

fol-

low-

ing

se-

ries

con-

verges

CHAPTER

2.

~~SERIES~~

or

di-

verges:

$$\sum_{n=1}^{\infty} g_n \text{ where } g_n = \frac{n2^n}{n!}.$$

16.

Write

down

the

MacLau-

rin

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ries

(i.e.

Tay-

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ries

about

$x =$

0)

for

the

ex-

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tial

func-

tion,

e^x .

Hence

find

the

dec-

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to

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dec-

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places.

17.

Find

2.6.

POWER

~~SERIES~~

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prox-
i-
ma-
tion

$$\sin x \approx x - \frac{x^3}{6},$$

given
that
 $|x| < \frac{1}{2}$.

18.

Approximate
 $\sin \frac{\pi}{5}$
to
four
dec-
i-
mal
places
us-
ing
an
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pro-
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ate
Tay-
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Chapter 3

Functions

You
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be
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the
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tion
of
a
func-
tion
in
math-
e-
mat-
ics
and
have
used
them
many
times
up
to
now.
But
in
this
chap-
ter
we
just
wish
to
for-
malise
the

CHAPTER
3.
FUNCTIONS

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of
a
func-
tion
and
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prop-
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of
func-
tions.
Through-
out
this
we
wish
to
em-
pha-
size
the
treat-
ment
of
func-
tions
as
math-
e-
mat-
i-
cal
ob-
jects
in
them-
selves.

Definition

3.1
(Func-
tion).
A
func-
tion
f

from
a
set
 X
to
a
set
 Y
is
a
rule
which
as-
signs
el-
e-
ments
of
the
set
 Y
to
el-
e-
ments
of
 X .
It
must
do
this
in
such
a
way
that
to
each
el-
e-
ment
of
 X
there
is
one
and
only
one
el-
e-
ment
of
 Y
as-
so-

CHAPTER
3.
FUNCTIONS
ci-
ated
to
it.

This
con-
di-
tion
is
some-
times
re-
ferred
to
as
say-
ing
that
func-
tions
are
*single-
valued*.

Definition

3.2
(Func-
tion
ter-
mi-
nol-
ogy
and
no-
ta-
tion).
We
can
write

$$f : X \rightarrow Y$$

to
de-
note
that
 f
is
a
func-
tion
from
the
set
 X
to

the
set
 Y .
The
set
 X
is
called
the
do-
main
of
 f
and
 Y
is
called
the
co-
domain
of
 f .
An
el-
e-
ment
 $x \in$
 X
is
called
an
ar-
gu-
ment
of
the
func-
tion
 f
and
we
use
 $f(x)$
to
de-
note
the
as-
so-
ci-
ated
el-
e-
ment
of
 Y ,

CHAPTER
 3.
 FUNCTIONS
 called
 the
value
 of
 the
 func-
 tion
 f
 at
 the
 ar-
 gu-
 ment
 x .

Note
 that
 there
 is
 no
 re-
 quire-
 ment
 that
all
 the
 el-
 e-
 ments
 of
 the
 co-
 domain
 oc-
 cur
 as
 val-
 ues
 of
 the
 func-
 tion
 but
 there
is
 a
 re-
 quire-
 ment
 that
 a
 value
 $f(x)$
 is
 de-
 fined

for
 ev-
 ery
 el-
 e-
 men-
 t/ar-
 gu-
 ment
 x
 from
 the
 do-
 main
 of
 f .

Example

3.1.

As
 men-
 tioned
 above
 you
 will
 be
 fa-
 mil-
 iar
 with
 these
 con-
 cepts
 from
 your
 study
 of
 math-
 e-
 mat-
 ics
 up
 to
 now.

1. The
 square
 root
 func-
 tion
 $s :$
 $\mathbb{R}^+ \rightarrow$
 $\mathbb{R}$
 is
 the
 func-

CHAPTER

3.

FUNCTIONS

tion
from
the
do-
main
 $\mathbb{R}^+$
of
pos-
i-
tive
real
num-
bers
to
the
co-
domain
 $\mathbb{R}$,
de-
fined
by
the
for-
mula

for all $x \in \mathbb{R}^+$, $s(x) = \sqrt{x}$.

2.

The
trigono-
met-
ric
sine
func-
tion,
 $\sin :$
 $\mathbb{R} \rightarrow$
 $[-1, 1]$.
This
can
be
de-
fined
in
many
ways.
One
way
is
to
de-
fine
 $\sin(\theta)$
as
the
 y -
coordinate

of
 the
 point
 on
 the
 unit
 circle
 in
 the
 plane
 which
 makes
 an
 angle
 of
 θ
 radians
 with
 the
 positive
 direction
 of
 the
 x -
 axis,
 see
 figure
[3.1](#).

Definition

3.3

(Range
 /
 image).
 The
range
 or
image
 of
 a
 function
 $g :$
 $V \rightarrow$
 W

CHAPTER
3.
FUNCTIONS

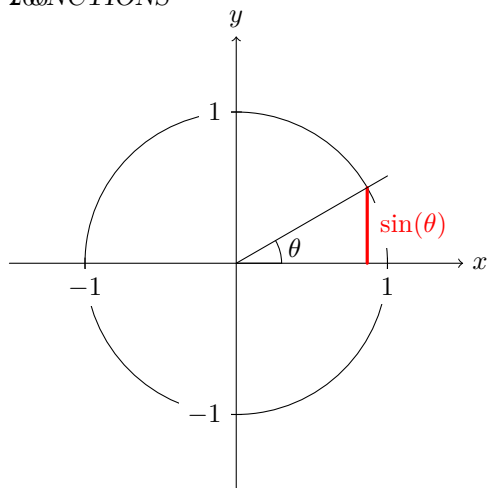


Figure 3.1:

A
def-
i-
ni-
tion
of
 $\sin(\theta)$.

is
the
sub-
set
of
the
co-
domain
 W
con-
sist-
ing
of
the
val-
ues
of
 g ,
i.e.

$$\text{im}(g) = \{ g(v) : v \in V \}.$$

When
dis-
cussing
par-
tic-
u-
lar

func-
tions
it
is
of-
ten
the
case
that
the
do-
main
and
co-
domain
will
not
be
ex-
plic-
itly
men-
tioned.
The
do-
main
for
the
func-
tion
should
be
clear
from
the
con-
text,
i.e.
to
what
set
of
ar-
gu-
ments
is
the
func-
tion
in-
tended
to
ap-
ply.
The
pre-

CHAPTER
3.
FUNCTIONS
cise
de-
scrip-
tion
of
the
co-
domain
is
a
bit
trick-
ier
as
the
choice
of
the
co-
domain
is
some-
what
ar-
bi-
trary
as
long
as
it
con-
tains
the
im-
age
of
the
func-
tion.
How-
ever
for
some
prop-
er-
ties
and
op-
er-
a-
tion
such
as
sur-
jec-
tiv-

ity
 and
com-
po-
si-
tion
 some
 con-
 sid-
 er-
 a-
 tion
 will
 need
 to
 be
 given
 to
 the
 ap-
 pro-
 pri-
 ate
 co-
 domain
 of
 the
 func-
 tion.

Definition

3.4

(In-
 jec-
 tive
 /
 one-
 to-one).

A
 func-
 tion
 $g :$
 $U \rightarrow$
 V
 is
in-
jec-
tive
 if
 and
 only
 if
 no
 two
 el-
 e-

CHAPTER
3.
FUNCTIONS

ments
from
the
do-
main
 U
get
mapped
to
the
same
value
in
the
co-
domain
 V .
This
con-
di-
tion
can
be
ex-
pressed
for-
mally
as:

$$\forall u_1, u_2 \in U \quad g(u_1) = g(u_2) \Rightarrow u_1 = u_2.$$

A
more
in-
for-
mal,
but
de-
scrip-
tive,
term
for
this
prop-
erty
is
to
say
that
such
a
func-
tion
 g
is
one-

*to-
one.*

Definition

3.5

(Sur-
jec-
tive
/
onto).

A
func-
tion

$g :$
 $U \rightarrow$
 V

is
*sur-
jec-
tive*

if
and
only

if
ev-
ery
el-
e-
ment

of
the
do-
main

oc-
curs

as
a
value

of
 g .

Put
more
for-
mally
this
con-
di-
tion
is:

$\forall v \in V \exists u \in U$ such that $g(u) = v$.

A
more
in-
for-
mal,

CHAPTER
3.
FUNCTIONS

but
de-
scrip-
tive,
term
for
this
prop-
erty
is
to
say
that
such
a
func-
tion
 g
is
onto.

These
prop-
er-
ties
can
of-
ten
de-
pend
on
the
ex-
act
choice
of
the
do-
main
and
co-
domain,
as
the
fol-
low-
ing
ex-
am-
ples
show.

Example

3.2.

The
func-
tion

$f_1 :$
 $(0, \frac{\pi}{2}) \rightarrow$
 $(0, 1)$
 de-
 fined
 by

$$f_1(t) = \sin t,$$

is
 both
 in-
 jec-
 tive
 and
 sur-
 jec-
 tive,
 i.e.
 bi-
 jec-
 tive.

However

the
 func-
 tion
 $f_2 :$
 $(0, \pi) \rightarrow$
 $(0, 2)$
 de-
 fined
 by

$$f_2(t) = \sin t,$$

is
 nei-
 ther
 in-
 jec-
 tive
 nor
 sur-
 jec-
 tive.

Since

$$\sin\left(\frac{\pi}{4}\right) =$$

$$\sin\left(\frac{3\pi}{4}\right) =$$

$$\frac{1}{\sqrt{2}},$$

f_2
 is
 not
 in-
 jec-
 tive
 and
 the

CHAPTER
3.
FUNCTIONS

fact
that
for
all
 t ,
 $|\sin t| \leq$
1
means
that
there
is
no
 t
for
which
 $\sin t =$
1.5,
and
so
 f_2
is
not
sur-
jec-
tive.

Definition

3.6
(Bi-
jec-
tive).
A
func-
tion
is
*bi-
jec-
tive*
if
and
only
if
it
is
both
in-
jec-
tive
and
sur-
jec-
tive.

3.1. REAL-VALUED FUNCTIONS

Real-valued functions

A lot of the time the functions we encounter in mathematics are *real-valued*, i.e. functions whose domains and codomains are subsets of the real numbers $\mathbb{R}$.

Algebra of real-valued functions

Using the familiar

CHAPTER
3.
FUNCTIONS
op-
er-
a-
tions
of
the
real
num-
bers
we
can
de-
fine
as-
so-
ci-
ated
op-
er-
a-
tions
amongst
real-
valued
func-
tions.
We
want
to
stress
here
the
con-
sid-
er-
a-
tion
of
func-
tions
as
math-
e-
mat-
i-
cal
ob-
jects
in
their
own
right
with
op-
er-
a-

3.1.
~~REAL-~~
~~VALUED~~
~~FUNCTIONS~~
~~are~~
~~de-~~
~~fin-~~
~~ed~~
~~on~~
~~them.~~

Definition

3.7

(Ad-
 di-
 tion).
 Given
 two
 real-
 valued
 func-
 tions
 f
 and
 g
 with
 the
 same
 do-
 main
 we
 can
 de-
 fined
 a
 new
 func-
 tion
 $f +$
 g ,
 called
 the
sum
 of
 f
 and
 g ,
 on
 this
 do-
 main
 us-
 ing
 the
 for-
 mula

$$(f + g)(x) = f(x) + g(x).$$

Definition

3.8

(Scalar

CHAPTER
3.
FUNCTIONS

mul-
ti-
pli-
ca-
tion).

Let
 f
be
a
real-
valued
func-
tion
and
 $\lambda \in$
 $\mathbb{R}$.

Then
we
can
form
a
new
func-
tion
 λf ,
which
has
the
same
do-
main
as
 f ,
and
is
de-
fined
by
the
for-
mula

$$(\lambda f)(x) = \lambda(f(x)).$$

Definition

3.9
(Mul-
ti-
pli-
ca-
tion).

Let
 f
and
 g
be
two

3.1. REAL- VALUED FUNCTIONS

real-
valued
func-
tions
with
the
same
do-
main.
Then
we
can
form
a
new
func-
tion
 fg ,
on
the
same
do-
main,
de-
fined
by
the
for-
mula

$$(fg)(x) = f(x)g(x).$$

Definition

3.10

(Re-
cip-
ro-
cal).
Let
 f
be
a
real-
valued
func-
tion
that
never
takes
the
value
zero.
Then
we
can
form
a

CHAPTER
3.
FUNCTIONS

new
func-
tion
 $\frac{1}{f}$,
on
the
same
do-
main
as
 f ,
de-
fined
by
the
for-
mula

$$\left(\frac{1}{f}\right)(x) = \frac{1}{f(x)}.$$

Note
that
we
do
not
like
to
use
the
no-
ta-
tion
 f^{-1}
for
the
re-
cip-
ro-
cal
func-
tion
as
this
is
nor-
mally
re-
served
for
the
*in-
verse*
func-
tion
of
 f ,

3.2.

COM-

PO-

see

SP-

INTER-

ITION

AND

INVERSE

3.2 Composition and in- verse

Composing

func-

tions

is

the

pro-

cess

of

ap-

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func-

tions

in

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quence,

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other.

You

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It

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CHAPTER
3.
FUNCTIONS

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con-
cept
and
we
can
study
it
even
be-
fore
ad-
mit-
ting
any
no-
tion
of
num-
ber
or
mea-
sure-
ment
etc.
It
will
play
an
im-
por-
tant
role
in
our
study
of
groups
in
the
next
chap-
ter.

Definition

3.11.

Let
 $f :$
 $X \rightarrow$
 Y
and
 $g :$
 $Y \rightarrow$
 Z
be

3.2.
 COM-
 PO-
 SITION
 AND
 INVERSE

that
 the
 co-
 domain
 of
 f
 is
 also
 the
 do-
 main
 of
 g).

Then
 we
 can
 form
 a
 new
 func-
 tion

$$g \circ f : X \rightarrow Z,$$

defined
 by
 the
 for-
 mula

$$(g \circ f)(x) = g(f(x)).$$

We
 can
 re-
 fer
 to
 $g \circ f$
 as
 g
com-
posed
with
 f ,
 or
 more
 in-
 for-
 mally,
 g
af-
ter
 f .

CHAPTER
3.
FUNCTIONS
Example

3.3.

Con-
sider
the
real-
valued
func-
tions
 f
and
 g
de-
fined
by
the
for-
mu-
las

$$f(x) = x^2 - 3 \text{ and } g(x) = 2x + 3.$$

Then
we
can
find
for-
mu-
las
defin-
ing
the
com-
po-
si-
tions
as
fol-
lows

$$(f \circ g)(x) = f(g(x)) = (2x + 3)^2 - 3 = 4x^2 + 12x + 6,$$

and

$$(g \circ f)(x) = g(f(x)) = 2(x^2 - 3) + 3 = 2x^2 - 3.$$

Note
that
these
for-
mu-
las
are
dif-
fer-
ent.
In

3.2.
COM-
PO-
SITION
AND
INVERSE
both
com-
po-
si-
tions
are
de-
fined,
com-
po-
si-
tion
is
not
com-
mu-
ta-
tive,
i.e.

$$f \circ g \neq g \circ f,$$

although
in
spe-
cial
cases
they
may
be
equal.

Example

3.4.
Con-
sider
the
real-
valued
func-
tions
 g, h, r
de-
fined
by
the
for-
mu-
las

$$g(x) = 2x - 7, \quad h(x) = x^2 - 1 \text{ and } r(x) = e^{2x}.$$

Obtain
for-

CHAPTER
3.
FUNCTIONS

mu-
las
for
each
of
the
fol-
low-
ing
com-
po-
si-
tions,

1.
 $g \circ$
 $h,$
2.
 $h \circ$
 $r,$
3.
 $g \circ$
 $h \circ$
 $r,$
4.
 $g \circ$
 $g,$
5.
 $g \circ$
 $g \circ$
 $g,$
6.
 $r \circ$
 $r.$

Definition

3.12

(Iden-
tity
func-
tion).

The
iden-
tity
func-
tion
id
on
the
do-
main
 X
is
the

3.2.

COM-

PO-

func-

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$\text{id} : X \rightarrow X,$

defined

by

the

for-

mula

$$\forall x \in X \quad \text{id}(x) = x.$$

We

shall

speak

of

the

iden-

tity

func-

tion

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cific

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CHAPTER
3.
FUNCTIONS

The
iden-
tity
func-
tion
has
the
prop-
erty
that
when
com-
posed
with
an-
other
func-
tion,
it
leaves
the
other
func-
tion
un-
changed,
i.e.

$$f \circ \text{id} = \text{id} \circ f = f.$$

Sometimes
the
ef-
fect
of
a
func-
tion
can
be
re-
versed,
as
in
the
fol-
low-
ing
def-
i-
ni-
tion.

Definition

3.13

(In-
verse

3.2.

COM-

PO-

func-

tion

AND

Let

$X \rightarrow$

Y

be

a

func-

tion

that

is

bi-

jec-

tive,

i.e.

is

in-

jec-

tive

and

sur-

jec-

tive.

Then

there

ex-

ists

a

func-

tion

$f^{-1} : Y \rightarrow X,$

called

the

in-

verse

of

$f,$

which

has

the

prop-

erty

that

$f^{-1} \circ f = f \circ f^{-1} = \text{id}.$

Or

in

other

words,

$\forall x \in X \quad f^{-1}(f(x)) = x,$

and

$\forall y \in Y \quad f(f^{-1}(y)) = y.$

CHAPTER
3.
FUNCTIONS
Example

3.5.

Given

a
func-
tion
de-
fined
by
a
for-
mula
it

is
of-
ten
pos-
si-
ble
to
ob-
tain
the
for-
mula
for
the
in-
verse
func-
tion
(if
it
ex-
ists).

Sup-
pose
that
the
real-
valued
func-
tion

f
is
de-
fined
by
the
for-
mula

$$f(x) = 2x - 5.$$

We
form
the

3.2. COM- PO- SITION AND INVERSE

and
ad-
just
it
so
that
 x
be-
comes
the
sub-
ject
of
the
equa-
tion,
which
is

$$x = \frac{y + 5}{2}.$$

This
now
gives
us
the
cor-
rect
for-
mula
for
the
in-
verse
func-
tion,
 f^{-1} ,
which
is
de-
fined
by

$$f^{-1}(y) = \frac{y + 5}{2}.$$

In
terms
of
graphs,
there
is

CHAPTER
3.
FUNCTIONS

a
nice
re-
la-
tion-
ship
be-
tween
the
graph
of
a
real-
valued
func-
tion
and
that
of
its
in-
verse.
The
graph
of
the
in-
verse
func-
tion
can
be
ob-
tained
by
re-
flect-
ing
the
graph
of
the
orig-
i-
nal
func-
tion
in
the
di-
ag-
o-
nal
line
in
the

3.2.
 COM-
 PO-
 SY-
 M-
 METR-
 ION
 de-
 AND
 FINED
 INVERSE
 by
 $y =$
 x .
 This
 has
 the
 ef-
 fect
 of
 in-
 vert-
 ing
 the
 re-
 la-
 tion-
 ship
 be-
 tween
 the
 do-
 main
 and
 co-
 domain.

Example

3.6.

Fig-
 ure
 3.2
 shows
 the
 graphs
 of
 the
 square
 func-
 tion
 $s :$
 $\mathbb{R}^+ \rightarrow$
 $\mathbb{R}^+$
 de-
 fined
 by
 $s(x) =$
 x^2
 and
 square
 root
 func-
 tion

CHAPTER
3.
FUNCTIONS

$r :$
 $\mathbb{R}^+ \rightarrow$
 $\mathbb{R}^+$
 de-
 fined
 by
 $r(x) =$
 $\sqrt{x},$
 along
 with
 the
 dot-
 ted
 line
 show-
 ing
 the
 axis
 of
 re-
 flec-
 tion
 $y =$
 $x.$

The
 next
 re-
 sult
 re-
 lates
 the
 in-
 verse
 of
 the
 com-
 po-
 si-
 tion
 of
 two
 func-
 tions
 to
 the
 com-
 po-
 si-
 tions
 of
 their
 in-
 verses.
 It
 should

3.2.
COM-
PO-

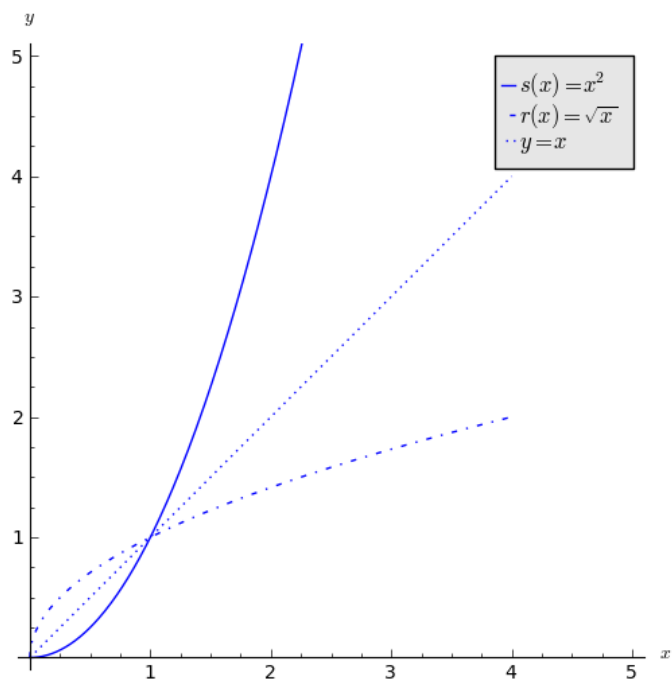


Figure 3.2:
The
square
func-
tion
and
its
in-
verse

re-
mind
you
of
a
sim-
i-
lar
re-
sult
about
ma-
trix
in-
verses.

Theorem
3.1.
*Sup-
pose
that*

CHAPTER
3.
FUNCTIONS

$f :$
 $X \rightarrow$
 Y

and

$g :$
 $Y \rightarrow$
 Z

are
two
bi-
jec-
tive
func-
tions.

Then
the
com-
po-
si-
tion

$g \circ$
 f
is
also
bi-
jec-
tive
and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

3.3 Rational func- tions and their par- tial frac- tion ex- pan- sions

Rational
func-
tions
are
those
func-
tions
that
are
quo-

3.3. RA- TIO- NALS OF FUNC- TIONS AND THEIR FAC- TIAL FRAC- TION EXPANSIONS

tion-
nal
func-
tion
 r ,
is
a
real-
valued
func-
tion
of
the
form

$$r = \frac{p}{q},$$

where
the
func-
tions
 p
and
 q
are
poly-
no-
mi-
als.

The
treat-
ment
of
ra-
tio-
nal
func-
tions
(for
in-
stance
when
in-
te-
grat-
ing

CHAPTER
3.
FUNCTIONS
them)
is
of-
ten
made
eas-
ier
by
re-
expressing
them
in
terms
of
par-
tial
frac-
tions.
This
is
a
tech-
nique
for
ex-
press-
ing
a
ra-
tio-
nal
func-
tion
as
a
sum
of
ra-
tio-
nal
func-
tions
of
a
very
sim-
ple
type.
An
ex-
am-
ple
of
a
par-
tial

3.3.
RA-
TIO-
NAL
FAC-
TION
FUNC-
TIONS
AND
THEIR
PAR-
TIAL
FRAC-
TION
EXPANSIONS
The
func-
tion
is:

$$\begin{aligned}\frac{x}{x^3 + x^2 + 2x - 4} &= \frac{x}{(x + 2)^2(x - 1)} \\ &= \frac{-1/9}{x + 2} + \frac{2/3}{(x + 2)^2} + \frac{1/9}{x - 1},\end{aligned}$$

where
the
fi-
nal
form
is
called
the
par-
tial
frac-
tion
ex-
pan-
sion
of
the
orig-
i-
nal
ra-
tio-
nal
func-
tion.
The
par-
tial
frac-
tions
tech-
nique
re-
lies
on
the
fol-
low-

CHAPTER
3.
FUNCTIONS

ing
fact
about
poly-
no-
mi-
als
with
real
co-
ef-
fi-
cients.

Theorem
3.2

(Poly-
no-
mial
Fac-
tori-
sa-
tion).

Let
p
be
a
poly-
no-
mial
with
real
co-
ef-
fi-
cients
and
lead-
ing
co-
ef-
fi-
cient
1,
i.e.

$$p(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_2x^2 + a_1x + a_0.$$

Then
p(x)
fac-
tor-
izes
into
a
prod-
uct

3.3. RA- TIO- NAL POLY- FUNG- TIONS AND THEIR PAR- TIAL FRAC- TION EXPANSIONS

cients,
each
of
which
is
ei-
ther
a
lin-
ear
fac-
tor
of
the
form
(x—
a)^m,
where
a ∈
 $\mathbb{R}$
and
m ∈
 $\mathbb{N}$
or
a
quadratic
fac-
tor
of
the
form
(x²+
ax+
b)^m,
where
a, b ∈
 $\mathbb{R}$
and
m ∈
 $\mathbb{N}$.
A
fac-
tor
(x—
a)^m

CHAPTER
3.
FUNCTIONS

cor-
re-
sponds
to
a
real
root
a
of
p
of
mul-
ti-
plic-
ity
m
and
a
quadratic
type
fac-
tor
 $(x^2 +$
 $ax +$
 $b)^m$
cor-
re-
sponds
to
a
pair
of
com-
plex
con-
ju-
gate
roots
of
p
of
mul-
ti-
plic-
ity
m.

Proof.
Omitted.
□

Below
is
a
com-
plete

3.3. RATIONAL DESCRIPTIONS OF THE METHOD OF FORMAL FRACTION EXPANSIONS

tial fraction expansion of a rational function. Following that we give some examples. You may find it easier to follow the examples first, referring back to the general

CHAPTER
3.
FUNCTIONS
de-
scrip-
tion
of
the
method
as
needed.

The Gen- eral Method

So
sup-
pose
that
 $r = \frac{p}{q}$
is
a
ra-
tio-
nal
func-
tion.
To
find
the
par-
tial
frac-
tion
ex-
pan-
sion
we
first
fac-
tor-
ize
 $q(x)$
in
the
way
de-
scribed
by
the
above
the-
o-
rem.
For
each

3.3.
 RA-
 TIO-
 NAL
 FAC-
 TOR
 FUNC-
 TIONS
 OF
 THE
 AND
 THEIR
 FORM
 OF
 THE
 PARTIAL
 FRACTION
 OF
 THE
 EXPANSIONS
 we
 in-
 clude
 the
 terms

$$\frac{\alpha_1 x + \beta_1}{x^2 + ax + b} + \frac{\alpha_2 x + \beta_2}{(x^2 + ax + b)^2} + \cdots + \frac{\alpha_m x + \beta_m}{(x^2 + ax + b)^m},$$

in
 the
 par-
 tial
 frac-
 tion
 ex-
 pan-
 sion.
 The
 real
 num-
 bers
 α_i, β_i
 will
 be
 de-
 ter-
 mined
 later.

For
 each
 fac-
 tor
 of
 the
 form
 $(x - a)^m$
 of
 $q(x)$
 we
 in-
 clude
 the
 terms

$$\frac{\alpha_1}{x - a} + \frac{\alpha_2}{(x - a)^2} + \cdots + \frac{\alpha_m}{(x - a)^m},$$

CHAPTER
3.
FUNCTIONS

in
the
par-
tial
frac-
tion
ex-
pan-
sion.
The
real
num-
bers
 α_i
will
be
de-
ter-
mined
later.

If
the
de-
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of
the
nu-
mer-
a-
tor
poly-
no-
mial
 $p(x)$
equals
or
ex-
ceeds
the
de-
gree
of
the
de-
nom-
i-
na-
tor
poly-
no-
mial
 $q(x)$
by
 m ,
where
 $m \geq$

3.3.

RA-

TIO-

QUAL

FUNCTIONS

AND

THEIR

PARTIAL

FRAC-

TION
 $\lambda_0 + \lambda_1 x + \cdots + \lambda_m x^m,$

EXPANSIONS

the

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tial

frac-

tion

ex-

pan-

sion,

the

num-

bers

λ_i

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be

de-

ter-

mined.

Following

these

steps

will

give

you

the

par-

tial

frac-

tion

ex-

pan-

sion

for

$r(x)$.

To

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ues

of

the

un-

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ef-

CHAPTER
3.
FUNCTIONS
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cients
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ti-
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of
the
par-
tial
frac-
tion
ex-
pan-
sion
by
 $q(x)$
which
will
have
the
ef-
fect
of
re-
mov-
ing
all
de-
nom-
i-
na-
tors
from
the
ex-
pres-
sion
(re-
mem-
ber
that
each
of
the
de-
nom-
i-
na-
tors
of
the
par-
tial

3.3. RA- TIO- NAL FAC- TION FUNCC- TIONS AND THEIR IS PAR- TIAL FRAC- TION EXPANSIONS

the
poly-
no-
mial
 $q(x)$

).
This
will
leave
us
with
an
equa-
tion
of
poly-
no-
mi-
als
in-
volv-
ing
the
un-
de-
ter-
mined
co-
ef-
fi-
cients.

We
can
de-
ter-
mine
the
val-
ues
of
these
co-
ef-
fi-
cients
now

CHAPTER
3.
FUNCTIONS

by
equat-
ing
co-
ef-
fi-
cients
of
equal
pow-
ers
of
 x
from
both
sides
of
the
equa-
tion.
(Re-
mem-
ber
that
two
poly-
no-
mi-
als
are
equal
if
and
only
if
their
cor-
re-
spond-
ing
co-
ef-
fi-
cients
are
equal).

Example

3.7.
Here
are
two
ex-
am-
ples
that

3.3.
RA-
TIO-
NAL
show
the
FUNC-
TIONS
method
work
AND
img
THEIR
PAR-
TIAL
FRAC-
TION¹
An
EXPANSIONS
am-

¹www.wolframalpha.com

will
pro-
vide
the
par-
tial
frac-
tion
ex-
pan-
sion
of
any
ra-
tio-
nal
func-
tion
that
is
sub-
mit-
ted
to
it.
In
ad-
di-
tion
by
click-
ing
the
show
steps
link
in
the
out-
put
it
will
dis-
play
a
step-
by-
step
deriva-
tion
of
the
ex-
pan-
sion

CHAPTER
3.
FUNCTIONS

ple
with
only
sim-
ple
fac-
tors
in
the
de-
nom-
i-
na-
tor

$$\begin{aligned}\frac{3x+1}{x^3-3x+2} &= \frac{3x+1}{(x-1)^2(x+2)} \\ &= \frac{\alpha_1}{x-1} + \frac{\alpha_2}{(x-1)^2} + \frac{\beta}{x+2}.\end{aligned}$$

Here
we
have
fac-
tor-
ized
the
de-
nom-
i-
na-
tor
and
formed
the
par-
tial
frac-
tion
ex-
pan-
sion.
To
de-
ter-
mine
the
par-
tial
frac-
tion
ex-
pan-
sion
we
mul-
ti-

3.3.

RA-

TIO-

NAL ply

FUNC both

TIONS sides

AND by

THEIR $(x -$

PAR- $1)^2(x +$

TIAL- $2)$

FRAC to

TION get

EXPANSIONS

$$\begin{aligned} 3x + 1 &= \alpha_1(x - 1)(x + 2) + \alpha_2(x + 2) + \beta(x - 1)^2 \\ &= (\alpha_1 + \beta)x^2 + (\alpha_1 + \alpha_2 - 2\beta)x + (-2\alpha_1 + 2\alpha_2 + \beta). \end{aligned}$$

Equating

co-

ef-

fi-

icients

from

both

sides

of

the

poly-

no-

mial

gives

us

the

si-

mul-

ta-

ne-

ous

equa-

tions

$$\alpha_1 + \beta = 0,$$

$$\alpha_1 + \alpha_2 - 2\beta = 3,$$

$$-2\alpha_1 + 2\alpha_2 + \beta = 1.$$

Solving

these

equa-

tions

leads

to

the

so-

lu-

tion

$$\alpha_1 = \frac{5}{9}, \alpha_2 = \frac{4}{3}, \beta = \frac{-5}{9}.$$

So

the

CHAPTER
3.
FUNCTIONS

partial
fraction
expansion
of
our
original
rational
function
is

$$\frac{3x+1}{x^3-3x+2} = \frac{5/9}{x-1} + \frac{4/3}{(x-1)^2} - \frac{5/9}{x+2}.$$

2.

An
example
with
a
quadratic
type
factor.

$$\begin{aligned} \frac{5x}{x^3-x^2-x-2} &= \frac{5x}{(x^2+x+1)(x-2)} \\ &= \frac{\alpha_1+\beta x}{x^2+x+1} + \frac{\alpha_2}{x-2} \end{aligned}$$

Multiplying
across
by
(x^2+
 $x+$
 1)($x-$
 2)
gives

$$5x = (\alpha_1+\beta x)(x-2)+\alpha_2(x^2+x+1).$$

Multiplying
out
the
right
hand
side
and
then

3.3.
RA-
TIO-
NAL equat-
FUNC-
TION^{ing}
AND^{co-}
THEIR^{ef-}
PAR-^f
TIAL^{icients}
FRAC-^{of}
TION^{the}
EXPAN^{two}
SIONS^{Doly}
no-
mi-
als
leads
to
the
val-
ues

$$\alpha_1 = \frac{5}{7}, \beta = \frac{-10}{7}, \alpha_2 = \frac{10}{7}.$$

And
so
the
par-
tial
frac-
tion
ex-
pan-
sion
is

$$\frac{5x}{x^3 - x^2 - x - 2} = \frac{\frac{5}{7} - \frac{10}{7}x}{x^2 + x + 1} + \frac{\frac{10}{7}}{x - 2}.$$

Exercises

3.1.

These
prob-
lems
re-
quire
the
ap-
pli-
ca-
tion
of
var-
i-
ous
con-
cepts
and
re-
sults

CHAPTER
3.
FUNCTIONS

from
this
chap-
ter.
We
can
use
these
as
a
ba-
sis
for
fur-
ther
dis-
cus-
sion
in
the
tu-
to-
ri-
als.

1. Consider
the
func-
tions
 u, v, w
de-
fined
by
 $u(x) =$
 $x +$
 $1,$
 $v(x) =$
 $x^2 +$
 1
and
 $w(x) =$
 $\sin x.$
Give
a
defin-
ing
for-
mula
for
the
fol-
low-
ing
func-
tions

3.3.
RA-
TIO-
NAL a)
FUNC- $u \circ$
TIONS v
AND b)
THEIR $v \circ$
PAR- w
TIAL c)
FRAC- $v \circ$
TION $u \circ$
EXPANSIONS

2.
Let
 S ,
 R ,
 Exp
and
 One
de-
note
the
func-
tions
de-
fined
by
 $S(x) =$
 x^2 ,
 $R(x) =$
 $\sqrt{x}$,
 $\text{Exp}(x) =$
 e^x
and
 $\text{One}(x) =$
1.
Con-
sider
the
func-
tions
 f
and
 g
de-
fined
by

$$f(x) = \left(3e^{2x^2}\right)^{\frac{1}{2}} \text{ and } g(x) = 2e^{(1+\sqrt{x})^2}.$$

Express
 f
and
 g
as
suit-
able

CHAPTER
3.
FUNCTIONS

com-
po-
si-
tions
of
com-
bi-
na-
tions
of
the
func-
tions
 S ,
 R
and
Exp.

3. Consider
the
func-
tion
 f, g, h
de-
fined
by
the
fol-
low-
ing
for-
mu-
lae:

$$f(x) = 3x+1, \quad g(x) = \frac{x}{x-3}, \quad h(x) = \frac{x+1}{x-1}.$$

What
are
ap-
pro-
pri-
ate
do-
mains
and
co-
domains
for
these
func-
tions?
Find
for-
mu-
lae
for

3.3.
RA-
TIO-
NAL their
in-
FUNC-
TIONS
AND
THEIR
PAR-
TIAL
FRAC-
TION
EXPANSIONS
domains
of
these
new
func-
tions.

4.
Let
the
func-
tions
 $\lambda :$
 $\mathbb{R} \rightarrow$
 $(-1, \infty)$
be
de-
fined
by

$$\lambda(t) = e^{\frac{t}{2}} - 1.$$

Find
the
for-
mula
for
the
func-
tion
 λ^{-1} ,
stat-
ing
its
do-
main
and
co-
domain.

5.
Let
the
func-
tions
 $\phi :$
 $(0, \infty) \rightarrow$

CHAPTER

3.

FUNCTIONS

$(-1, \infty)$

and

$\psi :$

$\mathbb{R} \rightarrow$

$(0, \infty)$

be

de-

fin-

ed

by

$$\phi(x) = x^2 - 1, \quad \psi(x) = e^{2x}.$$

Find

the

def-

i-

ni-

tions

of

the

func-

tions

$\phi \circ$

ψ

and

$(\phi \circ$

$\psi)^{-1}$

and

con-

firm

that

$$(\phi \circ \psi)^{-1} = \psi^{-1} \circ \phi^{-1}$$

in

this

case.

6.

Obtain

par-

tial

frac-

tion

ex-

pan-

sions

of

the

fol-

low-

ing

ra-

tio-

nal

func-

tions

3.4.

IN-

FI-

NITE a)

SETS

AND

CARDINALITY

255

b)

$$\frac{s^2 + s - 3}{s^2 + s},$$

c)

$$\frac{s^2 + 3}{s^3 - 5s^2 + s - 5}.$$

3.4 Infinite

sets

and

car-

di-

nal-

ity

We

in-

clude

here

a

short

sec-

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that

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tant

con-

cepts

that

are

con-

nected

to

deep

is-

sues

in

set

the-

ory,

logic

and

the

CHAPTER
3.
FUNCTIONS

foundations and philosophy of mathematics. The main question is how to speak precisely about the size of a set. We shall see how the answer to this question uses the concepts of injectivity and surjectivity of maps

3.4.
IN-
FI-
NITE
SETS
AND
CARDINALITY

**The
num-
ber
of
el-
e-
ments
in
a
set**

Intuitively,
to
work
out
the
size
of
a
set
we
should
sim-
ply
count
the
num-
ber
of
el-
e-
ments
that
it
has.
So
some
triv-
ial
ex-
am-
ples
are

- the
set
 $\{a, b, c\}$
has
size
3,

CHAPTER

3.

FUNCTIONS

(as-
sum-
ing
 a, b, c
are
dis-
tinct
ob-
jects),

- the
set
of
in-
te-
gers
 $\{10, 11, 12, 13, 14\}$
has
size
5,
- the
set
 $\{7\}$
has
size
1,
- the
empty
set
 $\{\}$
has
size
0.

So
surely,
given
any
set,
we
can
sim-
ply
count
the
num-
ber
of
el-
e-
ments
that
it

3.4.
IN-
FI-
NITE
has
and
SETS
AND
AND
CARDINALITY
we
know
its
size
and
can
com-
pare
sets
in
this
way.
This
ap-
proach
is
per-
fectly
cor-
rect
and
would
be
the
whole
story
if
one
was
only
con-
sid-
er-
ing
*fi-
nite*
sets.
But
in
math-
e-
mat-
ics
we
soon
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counter
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CHAPTER
3.
FUNCTIONS

finite
sets
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like
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speak
pre-
cisely
about
their
sizes
and
be
able
to
com-
pare
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The
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cept
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size
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fi-
nite
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Cen-
tury
Ger-
man
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Georg
Can-
tor.
The

3.4.
 IN-
 FI-
 tech-
 NITE
 di-
 SETS
 cal-
 AND
 form-
 CARDINALITY
 of
car-
di-
nal-
ity
 is
 used.

Definition

3.14

(Car-
 di-
 nal-
 ity).
 We
 say
 that
 two
 sets
 A
 and
 B
 have
 the
 same
 car-
 di-
 nal-
 ity
 if
 there
 ex-
 its
 a
 bi-
 jec-
 tion
 $\phi :$
 $A \rightarrow$
 B .

Notice
 that
 this
 def-
 i-
 ni-
 tion
 does
 not
 ex-
 plic-
 itly

CHAPTER
3.
FUNCTIONS

tell
you
what
the
car-
di-
nal-
ity
(or
size)
of
a
set
is,
rather
it
sup-
plies
a
cri-
te-
rion
(the
ex-
is-
tence
of
a
bi-
jec-
tion)
for
when
you
re-
gard
two
sets
as
hav-
ing
the
same
car-
di-
nal-
ity.

To
un-
der-
stand
this
def-
i-
ni-
tion

3.4.
IN-
FL-
think
NITE
of
SETS
the
AND
FIG
CARDINALITY

cess
of
count-
ing
the
el-
e-
ments
of
a
fi-
nite
set
with.
To
count
the
el-
e-
ments
of
a
set
is
re-
ally
to
set
up
a
pair-
ing
(or
bi-
jec-
tion)
of
the
el-
e-
ments
of
the
set
with
the
count-
ing
num-
bers
 $1, 2, 3, \dots, n,$
for

CHAPTER
3.
FUNCTIONS

some

ap-
pro-
pri-
ate
value

n .

Or

in

other

words,

when

we

count

the

el-

e-

ments

in

a

fi-

nite

set

X

we

are

re-

ally

con-

struct-

ing

a

bi-

jec-

tion

$$X \rightarrow \{1, 2, 3, \dots, n\}.$$

The

beauty

of

Can-

tor's

def-

i-

ni-

tion

is

that

it

sim-

ply

ex-

tends

this

idea

to

3.4.
 IN-
 FI-
 NITE
 all
 SETS
 both
 AND
 FIN-
 ITED CARDINALITY
 finite
 and
 in-
 fi-
 nite.

We
 will
 write
 $|A| =$
 $|B|$
 when
 A
 and
 B
 have
 the
 same
 car-
 di-
 nal-
 ity.
 The
 ex-
 is-
 tence
 of
 in-
 jec-
 tions
 be-
 tween
 sets
 also
 has
 im-
 pli-
 ca-
 tions
 for
 car-
 di-
 nal-
 ity.
 If
 there
 ex-
 ists
 an
 in-
 jec-
 tion
 $\psi :$

CHAPTER
3.
FUNCTIONS

$A \rightarrow$
 B

then

we

can

say

that

the

car-

di-

nal-

ity

of

A

is

no

big-

ger

than

B

and

write

$|A| \leq$

$|B|$.

Sim-

i-

larly

if

there

ex-

ists

a

sur-

jec-

tion

$\rho :$

$A \rightarrow$

B

we

can

say

that

the

car-

di-

nal-

ity

of

A

is

no

less

than

B

and

write

3.4. IN- FINITE SETS AND Example 3.8. CARDINALITY

We
can
see
this
def-
i-
ni-
tion
op-
er-
at-
ing
when
we
as-
sess
the
car-
di-
nal-
i-
ties
of
the
fa-
mil-
iar
ba-
sic
num-
ber
sys-
tems.

- the
nat-
u-
ral
num-
bers
 $\mathbb{N} =$
 $\{1, 2, 3, 4, \dots\}$

- the
in-
te-
gers
 $\mathbb{Z} =$
 $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

CHAPTER 3. FUNCTIONS

- the
ra-
tio-
nals
 $\mathbb{Q} =$
 $\{\frac{p}{q} :$
 $p, q \in$
 $\mathbb{Z}, q \neq$
 $0\}$
- the
real
num-
bers
 $\mathbb{R}$,
the
set
of
points
on
the
num-
ber
line
(or
*con-
tium*).

See
the
lec-
tures
for
de-
tails
of
these
ex-
am-
ples.

The
ba-
sic
def-
i-
ni-
tion
of
car-
di-
nal-
ity
al-
lows

3.4.
IN-
FI-
NITE
SETS
AND
CARDINALITY
in-
ter-
est-
ing
cri-
te-
rion
for
an
in-
fi-
nite
set.

Definition

3.15

(In-
fi-
nite
set).
A
set
 A
is
*in-
fi-
nite*
if
there
ex-
ists
a
bi-
jec-
tion

$$\phi : A \rightarrow C \subsetneq A,$$

from
 A
to
a
proper
sub-
set
of
it-
self.

CHAPTER
3.
FUNCTIONS
The
Con-
tin-
uum
Hy-
poth-
e-
sis

Sets
hav-
ing
the
same
car-
di-
nal-
ity
as
 $\mathbb{N}$
are
called
*count-
able*.
Hav-
ing
proved
that
 $\mathbb{R}$
is
un-
count-
able
the
ques-
tion
arises,
are
there
sets
with
in-
ter-
me-
di-
ate
size?
That
is,
does
there
ex-
ists
a
set

3.4. IN- FI- NITE SETS AND CARDINALITY

$|\mathbb{N}| < |X| < |\mathbb{R}|?$

The
Con-
tin-
uum
Hy-
poth-
e-
sis
(CH)
is
that
there
is
no
such
set
X.

This
may
seem
like
a
straight-
for-
ward
enough
ques-
tion,
at
first.
But
it
has
turned
out
to
be
ex-
tremely
elu-
sive
and
has
deep
con-
nec-
tions
to
the

CHAPTER
3.
FUNCTIONS
foun-
da-
tions
and
phi-
los-
o-
phy
of
math-
e-
mat-
ics.

**Status
of
CH
com-
pared
with
the
rest
of
set
the-
ory**

Zermelo-
Fraenkel
set
the-
ory
and
the
Ax-
iom
of
Choice
(ZFC)
is
the
com-
monly
used
ax-
iomatic
ba-
sis
for
set-
theory
(and
hence
most
math-

3.4. IN- FI- NITE SETS AND CARDINALITY

- Kurt Gödel proved in 1940 that CH can-not be dis-proved within ZFC.

- Paul Co-hen proved in 1963 that CH can-not be proved within ZFC.

These re-sults show that CH is in-de-pen-dent of ZFC, or rel-a-tively con-sis-tent with

CHAPTER
3.
FUNCTIONS

ZFC,
mean-
ing
that
as-
sum-
ing
CH
(or
its
con-
verse
 $\neg$ CH)
will
not
in-
tro-
duce
any
con-
tra-
dic-
tions
into
set
the-
ory
that
were
not
there
to
be-
gin
with.

Attitudes
to
the
ques-
tion
"Does
there
ex-
ists
a
set
 X
with
 $|N| <$
 $|X| <$
 $|\mathbb{R}|$?"
re-
flect
dif-
fer-
ent

3.4.
IN-
FI-
NITE
SETS
AND
CARDINALITY
the
phi-
los-
o-
phy
of
math-
e-
mat-
ics.

Resources

Here
are
some
links
to
fur-
ther
re-
sources
on
these
and
re-
lated
top-
ics.

- Hotel
Hilbert
([https://
www.
youtube.
com/
watch?
v=
revY6cmX5m8](https://www.youtube.com/watch?v=revY6cmX5m8)),
an
at-
mo-
spheric
BBC/Open
Uni-
ver-
sity
drama
in-
ves-
ti-

CHAPTER

3.

FUNCTIONS

gat-
ing
in-
fin-
ity
set
in
a
ho-
tel
with
an
in-
fi-
nite
num-
ber
of
rooms.

- To
In-
fin-
ity
and
Be-
yond
([http://
www.
bbc.
co.
uk/
programmes/
b00qszch](http://www.bbc.co.uk/programmes/b00qszch))
a
BBC
Hori-
zon
doc-
u-
men-
tary
on
in-
fin-
ity.

- Read
more
about
these
mat-
ters
in

3.4.

IN-

FI-

NITE^{the}

SETS^{Stan-}

AND^{ford}

CARDINALITY^{En-}

cy-

clo-

pe-

dia

of

Phi-

los-

o-

phy

en-

tries

on

set

the-

ory

([http:](http://plato.stanford.edu/entries/set-theory/)

[//](http://plato.stanford.edu/entries/set-theory/)

[plato.](http://plato.stanford.edu/entries/set-theory/)

[stanford.](http://plato.stanford.edu/entries/set-theory/)

[edu/](http://plato.stanford.edu/entries/set-theory/)

[entries/](http://plato.stanford.edu/entries/set-theory/)

[set-theory/](http://plato.stanford.edu/entries/set-theory/))

and

phi-

los-

o-

phy

of

math-

e-

mat-

ics

([http:](http://plato.stanford.edu/entries/philosophy-mathematics/)

[//](http://plato.stanford.edu/entries/philosophy-mathematics/)

[plato.](http://plato.stanford.edu/entries/philosophy-mathematics/)

[stanford.](http://plato.stanford.edu/entries/philosophy-mathematics/)

[edu/](http://plato.stanford.edu/entries/philosophy-mathematics/)

[entries/](http://plato.stanford.edu/entries/philosophy-mathematics/)

[philosophy-mathematics/](http://plato.stanford.edu/entries/philosophy-mathematics/)).

Chapter 4

Continuity and dif- fer- en- ti- a- tion

4.1 Reference Books

There
are
many
in-
tro-
duc-
tory
cal-
cu-
lus
books
that
you
could
use
for
this
part
of
the
course.
The

CHAPTER

4.

CON-
Schaums

Out-

line

of

Cal-

280 DIFFERENTIATION

cu-

lus

book

is

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one.

It

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CHAPTER

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4.2 Introduction

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the
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 $y = \frac{1}{x}$
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CHAPTER

4.

CON-

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DIFFERENTIATION

4.3. Limiting values of functions

We have met the concept of a limit in the context of sequences and series. The concept is also useful when discussing real-valued functions. Let a and l be two real numbers. Informally we say that

CHAPTER

4.

CON-

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of f

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$$\lim_{x \rightarrow a} f(x) = l,$$

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x

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a ,

4.3.

LIM-

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VAL-

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OF

FUNCTIONS

At a rigorous definition?

Just as in the case of the convergence for sequences, convergence and limiting values of functions also require a detailed ϵ -type definition. We shall postpone this for now in our discussion but we will meet the full detail when we discuss continuity later on.

$$f(x) \rightarrow l \text{ as } x \rightarrow a,$$

which

we

read

as

' $f(x)$

tends

to

l

as

x

tends

to

a '.

Notice

that

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4.3.
LIM-
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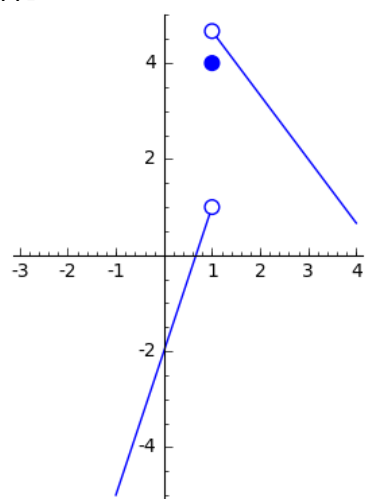


Figure 4.1:
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 of
 a
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 tion
 at
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 1.

ent
 lim-
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CHAPTER

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limit

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f

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tends

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$$\lim_{x \rightarrow a^-} f(x),$$

and

the

limit

of

f

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x

tends

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as

$$\lim_{x \rightarrow a^+} f(x).$$

4.3.
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CHAPTER

4.

CON-

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&

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we

can

write

$$\lim_{x \rightarrow a} f(x) = \infty.$$

For

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ple

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty,$$

and

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty,$$

If

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4.3.

LIM-

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UES $\lim_{x \rightarrow +\infty} f(x) = l$.

OF

FUNCTIONS

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ple

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0,$$

and

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

These

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$y =$

$\frac{1}{x}$

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CHAPTER
4.
CON-

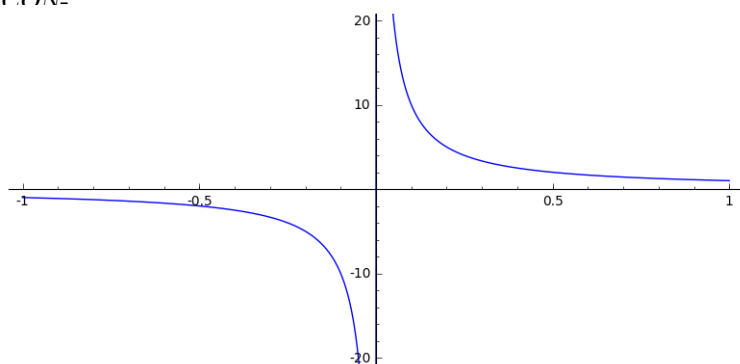


Figure 4.2:
The
func-
tion
 $g(x) =$
 $1/x$.

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4.3.

LIM-

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Theorem

4.4

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its).

Let

$l, m \in$

$\mathbb{R}$

and

sup-

pose

that

$\lim_{x \rightarrow a} f(x) =$

l

and

$\lim_{x \rightarrow a} g(x) =$

$m,$

then

$\lim_{x \rightarrow a} (\alpha f(x) + \beta g(x)) = \alpha l + \beta m,$

$\lim_{x \rightarrow a} (f(x)g(x)) = lm,$

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m},$ providing $m \neq 0.$

Of

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l'Hopital's

rule.

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$\lim_{x \rightarrow a} \frac{f(x)}{g(x)},$

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func-

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f

and

$g.$

CHAPTER

4.

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$\frac{0}{0}, \frac{\infty}{\infty}$.

In

these

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4.3.
LIM-
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the
orig-
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limit.

Theorem

4.2

(l'Hopital's
rule).

If

$$\lim_{x \rightarrow a} f(x) =$$

$$\lim_{x \rightarrow a} g(x) =$$

0

or

$$\lim_{x \rightarrow a} f(x) =$$

$$\lim_{x \rightarrow a} g(x) =$$

∞

then

pro-

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the

limit

ex-

ists

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

where

f'

and

g'

are

the

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f

and

g .

It
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CHAPTER

4.

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Example

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4.4.

CONTINUITY

illustrate the use of these techniques.

1.

$$\lim_{x \rightarrow \infty} \frac{3x+1}{2x} = \lim_{x \rightarrow \infty} \frac{3+\frac{1}{x}}{2} = \frac{3}{2}.$$

2.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1.$$

3.

$$\lim_{x \rightarrow \infty} \frac{x+1}{x} = \lim_{x \rightarrow \infty} \frac{1+\frac{1}{x}}{1} = 1.$$

4.

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \lim_{x \rightarrow 0} \frac{2 \cos 2x}{3} = \frac{2}{3}.$$

5.

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x} = \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{1} = 2.$$

6.

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{-\sin x}{6x} = \lim_{x \rightarrow 0} \frac{-\cos x}{6} = \frac{-1}{6}.$$

4.4 Continuity

In the previous section we made the point that the limiting value, $\lim_{x \rightarrow a} f(x)$,

CHAPTER

4.

CON-

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DIFFERENTIATION

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Definition

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$f :$
 $\mathbb{R} \rightarrow$
 $\mathbb{R}$

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$x =$
 a
if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

If

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CHAPTER

4.

CON-

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CONTINUITY

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CHAPTER
4.
CON-

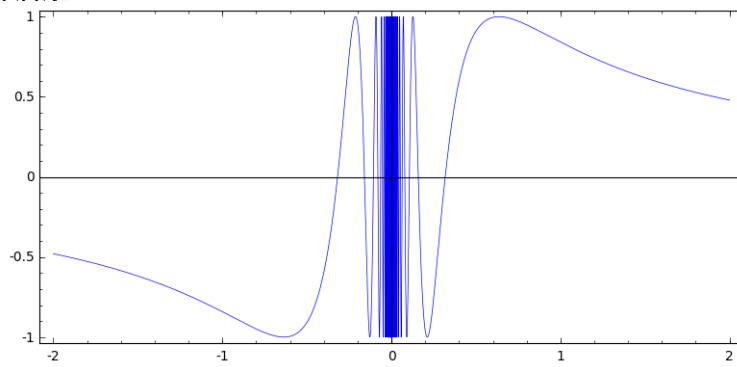


Figure 4.3:
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 $\sin(1/x)$
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 $\sin(\frac{1}{x})$
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4.4.
CONTINUITY

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 $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$
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CONTINUITY

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CHAPTER

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4.4. CONTINUITY

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 $x =$
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 at
 the
 point
 $x =$
 $f(a)$
 then
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 at
 $x =$
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CHAPTER

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4.4. CONTINUITY

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unit.
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CHAPTER

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Definition

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$\mathbb{R} \rightarrow$

$\mathbb{R}$

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$x =$

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and

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$\epsilon >$

0

4.4. CONTINUITY

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that

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$|x -$

$a| <$

δ

then

$|f(x) -$

$f(a)| <$

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$$\forall \epsilon > 0 \exists \delta > 0 \ |x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon.$$

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CHAPTER

4.

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DIFFERENTIATION

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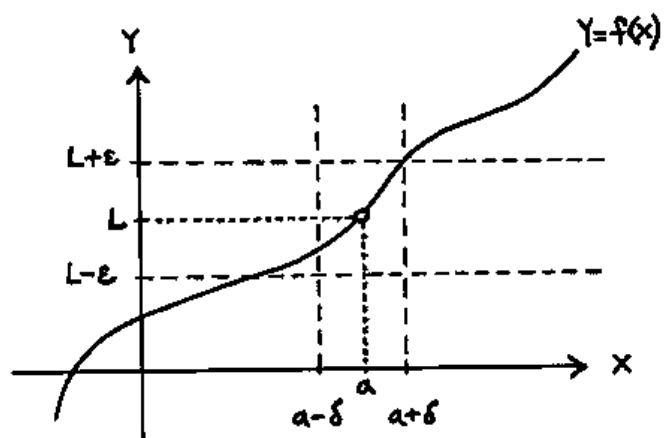
of

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4.4. CONTINUITY

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CHAPTER

4.

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DIFFERENTIATION

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Theorem

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4.4. CONTINUITY

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CHAPTER

4.

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Proof.

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$\delta_1, \delta_2 >$

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$|x -$

$a| <$

δ_1

we

have

$|f(x) -$

$f(a)| <$

$\epsilon/(2|\alpha|),$

and

for

all

$x -$

$a| <$

δ_2

we

have

$|g(x) -$

$g(a)| <$

$\epsilon/(2|\beta|).$

Let

$\delta =$

4.4.

CONTINUITY

$\min(\delta_1, \delta_2)$

and

then

for

all

$|x -$

$a| <$

δ

we

have

$$\begin{aligned} |(\alpha f + \beta g)(x) - (\alpha f + \beta g)(a)| &\leq |\alpha||f(x) - f(a)| + |\beta||g(x) - g(a)| \\ &< |\alpha|\epsilon/(2|\alpha|) + |\beta|\epsilon/(2|\beta|) \\ &= \epsilon. \end{aligned}$$

Therefore

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Next

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CHAPTER

4.

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$a| <$

δ_1

we

have

$|f(x) -$

$f(a)| <$

$\min(\sqrt{\epsilon}/\sqrt{3}, \epsilon/(3|g(a)|))$

and

$|x -$

$a| <$

δ_2

we

have

$|g(x) -$

$g(a)| <$

$\min(\sqrt{\epsilon}/\sqrt{3}, \epsilon/(3|f(a)|)).$

Letting

$\delta =$

$\min(\delta_1, \delta_2)$

and

then

for

all

$|x -$

$a| <$

δ

we

have

$$\begin{aligned}
 & |(fg)(x) - (fg)(a)| \\
 & \leq |f(x) - f(a)||g(x) - g(a)| + |g(a)||f(x) - f(a)| + |f(a)||g(x) - g(a)| \\
 & < \sqrt{\frac{\epsilon}{3}}\sqrt{\frac{\epsilon}{3}} + |g(a)|\frac{\epsilon}{3|g(a)|} + |f(a)|\frac{\epsilon}{3|f(a)|} \\
 & = \epsilon.
 \end{aligned}$$

Therefore

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CHAPTER

4.

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Theorem

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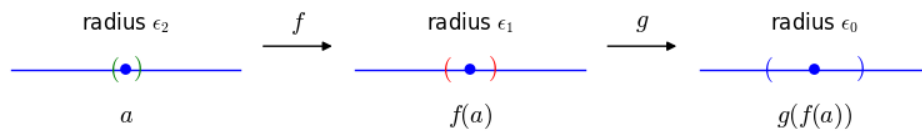


Figure 4.4:
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for
all
 $|y -$
 $f(a)| <$
 ϵ_1
we
have
 $|g(y) -$
 $g(f(a))| <$
 ϵ_0 .
But
since
 f
is

CHAPTER

4.

CON-

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DIFFERENTIATION

~~328~~

there

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ϵ_2

such

that

for

all

$|x -$

$a| <$

ϵ_2

we

have

$|f(x) -$

$f(a)| <$

ϵ_1 .

Combining

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$|g(f(x)) -$

$g(f(a))| <$

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 $\mathbb{R}$.
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CHAPTER

4.

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CONTINUITY

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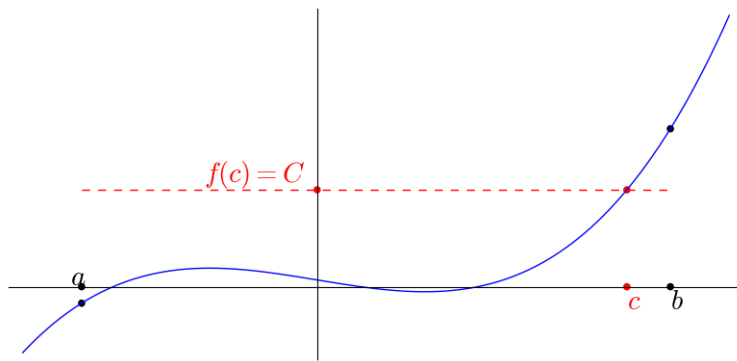


Figure 4.5:
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Theorem 4.5

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 $[a, b]$.
Sup-
pose
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 $f(a) \neq$
 $f(b)$.
Then
for
any
num-
ber
 C
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tween
 $f(a)$
and
 $f(b)$,
there
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ber
 $c \in$
 $[a, b]$
such
that
 $f(c) =$
 C .

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CHAPTER

4.

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DIFFERENTIATION

4.5 Differentiation

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4.5.

~~329~~ DIFFERENTIATION

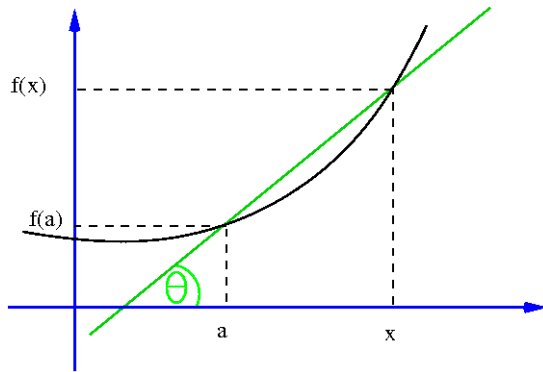
of
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CHAPTER
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CON-
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&
DIFFERENTIATION

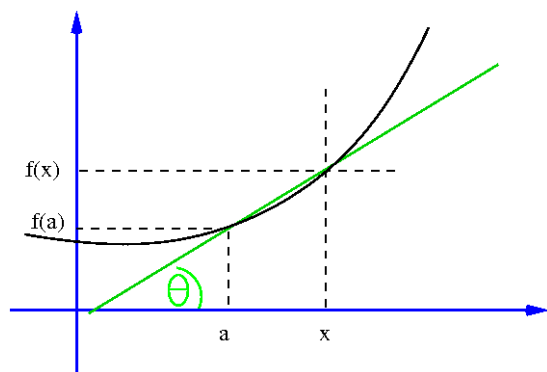


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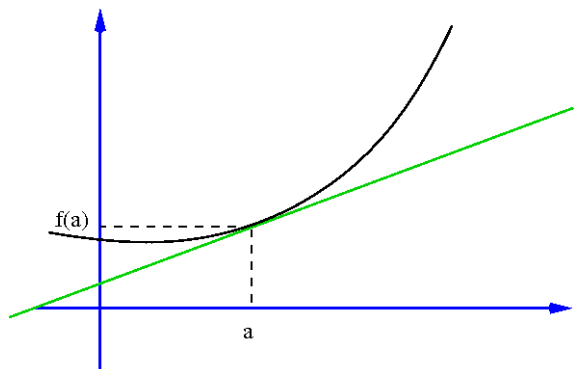
$$\text{gradient} = \tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{f(x) - f(a)}{x - a}$$

4.5.

DIFFERENTIATION



As
 $x \rightarrow$
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CHAPTER

4.

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4.5.

DIFFERENTIATION

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Definition

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$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

If

CHAPTER

4.

CON-

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$x =$

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$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

or

$$\left. \frac{df}{dx} \right|_{x=a} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Example

4.2.

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4.5.

DIFFERENTIATION

$$f(x) =$$

$$kx$$

and

$$g(x) =$$

$$kx^2$$

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CHAPTER

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$$\begin{aligned} \left. \frac{df}{dx} \right|_{x=a} &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{k(x - a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{k(x - a)}{x - a} = k. \end{aligned}$$

The

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$$\begin{aligned} \left. \frac{dg}{dx} \right|_{x=a} &= \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{kx^2 - ka^2}{x - a} \\ &= \lim_{x \rightarrow a} \frac{k(x^2 - a^2)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{k(x - a)(x + a)}{x - a} = 2ka. \end{aligned}$$

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4.5.

~~DIFF~~ DIFFERENTIATION

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So
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CHAPTER

4.

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4.5.

DIFFERENTIATION

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tives

$$\frac{d^2 f}{dx^2} = \frac{d}{dx} \left(\frac{df}{dx} \right),$$

$$\frac{d^3 f}{dx^3} = \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right), \text{ etc....}$$

Standard deriva- tives

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DIFFERENTIATION

1.4 Calculus

The constant of integration has been omitted.

$\int y \, dx$	y	$\frac{dy}{dx}$
$\left. \begin{array}{l} \frac{x^{n+1}}{n+1} \quad n \neq -1 \\ \log_e(x) \quad n = -1 \end{array} \right\}$	x^n	nx^{n-1}
$x \log_e(x) - x$	$\log_e(x)$	$\frac{1}{x}$
$\frac{1}{a} e^{ax}$	e^{ax}	ae^{ax}
$\frac{a^x}{\log_e(a)}$	$a^x \quad (a > 0)$	$a^x \log_e(a)$
$-\frac{1}{a} \cos(ax)$	$\sin(ax)$	$a \cos(ax)$
$\frac{1}{a} \sin(ax)$	$\cos(ax)$	$-a \sin(ax)$
$\frac{1}{a} \log_e[\sec(ax)]$	$\tan(ax)$	$a \sec^2(ax)$
$\log_e \left[\tan \left(\frac{x}{2} \right) \right]$	$\operatorname{cosec}(x)$	$-\operatorname{cosec}(x) \cot(x)$
$\log_e[\sec(x) + \tan(x)]$	$\sec(x)$	$\sec(x) \tan(x)$
$\log_e[\sin(x)]$	$\cot(x)$	$-\operatorname{cosec}^2(x)$
$\cosh(x)$	$\sinh(x)$	$\cosh(x)$
$\sinh(x)$	$\cosh(x)$	$\sinh(x)$
$\sin^{-1}(x)$	$\frac{1}{\sqrt{1-x^2}}$	
$\cos^{-1}(x)$	$\frac{1}{\sqrt{x^2-1}}$	
$\tan^{-1}(x)$	$\frac{1}{1+x^2}$	
$\sec^{-1}(x)$	$\frac{1}{x\sqrt{x^2-1}}$	
$\left. \begin{array}{l} \sinh^{-1}(x) \\ \log_e(x + \sqrt{x^2+1}) \end{array} \right\}$	$\frac{1}{\sqrt{x^2+1}}$	
$\left. \begin{array}{l} \cosh^{-1}(x) \\ \log_e(x + \sqrt{x^2-1}) \end{array} \right\}$	$\frac{1}{\sqrt{x^2-1}}$	
$\tanh^{-1}(x)$	$\frac{1}{1-x^2}$	

$\int y \, dx$	y
$\sin^{-1} \left(\frac{x}{a} \right)$	$\frac{1}{\sqrt{a^2 - x^2}}$
$\cos^{-1} \left(\frac{x}{a} \right)$	$-\frac{1}{\sqrt{a^2 - x^2}}$
$\tan^{-1} \left(\frac{x}{a} \right)$	$\frac{a}{a^2 + x^2}$
$\sec^{-1} \left(\frac{x}{a} \right)$	$-\frac{a}{x\sqrt{a^2 - x^2}}$
$\sinh^{-1} \left(\frac{x}{a} \right)$ or $\log_e \left(\frac{x + \sqrt{x^2 + a^2}}{a} \right)$	$\frac{1}{\sqrt{a^2 + x^2}}$
$\cosh^{-1} \left(\frac{x}{a} \right)$ or $\log_e \left(\frac{x + \sqrt{x^2 - a^2}}{a} \right)$	$-\frac{1}{\sqrt{a^2 - x^2}}$
$\tanh^{-1} \left(\frac{x}{a} \right)$	$\frac{a}{a^2 - x^2}$

Multiplication by a scalar

If $y = ku(x)$ where k is a scalar then

$$\frac{dy}{dx} = k \frac{du}{dx}$$

Product rule

If $y = u(x)v(x)$ then

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Quotient rule

If $y = \frac{u(x)}{v(x)}$ then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain rule

If y is a function of u and u is the function of x then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Sum/difference rules

$$\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

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4.6 Properties of the deriva- tive

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$$\frac{df}{dx} = \alpha \frac{dg}{dx} + \beta \frac{dh}{dx}.$$

Proof.

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PROPOSITION 4.6.
Suppose that the function f is differentiable at a point $x = a$.

and h are both differentiable at a point $x = a$.

$$\begin{aligned}\alpha g'(a) + \beta h'(a) &= \alpha \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} + \beta \lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a}, \text{ by defn.} \\ &= \lim_{x \rightarrow a} \left(\alpha \frac{g(x) - g(a)}{x - a} + \beta \frac{h(x) - h(a)}{x - a} \right), \text{ by algebra of limits} \\ &= \lim_{x \rightarrow a} \left(\frac{(\alpha g + \beta h)(x) - (\alpha g + \beta h)(a)}{x - a} \right), \text{ rearranging fractions} \\ &= (\alpha g + \beta h)'(a), \text{ by definition}\end{aligned}$$

The above equations demonstrate the linearity of the derivative.

□

Theorem

4.7

(Product rule).

Consider

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$$\frac{df}{dx} = g \frac{dh}{dx} + h \frac{dg}{dx}.$$

Proof.

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es indeed exist! Do

$$f'(a) = \lim_{x \rightarrow a} \frac{g(x)h(x) - g(a)h(a)}{x - a}, \text{ def. of derivative and } f$$

$$= \lim_{x \rightarrow a} \frac{g(x)(h(x) - h(a)) + h(a)(g(x) - g(a))}{x - a}, \text{ preparatory manipulation}$$

$$= \left(\lim_{x \rightarrow a} g(x) \right) \left(\lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a} \right) + h(a) \left(\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \right), \text{ alg. of limits}$$

$$= g(a)h'(a) + h(a)g'(a), \text{ known derivs.}$$

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Example 4.3. 1.

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 $(x^2 -$
 $1) \sin x.$
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$$u(x) = x^2 - 1 \text{ and } v(x) = \sin x,$$

whose
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are

$$\frac{du}{dx} = 2x \text{ and } \frac{dv}{dx} = \cos x,$$

and
so
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is

$$\begin{aligned} \frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} \\ &= (x^2 - 1) \cos x + 2x \sin x. \end{aligned}$$

2.

Find
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$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

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$$\lim_{k \rightarrow 0} \epsilon(k) = 0.$$

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$$f(a+k) = f(a) + f'(a)k + \epsilon(k).$$

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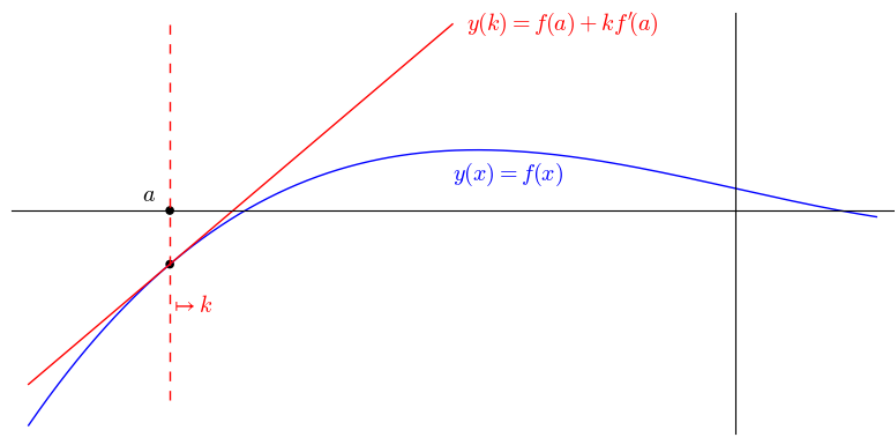


Figure 4.6:
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Let

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$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx},$$

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$$y'(a) = f'(g(a))g'(a).$$

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$$g(a+k) = g(a) + kg'(a) + k\epsilon(k)$$
$$f(g(a)+l) = f(g(a)) + lf'(g(a)) + l\mu(l),$$

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$\epsilon(k) \rightarrow$

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$y(a+k) = f(g(a+k))$, defn. of composition

$= f(g(a) + kg'(a) + k\epsilon(k))$, from above for $g(a+k)$

$= f(g(a)) + lf'(g(a)) + l\mu(l)$, from above with $l = kg'(a) + k\epsilon(k)$

$= f(g(a)) + k(g'(a)f'(g(a))) + \lambda(k)$,

where

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$\lambda(k)$

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$\lambda(k) = k\epsilon(k)f'(g(a)) + [kg'(a) + k\epsilon(k)]\mu(kg'(a) + k\epsilon(k))$.

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$$\frac{\lambda(k)}{k} = \epsilon(k)f'(g(a)) + [g'(a) + \epsilon(k)]\mu(kg'(a) + k\epsilon(k)).$$

Check the details

We do take the time to confirm for your-
self that $\lambda(k)/k$ has the required con-
vergence property.
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1.

Differentiate

$y =$

$(x^2 -$

$1)^5.$

Let

$u =$

$x^2 -$

1

so

that

$y =$

$u^5.$

Then

$\frac{du}{dx} =$

$2x$

and

$\frac{dy}{du} =$

$5u^4$

and

so

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = 5u^4 2x = 5(x^2 - 1)^4 2x = 10x(x^2 - 1)^4.$$

4.6.

PROP-

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OF Differentiate

THE $y =$

~~DERIVATIVE~~ $\sin(2x^2 - x)$.

Let

$u =$

$2x^2 -$

x

so

that

$y =$

$\sin u$.

Then

$\frac{du}{dx} =$

$4x -$

1

and

$\frac{dy}{du} =$

$\cos u$

and

so

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \cos(u)(4x-1) = (4x-1) \cos(2x^2 - x).$$

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$y =$
 $\frac{u}{v}$
 then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

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$$y'(a) = \frac{v(a)u'(a) - u(a)v'(a)}{v(a)^2}.$$

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4.6.
 PROP-
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 OF
 THE
 DERIVATIVE
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$a \neq$

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by

$$r'(a) = \frac{-1}{a^2}.$$

Proof.

Exercise.

□

Example

4.5.

These

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rule.

1.

Differentiate

$$y = \frac{x^2 - 1}{\sin x}.$$

Let

$$u =$$

$$x^2 -$$

$$1$$

and

$$v =$$

$$\sin x$$

then

$$\frac{du}{dx} =$$

$$2x$$

and

$$\frac{dv}{dx} =$$

$$\cos x.$$

Then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} = \frac{2x \sin x - (x^2 - 1) \cos x}{\sin^2 x}.$$

2.

Differentiate

CHAPTER

4.

CON-

TI- $y =$

NU- $\frac{2x-1}{3x^2+x}$.

ITY Let

& $u =$

~~DIFFERENTIATION~~

1

and

$v =$

$3x^2 +$

x

then

$\frac{du}{dx} =$

2

and

$\frac{dv}{dx} =$

$6x +$

1.

Then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} = \frac{2(3x^2 + x) - (2x - 1)(6x + 1)}{(3x^2 + x)^2} = \frac{-6x^2 + 6x + 1}{(3x^2 + x)^2}.$$

Derivatives

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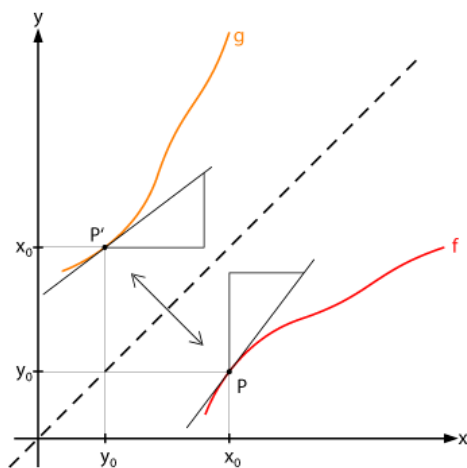
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4.6.
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 $f(x)$
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CHAPTER

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CON



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in-
verse

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}.$$

Parametric dif- fer- en- ti- a- tion

Curves
that
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graphs

4.6.
PROPERTIES
OF
THE
DERIVATIVE

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 tions

$$x = x(t), y = y(t),$$

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These
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CHAPTER

4.

CON-

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$(x(t), y(t))$

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DIFFERENTIATION

Example

4.6.

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1.

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curve

$x =$

$t^2,$

$y =$

4.6.

PROP-

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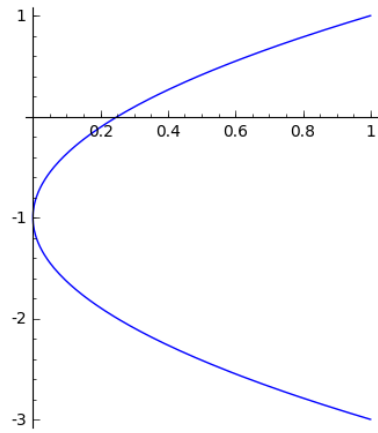


Figure 4.7:

The

para-

met-

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curve

$(t^2, 2t -$

$1)$.

$2t -$

1

with

$t \in$

$[-1, 1]$

is

shown

in

fig-

ure

[4.7](#).

2.

The

para-

met-

ric

curve

$x = \cos t - \cos(2t), y = \sin t - \sin(2t), t \in [-\pi, \pi]$

is

shown

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fig-

ure

[4.8](#).

We

can

CHAPTER
4.
CON-

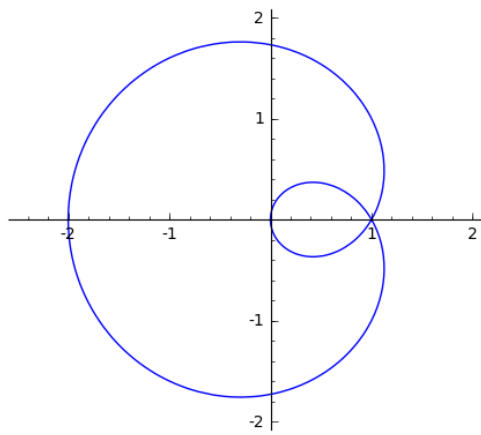


Figure 4.8:
The
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curve
 $(\cos(t) - \cos(2t), \sin(t) - \sin(2t))$.

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4.6.

PROP-

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$y =$

$y(x)$

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$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt},$$

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$$\frac{dy}{dx} = \frac{dy}{dt} \frac{1}{\frac{dx}{dt}},$$

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CHAPTER

4.

CON-

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DIFFERENTIATION

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}.$$

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that

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{1}{\frac{dx}{dt}}.$$

Example

4.7.

More

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$$\frac{dy}{dx} = \frac{dy}{dt} \frac{1}{\frac{dx}{dt}} = \frac{\cos t - 2 \cos(2t)}{-\sin t + 2 \sin(2t)}.$$

2.

If

4.6.
 PROP-
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 and
 $\frac{d^2y}{dx^2}$
 in
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 of
 t .

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{1}{\frac{dx}{dt}} = \frac{e^t}{te^t + e^t} = \frac{1}{t+1},$$

and

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{1}{\frac{dx}{dt}} = \frac{-1}{(t+1)^2} \frac{1}{te^t + e^t} = \frac{-1}{e^t(t+1)^3}.$$

Implicit dif- fer- en- ti- a- tion

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CHAPTER

4.

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4.6.

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Example

4.8.

Find

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4.6.
PROPERTIES OF THE DERIVATIVE

both sides of the equation with respect to x and carefully using the chain rule and product rule.

$$\begin{aligned} x^2 + 2y^2 - xy + 2x &= 4 \\ \Rightarrow \frac{d}{dx}(x^2 + 2y^2 - xy + 2x) &= \frac{d}{dx}(4) \\ \Rightarrow 2x + 4y \frac{dy}{dx} - x \frac{dy}{dx} - y + 2 &= 0 \\ \Rightarrow \frac{dy}{dx}(4y - x) &= y - 2 - 2x \\ \Rightarrow \frac{dy}{dx} &= \frac{y - 2 - 2x}{4y - x}. \end{aligned}$$

Logarithmic differentiation

This refers to the technique we use to

CHAPTER

4.

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$y(x)$

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Example

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4.6. *PROPERTIES OF THE DERIVATIVE*

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$$\begin{aligned}
 y &= x^x \\
 \Rightarrow \ln y &= \ln (x^x) \\
 \Rightarrow \ln y &= x \ln x \\
 \Rightarrow \frac{d}{dx}(\ln y) &= \frac{d}{dx}(x \ln x) \\
 \Rightarrow \frac{1}{y} \frac{dy}{dx} &= x \frac{1}{x} + \ln x = 1 + \ln x \\
 \Rightarrow \frac{dy}{dx} &= y(1 + \ln x) = x^x(1 + \ln x)
 \end{aligned}$$

Applications of differentiation

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CHAPTER

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$$f(a+x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} x^n.$$

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4.7.
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$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$$

**Maclaurin
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tions**

$$(1+x)^r = 1 + rx + \frac{r(r-1)}{2!}x^2 + \frac{r(r-1)(r-2)}{3!}x^3 + \cdots + \frac{r(r-1)\cdots(r-n+1)}{n!}x^n + \cdots$$

(where $-1 < x < 1$, $r \in \mathbb{R}$)

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{n!}x^n + \cdots, \quad (x \in \mathbb{R})$$

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots + \frac{(-1)^n}{(2n+1)!}x^{2n+1} + \cdots, \quad (x \in \mathbb{R})$$

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots + \frac{(-1)^n}{(2n)!}x^{2n} + \cdots, \quad (x \in \mathbb{R})$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots + \frac{(-1)^n}{n+1}x^{n+1} + \cdots, \quad (-1 < x \leq 1)$$

$$\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \cdots, \quad \left(-\frac{\pi}{2} < x < \frac{\pi}{2}\right)$$

Exercises

4.1.

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CHAPTER

4.

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DIFFERENTIATION

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$e^x \sin x$

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$\sin x$.

3.

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4.8.
 STA-
 TION-
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 POINTS ^{of}
 OF ⁽¹⁻
~~FUNCTIONS~~ ^{x²)-1/2}
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 sion
 for
 $\sin^{-1}(x)$.

4.8 Stationary points of func- tions

Definition

4.5
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 point).

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 $y =$
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CHAPTER

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$$\frac{dy}{dx} = 0.$$

Stationary

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4.8.
STA-
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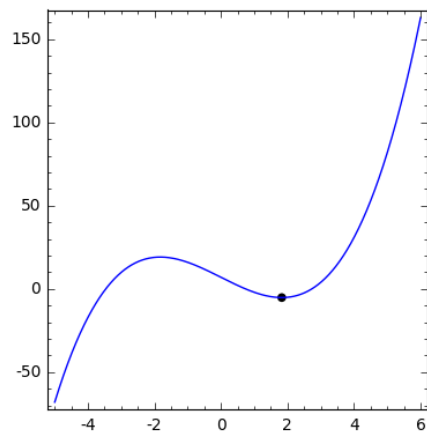


Figure 4.9:

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[4.10](#)
and
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CHAPTER
4.
CON-
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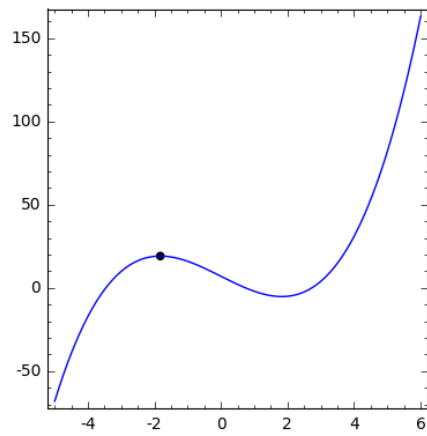


Figure 4.10:
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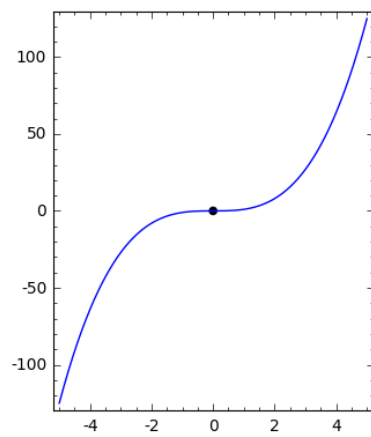


Figure 4.11:
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4.8.
 STA-
 TION-
 ARY
 POINTS
 OF
 FUNCTIONS
 turning
 points
 of
 the
 func-
 tion.

Classification of sta- tion- ary points us- ing deriva- tives

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 $x =$
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 $y =$
 $f(x)$
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 ing

$$\left. \frac{dy}{dx} \right|_{x=a} = 0.$$

The
 point
 $x =$
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CHAPTER

4.

CON-

TI-

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ITY **max-**

& **i-**

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DIFFERENTIATION

$$\left. \frac{dy}{dx} \right|_{x < a} > 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} < 0,$$

or

equiv-

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,

$$\left. \frac{d^2y}{dx^2} \right|_{x=a} < 0.$$

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$$\left. \frac{dy}{dx} \right|_{x < a} < 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} > 0,$$

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$$\left. \frac{d^2y}{dx^2} \right|_{x=a} > 0.$$

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$$\left. \frac{dy}{dx} \right|_{x < a} < 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} < 0,$$

or

,

$$\left. \frac{dy}{dx} \right|_{x < a} > 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} > 0.$$

Example

4.10.

4.8.
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1. Determine
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 $y =$
 $x^3 -$
 x .

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$$\frac{dy}{dx} = 3x^2 - 1 = 0.$$

This
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$$x_1 = +\sqrt{\frac{1}{3}} \text{ and } x_2 = -\sqrt{\frac{1}{3}}.$$

CHAPTER

4.

CON-

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DERIVATIVE

4. DIFFERENTIATION

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$$\frac{d^2y}{dx^2} = 6x,$$

and

so

$$\left. \frac{d^2y}{dx^2} \right|_{x=x_1} = 6x_1 \approx 3.462 > 0$$

and

$$\left. \frac{d^2y}{dx^2} \right|_{x=x_2} = 6x_2 \approx -3.462 < 0.$$

So

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4.8.
 STA-
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 $y =$
 $x^3 -$
 $x.$

2.
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 of
 the
 func-
 tion
 $y =$
 $3x^4 -$
 $4x^3.$

The
 equa-
 tion

$$\frac{dy}{dx} = 12x^2(x-1) = 0$$

has
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 $x_1 =$
 0
 and
 $x_2 =$
 1.

Now
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$$\left. \frac{dy}{dx} \right|_{x < 0} < 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > 0} < 0$$

and
 there-
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 x_1
 is

CHAPTER

4.

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$\left. \frac{d^2 y}{dx^2} \right|_{x=x_2}$

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4.9.
CURVE
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$$f(x) = \frac{3x^2 - 7}{2x + 1}.$$

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$f(x)$

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4.9.

CURVE

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points.
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that
 $f(x)$
tends
to
 $\pm\infty$
as
 x
tends
to
one
of
these
un-
de-
fined
points.

•

The
lo-
ca-
tions
where
the
graph
of
 f
cuts
the
co-
or-
di-
nate
axes,
i.e.
the
value
 $f(0)$
and
the
so-
lu-
tion
 x
to
 $f(x) =$

CHAPTER

4.

CON-
TI-

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NU-

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&

of

DIFFERENTIATION

Asymptotes

These

are

lines

(or

other

curves)

that

$f(x)$

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to

$\pm\infty$.

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Stationary

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of

f .

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or

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points

and

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points

of

in-

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tions.

•

Whether

$f(x)$

is

pos-

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4.9.

CURVE

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or

neg-

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on

var-

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parts

of

the

do-

main.

Whether

$f(x)$

lies

above

or

be-

low

any

asymptotic

lines/curves

at

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of

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Example

4.11.

Let

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func-

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f

de-

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above

CHAPTER

4.

CON-

by

FI-

NU-

$$f(x) = \frac{3x^2 - 7}{2x + 1}.$$

&

DIFFERENTIATION

f

will

be

un-

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ed

when

the

de-

nom-

i-

na-

tor

is

zero,

i.e.

when

$x =$

$-\frac{1}{2}$.

At

all

other

points

$f(x)$

is

de-

fin-

ed.

So

the

do-

main

of

this

func-

tion

is

$\mathbb{R} \setminus \{-\frac{1}{2}\}$.

•

If

x

is

close

to

$-\frac{1}{2}$

and

less

than

$-\frac{1}{2}$

then

the

4.9.

CURVE

SKETCHING

de-
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na-
tor
is
neg-
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tive
and
the
nu-
mer-
a-
tor
is
neg-
a-
tive
so
 $f(x)$
will
be
pos-
i-
tive
and
as
 x
tends
to
 $\frac{-1}{2}$
then
 $f(x)$
tends
to
 $+\infty$.
On
the
other
hand
if
 x
is
close
to
 $\frac{-1}{2}$
and
greater
than
 $\frac{-1}{2}$
then
the
de-
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CHAPTER

4.

CON-

TI- na-

NU- tor

ITY is

& pos-

itive

DIFFERENTIATION

and

the

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tor

is

neg-

a-

tive

so

$f(x)$

will

be

neg-

a-

tive

and

as

x

tends

to

$-\frac{1}{2}$

then

$f(x)$

tends

to

$-\infty$.

•

$f(0) =$

-7

and

$f(x) =$

0

when

$x =$

$\pm\sqrt{\frac{7}{3}} \approx$

± 1.53 .

•

Consider

the

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tion

of

f ,

$$f(x) = \frac{3x^2 - 7}{2x + 1}.$$

4.9.

CURVE

~~SKETCHING~~

Dividing

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to

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sion

$$f(x) = \frac{3}{2}x - \frac{3}{4} - \frac{\frac{25}{4}}{2x+1}.$$

So

we

see

that

the

line

$y =$

$\frac{3}{2}x -$

$\frac{3}{4}$

is

an

asympt-

ote

of

f .

As

x

tends

to

$\pm\infty$

the

term

$\frac{\frac{25}{4}}{2x+1}$

tends

to

zero

and

so

CHAPTER

4.

CON-

TI- the

NU- points

ITY $y =$

& $f(x)$

121 ~~122~~ ¹²³ ~~124~~ ¹²⁵ ~~126~~ ¹²⁷ ~~128~~ ¹²⁹ ~~130~~ ¹³¹ ~~132~~ ¹³³ ~~134~~ ¹³⁵ ~~136~~ ¹³⁷ ~~138~~ ¹³⁹ ~~140~~ ¹⁴¹ ~~142~~ ¹⁴³ ~~144~~ ¹⁴⁵ ~~146~~ ¹⁴⁷ ~~148~~ ¹⁴⁹ ~~150~~ ¹⁵¹ ~~152~~ ¹⁵³ ~~154~~ ¹⁵⁵ ~~156~~ ¹⁵⁷ ~~158~~ ¹⁵⁹ ~~160~~ ¹⁶¹ ~~162~~ ¹⁶³ ~~164~~ ¹⁶⁵ ~~166~~ ¹⁶⁷ ~~168~~ ¹⁶⁹ ~~170~~ ¹⁷¹ ~~172~~ ¹⁷³ ~~174~~ ¹⁷⁵ ~~176~~ ¹⁷⁷ ~~178~~ ¹⁷⁹ ~~180~~ ¹⁸¹ ~~182~~ ¹⁸³ ~~184~~ ¹⁸⁵ ~~186~~ ¹⁸⁷ ~~188~~ ¹⁸⁹ ~~190~~ ¹⁹¹ ~~192~~ ¹⁹³ ~~194~~ ¹⁹⁵ ~~196~~ ¹⁹⁷ ~~198~~ ¹⁹⁹ ~~200~~ ²⁰¹ ~~202~~ ²⁰³ ~~204~~ ²⁰⁵ ~~206~~ ²⁰⁷ ~~208~~ ²⁰⁹ ~~210~~ ²¹¹ ~~212~~ ²¹³ ~~214~~ ²¹⁵ ~~216~~ ²¹⁷ ~~218~~ ²¹⁹ ~~220~~ ²²¹ ~~222~~ ²²³ ~~224~~ ²²⁵ ~~226~~ ²²⁷ ~~228~~ ²²⁹ ~~230~~ ²³¹ ~~232~~ ²³³ ~~234~~ ²³⁵ ~~236~~ ²³⁷ ~~238~~ ²³⁹ ~~240~~ ²⁴¹ ~~242~~ ²⁴³ ~~244~~ ²⁴⁵ ~~246~~ ²⁴⁷ ~~248~~ ²⁴⁹ ~~250~~ ²⁵¹ ~~252~~ ²⁵³ ~~254~~ ²⁵⁵ ~~256~~ ²⁵⁷ ~~258~~ ²⁵⁹ ~~260~~ ²⁶¹ ~~262~~ ²⁶³ ~~264~~ ²⁶⁵ ~~266~~ ²⁶⁷ ~~268~~ ²⁶⁹ ~~270~~ ²⁷¹ ~~272~~ ²⁷³ ~~274~~ ²⁷⁵ ~~276~~ ²⁷⁷ ~~278~~ ²⁷⁹ ~~280~~ ²⁸¹ ~~282~~ ²⁸³ ~~284~~ ²⁸⁵ ~~286~~ ²⁸⁷ 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~~786~~ ⁷⁸⁷ ~~788~~ ⁷⁸⁹ ~~790~~ ⁷⁹¹ ~~792~~ ⁷⁹³ ~~794~~ ⁷⁹⁵ ~~796~~ ⁷⁹⁷ ~~798~~ ⁷⁹⁹ ~~800~~ ⁸⁰¹ ~~802~~ ⁸⁰³ ~~804~~ ⁸⁰⁵ ~~806~~ ⁸⁰⁷ ~~808~~ ⁸⁰⁹ ~~810~~ ⁸¹¹ ~~812~~ ⁸¹³ ~~814~~ ⁸¹⁵ ~~816~~ ⁸¹⁷ ~~818~~ ⁸¹⁹ ~~820~~ ⁸²¹ ~~822~~ ⁸²³ ~~824~~ ⁸²⁵ ~~826~~ ⁸²⁷ ~~828~~ ⁸²⁹ ~~830~~ ⁸³¹ ~~832~~ ⁸³³ ~~834~~ ⁸³⁵ ~~836~~ ⁸³⁷ ~~838~~ ⁸³⁹ ~~840~~ ⁸⁴¹ ~~842~~ ⁸⁴³ ~~844~~ ⁸⁴⁵ ~~846~~ ⁸⁴⁷ ~~848~~ ⁸⁴⁹ ~~850~~ ⁸⁵¹ ~~852~~ ⁸⁵³ ~~854~~ ⁸⁵⁵ ~~856~~ ⁸⁵⁷ ~~858~~ ⁸⁵⁹ ~~860~~ ⁸⁶¹ ~~862~~ ⁸⁶³ ~~864~~ ⁸⁶⁵ ~~866~~ ⁸⁶⁷ ~~868~~ ⁸⁶⁹ ~~870~~ ⁸⁷¹ ~~872~~ ⁸⁷³ ~~874~~ ⁸⁷⁵ ~~876~~ ⁸⁷⁷ ~~878~~ ⁸⁷⁹ ~~880~~ ⁸⁸¹ ~~882~~ ⁸⁸³ ~~884~~ ⁸⁸⁵ ~~886~~ ⁸⁸⁷ ~~888~~ ⁸⁸⁹ ~~890~~ ⁸⁹¹ ~~892~~ ⁸⁹³ ~~894~~ ⁸⁹⁵ ~~896~~ ⁸⁹⁷ ~~898~~ ⁸⁹⁹ ~~900~~ ⁹⁰¹ ~~902~~ ⁹⁰³ ~~904~~ ⁹⁰⁵ ~~906~~ ⁹⁰⁷ ~~908~~ ⁹⁰⁹ ~~910~~ ⁹¹¹ ~~912~~ ⁹¹³ ~~914~~ ⁹¹⁵ ~~916~~ ⁹¹⁷ ~~918~~ ⁹¹⁹ ~~920~~ ⁹²¹ ~~922~~ ⁹²³ ~~924~~ ⁹²⁵ ~~926~~ ⁹²⁷ ~~928~~ ⁹²⁹ ~~930~~ ⁹³¹ ~~932~~ ⁹³³ ~~934~~ ⁹³⁵ ~~936~~ ⁹³⁷ ~~938~~ ⁹³⁹ ~~940~~ ⁹⁴¹ ~~942~~ ⁹⁴³ ~~944~~ ⁹⁴⁵ ~~946~~ ⁹⁴⁷ ~~948~~ ⁹⁴⁹ ~~950~~ ⁹⁵¹ ~~952~~ ⁹⁵³ ~~954~~ ⁹⁵⁵ ~~956~~ ⁹⁵⁷ ~~958~~ ⁹⁵⁹ ~~960~~ ⁹⁶¹ ~~962~~ ⁹⁶³ ~~964~~ ⁹⁶⁵ ~~966~~ ⁹⁶⁷ ~~968~~ ⁹⁶⁹ ~~970~~ ⁹⁷¹ ~~972~~ ⁹⁷³ ~~974~~ ⁹⁷⁵ ~~976~~ ⁹⁷⁷ ~~978~~ ⁹⁷⁹ ~~980~~ ⁹⁸¹ ~~982~~ ⁹⁸³ ~~984~~ ⁹⁸⁵ ~~986~~ ⁹⁸⁷ ~~988~~ ⁹⁸⁹ ~~990~~ ⁹⁹¹ ~~992~~ ⁹⁹³ ~~994~~ ⁹⁹⁵ ~~996~~ ⁹⁹⁷ ~~998~~ ⁹⁹⁹ ~~1000~~ ¹⁰⁰¹ ~~1002~~ ¹⁰⁰³ ~~1004~~ ¹⁰⁰⁵ ~~1006~~ ¹⁰⁰⁷ ~~1008~~ ¹⁰⁰⁹ ~~1010~~ ¹⁰¹¹ ~~1012~~ ¹⁰¹³ ~~1014~~ ¹⁰¹⁵ ~~1016~~ ¹⁰¹⁷ ~~1018~~ ¹⁰¹⁹ ~~1020~~ ¹⁰²¹ ~~1022~~ ¹⁰²³ ~~1024~~ ¹⁰²⁵ ~~1026~~ ¹⁰²⁷ ~~1028~~ ¹⁰²⁹ ~~1030~~ ¹⁰³¹ ~~1032~~ ¹⁰³³ ~~1034~~ ¹⁰³⁵ ~~1036~~ ¹⁰³⁷ ~~1038~~ ¹⁰³⁹ ~~1040~~ ¹⁰⁴¹ ~~1042~~ ¹⁰⁴³ ~~1044~~ ¹⁰⁴⁵ ~~1046~~ ¹⁰⁴⁷ ~~1048~~ ¹⁰⁴⁹ ~~1050~~ ¹⁰⁵¹ ~~1052~~ ¹⁰⁵³ ~~1054~~ ¹⁰⁵⁵ ~~1056~~ ¹⁰⁵⁷ ~~1058~~ ¹⁰⁵⁹ ~~1060~~ ¹⁰⁶¹ ~~1062~~ ¹⁰⁶³ ~~1064~~ ¹⁰⁶⁵ ~~1066~~ ¹⁰⁶⁷ ~~1068~~ ¹⁰⁶⁹ ~~1070~~ ¹⁰⁷¹ ~~1072~~ ¹⁰⁷³ ~~1074~~ ¹⁰⁷⁵ ~~1076~~ ¹⁰⁷⁷ ~~1078~~ ¹⁰⁷⁹ ~~1080~~ ¹⁰⁸¹ ~~1082~~ ¹⁰⁸³ ~~1084~~ ¹⁰⁸⁵ ~~1086~~ ¹⁰⁸⁷ ~~1088~~ ¹⁰⁸⁹ ~~1090~~ ¹⁰⁹¹ ~~1092~~ ¹⁰⁹³ ~~1094~~ ¹⁰⁹⁵ ~~1096~~ ¹⁰⁹⁷ ~~1098~~ ¹⁰⁹⁹ ~~1100~~ ¹¹⁰¹ ~~1102~~ ¹¹⁰³ ~~1104~~ ¹¹⁰⁵ ~~1106~~ ¹¹⁰⁷ ~~1108~~ ¹¹⁰⁹ ~~1110~~ ¹¹¹¹ ~~1112~~ ¹¹¹³ ~~1114~~ ¹¹¹⁵ ~~1116~~ ¹¹¹⁷ ~~1118~~ ¹¹¹⁹ ~~1120~~ ¹¹²¹ ~~1122~~ ¹¹²³ ~~1124~~ ¹¹²⁵ ~~1126~~ ¹¹²⁷ ~~1128~~ ¹¹²⁹ ~~1130~~ ¹¹³¹ ~~1132~~ ¹¹³³ ~~1134~~ ¹¹³⁵ ~~1136~~ ¹¹³⁷ ~~1138~~ ¹¹³⁹ ~~1140~~ ¹¹⁴¹ ~~1142~~ ¹¹⁴³ ~~1144~~ ¹¹⁴⁵ ~~1146~~ ¹¹⁴⁷ ~~1148~~ ¹¹⁴⁹ ~~1150~~ ¹¹⁵¹ ~~1152~~ ¹¹⁵³ ~~1154~~ ¹¹⁵⁵ ~~1156~~ ¹¹⁵⁷ ~~1158~~ ¹¹⁵⁹ ~~1160~~ ¹¹⁶¹ ~~1162~~ ¹¹⁶³ ~~1164~~ ¹¹⁶⁵ ~~1166~~ ¹¹⁶⁷ ~~1168~~ ¹¹⁶⁹ ~~1170~~ ¹¹⁷¹ ~~1172~~ ¹¹⁷³ ~~1174~~ ¹¹⁷⁵ ~~1176~~ ¹¹⁷⁷ ~~1178~~ ¹¹⁷⁹ ~~1180~~ ¹¹⁸¹ ~~1182~~ ¹¹⁸³ ~~1184~~ ¹¹⁸⁵ ~~1186~~ ¹¹⁸⁷ ~~1188~~ ¹¹⁸⁹ ~~1190~~ ¹¹⁹¹ ~~1192~~ ¹¹⁹³ ~~1194~~ ¹¹⁹⁵ ~~1196~~ ¹¹⁹⁷ ~~1198~~ ¹¹⁹⁹ ~~1200~~ ¹²⁰¹ ~~1202~~ ¹²⁰³ ~~1204~~ ¹²⁰⁵ ~~1206~~ ¹²⁰⁷ ~~1208~~ ¹²⁰⁹ ~~1210~~ ¹²¹¹ ~~1212~~ ¹²¹³ ~~1214~~ ¹²¹⁵ ~~1216~~ ¹²¹⁷ ~~1218~~ ¹²¹⁹ ~~1220~~ ¹²²¹ ~~1222~~ ¹²²³ ~~1224~~ ¹²²⁵ ~~1226~~ ¹²²⁷ ~~1228~~ ¹²²⁹ ~~1230~~ ¹²³¹ ~~1232~~ ¹²³³ ~~1234~~ ¹²³⁵ ~~1236~~ ¹²³⁷ ~~1238~~ ¹²³⁹ ~~1240~~ ¹²⁴¹ ~~1242~~ ¹²⁴³ ~~1244~~ ¹²⁴⁵ ~~1246~~ ¹²⁴⁷ ~~1248~~ ¹²⁴⁹ ~~1250~~ ¹²⁵¹ ~~1252~~ ¹²⁵³ ~~1254~~ ¹²⁵⁵ ~~1256~~ ¹²⁵⁷ ~~1258~~ ¹²⁵⁹ ~~1260~~ ¹²⁶¹ ~~1262~~ ¹²⁶³ ~~1264~~ ¹²⁶⁵ ~~1266~~ ¹²⁶⁷ ~~1268~~ ¹²⁶⁹ ~~1270~~ ¹²⁷¹ ~~1272~~ ¹²⁷³ ~~1274~~ ¹²⁷⁵ ~~1276~~ ¹²⁷⁷ ~~1278~~ ¹²⁷⁹ ~~1280~~ ¹²⁸¹ ~~1282~~ ¹²⁸³ ~~1284~~ ¹²⁸⁵ ~~1286~~ ¹²⁸⁷ ~~1288~~ ¹²⁸⁹ ~~1290~~ ¹²⁹¹ ~~1292~~ ¹²⁹³ ~~1294~~ ¹²⁹⁵ ~~1296~~ ¹²⁹⁷ ~~1298~~ ¹²⁹⁹ ~~1300~~ ¹³⁰¹ ~~1302~~ ¹³⁰³ ~~1304~~ ¹³⁰⁵ ~~1306~~ ¹³⁰⁷ ~~1308~~ ¹³⁰⁹ ~~1310~~ ¹³¹¹ ~~1312~~ ¹³¹³ ~~1314~~ ¹³¹⁵ ~~1316~~ ¹³¹⁷ ~~1318~~ ¹³¹⁹ ~~1320~~ ¹³²¹ ~~1322~~ ¹³²³ ~~1324~~ ¹³²⁵ ~~1326~~ ¹³²⁷ ~~1328~~ ¹³²⁹ ~~1330~~ ¹³³¹ ~~1332~~ ¹³³³ ~~1334~~ ¹³³⁵ ~~1336~~ ¹³³⁷ ~~1338~~ ¹³³⁹ ~~1340~~ ¹³⁴¹ ~~1342~~ ¹³⁴³ ~~1344~~ ¹³⁴⁵ ~~1346~~ ¹³⁴⁷ ~~1348~~ ¹³⁴⁹ ~~1350~~ ¹³⁵¹ ~~1352~~ ¹³⁵³ ~~1354~~ ¹³⁵⁵ ~~1356~~ ¹³⁵⁷ ~~1358~~ ¹³⁵⁹ ~~1360~~ ¹³⁶¹ ~~1362~~ ¹³⁶³ ~~1364~~ ¹³⁶⁵ ~~1366~~ ¹³⁶⁷ ~~1368~~ ¹³⁶⁹ ~~1370~~ ¹³⁷¹ ~~1372~~ ¹³⁷³ ~~1374~~ ¹³⁷⁵ ~~1376~~ ¹³⁷⁷ ~~1378~~ ¹³⁷⁹ ~~1380~~ ¹³⁸¹ ~~1382~~ ¹³⁸³ ~~1384~~ ¹³⁸⁵ ~~1386~~ ¹³⁸⁷ ~~1388~~ ¹³⁸⁹ ~~1390~~ ¹³⁹¹ ~~1392~~ ¹³⁹³ ~~1394~~ ¹³⁹⁵ ~~1396~~ ¹³⁹⁷ ~~1398~~ ¹³⁹⁹ ~~1400~~ ¹⁴⁰¹ ~~1402~~ ¹⁴⁰³ ~~1404~~ ¹⁴⁰⁵ ~~1406~~ ¹⁴⁰⁷ ~~1408~~ ¹⁴⁰⁹ ~~1410~~ ¹⁴¹¹ ~~1412~~ ¹⁴¹³ ~~1414~~ ¹⁴¹⁵ ~~1416~~ ¹⁴¹⁷ ~~1418~~ ¹⁴¹⁹

4.9.

CURVE

~~SKETCHING~~

0.

The
deriva-
tive
is
(us-
ing
the
quo-
tient
rule)

$$\frac{df}{dx} = \frac{(2x+1)(6x) - (3x^2-7)(2)}{(2x+1)^2} = \frac{6x^2+6x+14}{(2x+1)^2}.$$

So
we
need
to
solve
 $6x^2+$
 $6x+$
 $14 =$
0.

But
the
dis-
crim-
i-

nant
 $b^2 -$
 $4ac$

of
this
equa-
tion
is

neg-
a-
tive
and
so

this
equa-
tion
has

no
real
roots.

There-
fore

f
has

no
turn-
ing
points.

CHAPTER

4.

CON-
Finally

II-
We

can

III

put

&
all

this

in-

for-

ma-

tion

to-

gether

to

pro-

duce

a

sketch

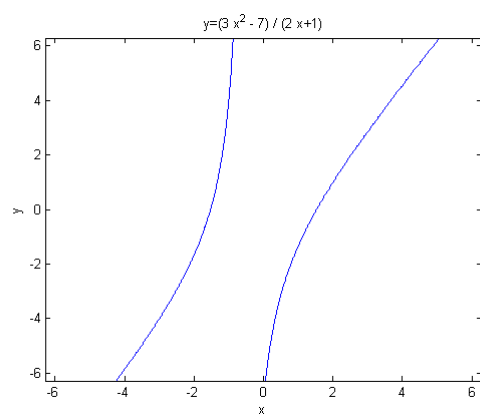
of

the

func-

tion

DIFFERENTIATION



Exercises

4.2.

Here
are
some
more
ex-
am-
ples
to
in-
ves-
ti-
gate.

1.

$$y = \frac{x^2 + 1}{x^2 - 3x + 2}.$$

4.9.

CURVE

~~SKETCHING~~

2.

$$y = \frac{x}{1+x^2}.$$

3.

$$y = \frac{x^3}{2x-1}.$$

4.

This

last

curve

is

not

the

graph

of

a

ra-

tio-

nal

func-

tion.

But

it

il-

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some

more

sketch-

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tech-

niques.

Con-

sider

the

curve

de-

fin-

ed

by

the

equa-

tion

$$x^2+xy+y^2=1.$$

Show

that

the

max-

i-

mum

and

min-

i-

mum

CHAPTER

4.

CON-

TI- val-

NU- ues

ITY taken

& by

~~126~~ DIFFERENTIATION

are

± 2

$\sqrt{3}$.

By

chang-

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or-

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nates

sketch

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curve.

Po-

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di-

nates

are

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pair

of

co-

or-

di-

nates

(r, θ)

which

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ca-

tion

of

a

point

in

the

plane

by

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fy-

ing

the

points

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4.9.

CURVE

~~SKETCHING~~

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i.e.

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the

x -

axis.

The

equa-

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Carte-

sian

co-

or-

di-

nates

(x, y)

and

the

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co-

or-

CHAPTER

4.

CON-

TI- di-

NU- nates

ITY (r, θ)

& are

~~DIFFERENTIATION~~

$x = r \cos \theta, \quad y = r \sin \theta.$

4.10 Kinematics

Kinematics

is

the

study

of

the

mo-

tion

of

a

body

with-

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to

the

forces

act-

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on

the

body.

We

study

the

mo-

tion

of

a

body

in

1-

dimension

us-

ing

the

three

func-

tions:

x

dis-

place-

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v

ve-

4.10.
~~4.9~~ KINEMATICS

loc-
 ity
 and
 a
 ac-
 cel-
 er-
 a-
 tion.

- $x(t)$
 is
 the
 po-
 si-
 tion
 of
 the
 body
 at
 time
 t ,

- $v(t)$
 is
 the
 ve-
 loc-
 ity
 of
 the
 body
 at
 time
 t ,

- $a(t)$
 is
 the
 ac-
 cel-
 er-
 a-
 tion
 of
 the
 body
 at
 time
 t .

These
 quan-
 ti-
 ties

CHAPTER

4.

CON-

AP-

IF-

NU-

lated

III

by

&

DIFFERENTIATION

$v(t) = \frac{dx}{dt}$ and $a(t) = \frac{dv}{dt}$.

Various

equa-

tions

re-

lat-

ing

the

$x(t)$,

$v(t)$,

$a(t)$

and

t

can

be

found

by

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te-

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ing

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above

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tions.

The

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cel-

er-

a-

tion

prob-

lems.

Here

the

ac-

cel-

er-

4.10.

KINEMATICS

a-
tion
is
con-
stant,
e.g.
due
to
the
in-
flu-
ence
of
grav-
ity
(9.8ms^{-2}
down-
wards),
or
a
car
ac-
cel-
er-
at-
ing
at
a
con-
stant
rate,
etc.

Example

4.12.

These
sim-
ple
kine-
mat-
ics
prob-
lems
can
be
solved
by
ap-
ply-
ing
the
re-
la-
tion-
ships
be-

CHAPTER

4.

CON-
tween

TY-

NU-

SITY

tion,

&

loc-

ity

and

ac-

cel-

er-

a-

tion.

DIFFERENTIATION

1.

A

car

waits

at

a

traf-

fic

light.

When

the

light

turns

green

the

car

ac-

cel-

er-

ates

at

a

con-

stant

rate

of

6ms^{-2}

for

4

sec-

onds.

What

ve-

loc-

ity

will

the

car

have

reached

af-

ter

4.10.

~~4.10~~ KINEMATICS

4
sec-
onds
and
how
far
will
it
have
trav-
eled.

2.

A
stu-
dent,
who
is
run-
ning
to
catch
his
bus,
sees
the
bus
start
to
pull
away
from
the
stop.
He
is
20
m
away
from
the
stop
and
run-
ning
at
his
top
speed
of
 5ms^{-1} .
If
the
bus
ac-
cel-

CHAPTER

4.

CON-

TI- er-

NU- ates

ITY at

& 2ms^{-2} ,

can

~~DIFFERENTIATION~~

he

catch

the

bus?

3.

Suppose

you

are

at

a

win-

dow

18m

above

the

ground

and

you

want

to

throw

a

stone

out

the

win-

dow

so

that

it

will

be

in

the

air

for

ex-

actly

3

sec-

onds.

How

should

you

throw

it?

Exercises**4.3.**

These

ex-

er-

cises

in-

volve

the

def-

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tions,

prop-

er-

ties

and

ap-

pli-

ca-

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1.

Decide

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and

if

so

eval-

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ate

them.

a)

$$\lim_{x \rightarrow 3} \frac{2x - 1}{3x + 2}$$

b)

$$\lim_{x \rightarrow 2} \frac{3x - 6}{2x}$$

CHAPTER

4.

CON-

TI- c)

NU-

$$\lim_{x \rightarrow 0} \frac{2 \sinh x - 2 \sin x}{\cosh x - 1}$$

&

DIFFERENTIATION

$$\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{1 - \cos(2x)}$$

e)

$$\lim_{x \rightarrow 0} \frac{\sin(4x)}{\sin(6x)}$$

f)

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$$

g)

$$\lim_{x \rightarrow \infty} \frac{3x^3 - 6x}{3x + 2x^3}$$

h)

$$\lim_{x \rightarrow 0} \frac{5x - 6}{2x^3}$$

i)

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$$

2.

Using
the
for-
mal
def-
i-
ni-
tion
of
con-
ti-
nu-
ity
prove
that
if
two
func-
tions
 f
and
 g
are
both

4.10.

437 KINEMATICS

con-
tin-
u-
ous
at
the
point
 $x =$
 a
then
so
is
the
func-
tion
 $f +$
 $g \cdot$

3.

Prove
from
first
prin-
ci-
ples
that
 $\frac{d}{dx}(x^3) =$
 $3x^2$.

4.

Find
 $\frac{dy}{dx}$
for
each
of
the
fol-
low-
ing
func-
tions

a)

$$y = 5 \cos(2x) + 3e^{2x} - 1$$

b)

$$y = e^{4x} \cos(2x)$$

c)

$$y = \frac{\sin(2x)}{xe^x}$$

d)

$$y = e^{\cos(3x)}$$

CHAPTER

4.

CON-

TI- e)

NU-

ITY $y =$

& $(2x)^{3x} +$

~~DIFFERENTIATION~~ $\sin(2x^2 - 7e^x)$

$y =$

$\cos^{-1} x$

5.

Find

the

gra-

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of

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fol-

low-

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itly

de-

fin-

func-

tions

y

at

the

points

in-

di-

cated

a)

$3x^2 +$

$y^2 +$

$2xy -$

$6 =$

0

at

the

point

$(1, 1)$

b)

$x^2 +$

$y^2 -$

$6xy +$

$4x +$

$4 =$

0

at

the

point

$(1, 3)$

4.10.

~~4.9~~ KINEMATICS

6.

Given
that

a)

$$\begin{aligned}x &= \\&5 \cosh t \\&\text{and} \\y &= \\&4 \sinh t\end{aligned}$$

b)

$$\begin{aligned}x &= \\&2 \cos t \\&\text{and} \\y &= \\&\sin^2 t\end{aligned}$$

c)

$$\begin{aligned}x &= \\&t^2 - \\&4 \\&\text{and} \\y &= \\&t^3 - \\&4t \\&\text{for} \\&-3 \leq \\&t \leq \\&3\end{aligned}$$

determine

$$\frac{dy}{dx}$$

and

$$\frac{d^2y}{dx^2}$$

in

terms

of

t

for

each

para-

met-

ric

curve.

Sketch

these

curves

us-

ing

MAT-

LAB.

7.

Find

$$\frac{dy}{dx}$$

where

$$y =$$

CHAPTER

4.

CON-
TI-
NU-

ITY

&

DIFFERENTIATION

Find
 $\frac{d}{dx}\left(\frac{u}{v}\right)$

where

u, v, w

and

z

are

func-

tions

of

x .

9.

Suppose

that

an

ath-

let-

ics

field

con-

sists

of

a

rect-

an-

gle

with

semi-

circular

ends

as

shown.

De-

ter-

mine

the

max-

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mum

pos-

si-

ble

area

of

the

rect-

an-

gu-

lar

part

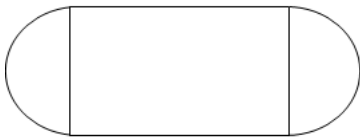
of

the

4.10.

KINEMATICS

field
if
the
run-
ning
track
around
the
perime-
ter
of
the
whole
field
is
to
have
a
length
of
ex-
actly
400m.



10.

Determine
the
di-
men-
sions
of
a
cylin-
dri-
cal
food
tin
of
vol-
ume
330ml
that
has
min-
i-
mum

CHAPTER

4.

CON-

TI- sur-

NU- face

ITY area.

&11.

DIFFERENTIATION

Sketch

graphs

of

the

fol-

low-

ing

func-

tions,

a)

$$y = \frac{2+x}{1+x},$$

b)

$$y = \frac{1}{2} \left(x + \frac{2}{x} \right),$$

c)

$$y = \frac{3x^4 + 12x^2 - 4}{8x^3},$$

d)

$$y = \frac{(x-1)(x-2)}{(x+1)(x-3)}.$$

12.

By

lo-

cat-

ing

their

turn-

ing

points

and

asymptotes

sketch

graphs

of

the

fol-

low-

ing

func-

tions,

a)

$$y = \frac{x^2 - 8x + 15}{x},$$

4.10.

~~4.10~~ KINEMATICS

b)

$$y = \frac{x+1}{x-1},$$

c)

$$y = \frac{x^2 + 5x - 14}{x + 5}.$$

13.

Consider
the
func-
tion
 f
de-
fined
by

$$f(x) = e^{\cos x}.$$

Show
that
 $f'(x) =$
 $-f(x) \sin x$
and
hence
ob-
tain
a
for-
mula
for
 $f''(x)$
in
terms
of
 $f(x)$
and
 $f'(x)$.
Dif-
fer-
en-
ti-
ate
again
to
find
a
for-
mula
for
 $f^{(3)}(x)$.
Use
these
to
find

CHAPTER

4.

CON-

TI- the

NU- first

ITY few

& terms

of

DIFFERENTIATION

the

Maclau-

rin

se-

ries

for

f .

Confirm

that

your

an-

swer

is

cor-

rect

by

ob-

tain-

ing

the

Maclau-

rin

se-

ries

for

f

us-

ing

the

se-

ries

for

e^x

and

$\cos x$.

14.

Show

that

the

bi-

no-

mial

ex-

pan-

sion

of

$(1+$

$x)^{-1}$

is

$$(1+x)^{-1} = 1-x+x^2-x^3+x^4-\dots, \quad (-1 < x < 1).$$

4.10.

~~4.10~~ KINEMATICS

Use
this
to
ob-
tain
the
Maclau-
rin
se-
ries
for
 $\tan^{-1} x$
which
has
can
be
de-
fined
by

$$\tan^{-1} x = \int_0^x \frac{1}{1+z^2} dz.$$

15.

Use
the
Maclau-
rin
se-
ries
for
 $\sin x$
and
 $\cos x$
to
ob-
tain
the
se-
ries
for
 $\tan x$.

16.

Deduce
the
se-
ries
for
 $\ln(\cos x)$
us-
ing
the
se-
ries
for
 $\ln x$

CHAPTER

4.

CON-

TI- and

NU- $\cos x$.

ITY.

17.

& Use

DIFFERENTIATION

the

se-

ries

for

e^x

and

$\cos x$

to

show

that

$$\lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - 2}{2 \cos 2x - 2} \right) = \frac{-1}{4}.$$

18.

Use

the

se-

ries

for

e^{-x^2}

to

show

that

$$\int_0^{1/2} e^{-x^2} dx \approx 0.461.$$

For

these

kine-

mat-

ics

prob-

lems

as-

sume

that

the

ef-

fects

of

air

re-

sis-

tance

can

be

ig-

nored

and

4.10.

~~4.10~~ KINEMATICS

*that
ac-
cel-
er-
a-
tion
due
to
grav-
ity
is
9.8
 ms^{-2} .*

19.

A
stone
is
dropped
from
a
height
of
5m
above
the
ground.
How
long
will
it
take
to
drop
to
the
ground?
What
is
it's
speed
upon
im-
pact?

20.

A
car
starts
from
rest
and
ac-
cel-
er-
ates
in

CHAPTER

4.

CON-

TI- a

NU- straight

ITY line

& with

copy

DIFFERENTIATION

stant

ac-

cel-

er-

a-

tion.

If

it

reaches

a

speed

of

20ms^{-1}

af-

ter

5s

what

is

its

ac-

cel-

er-

a-

tion

and

how

far

will

it

have

trav-

elled?

21.

A

pro-

jec-

tile

is

fired

straight

up

into

the

air.

If

the

pro-

jec-

tile

takes

4.10.

~~4.10~~ KINEMATICS

2.4s
to
reach
its
high-
est
point
what
was
its
ini-
tial
speed
and
how
high
did
it
reach?

*Finding
op-
ti-
mal
so-
lu-
tions
to
prob-
lems
by
find-
ing
ap-
pro-
pri-
ate
sta-
tion-
ary
points
of
func-
tions
is
an
im-
por-
tant
ap-
pli-
ca-
tion
of
the*

CHAPTER

4.

CON-

TI- *tech-*

NU- *nique*

ITY *of*

& *dif-*

~~DIFF~~ *fer-*

ERENTIATION

en-

ti-

a-

tion.

Here

are

some

fur-

ther

ex-

er-

cises

for

you

to

try.

22.

Of

all

the

rect-

an-

gles

hav-

ing

a

par-

tic-

u-

lar

area,

which

one

has

the

small-

est

perime-

ter?

23.

Of

all

the

rect-

an-

gles

hav-

ing

a

par-

4.10.

~~45~~ KINEMATICS

tic-
u-
lar
perime-
ter,
which
one
has
the
great-
est
area?

24.

A
box
with
a
square
base
and
no
top
is
to
have
a
vol-
ume
of
 3m^3 .
Find
the
di-
men-
sions
of
the
box
which
uses
the
least
amount
of
ma-
te-
rial
in
its
con-
struc-
tion.

25.

Suppose
that

CHAPTER

4.

CON-

TI- a

NU- rect-

ITY an-

& gle

has

DIFFERENTIATION

two

cor-

ners

ly-

ing

on

the

x -

axis

and

the

other

two

ly-

ing

on

the

parabola

$y =$

$8 -$

x^2 .

Find

such

a

rect-

an-

gle

with

max-

i-

mal

area.

26.

What

is

the

largest

rect-

an-

gle

one

can

in-

scribe

in

a

cir-

cle?

That

is,

4.10.

~~45~~ KINEMATICS

find
the
rect-
an-
gle
of
largest
area
that
can
fit
in-
side
the
cir-
cle
of
ra-
dius
 r .

27.

Consider
a
river
that
is
0.5
miles
wide
and
runs
in
a
straight
line
for
the
pur-
poses
of
this
prob-
lem.
A
pipeline
is
to
be
laid
across
this
river
and
run
to

CHAPTER

4.

CON-

TI- a

NU- point

ITY 3

& miles

down

DIFFERENTIATION

stream

on

the

op-

po-

site

bank.

Lay-

ing

pipe

un-

der

the

wa-

ter

is

ex-

pen-

sive,

in

fact

25%

more

ex-

pen-

sive

per

mile

than

lay-

ing

pipe

on

dry

land.

How

should

the

pipeline

be

lain

across

the

river

in

or-

der

to

min-

i-

mize

4.10.

~~45~~ KINEMATICS

the
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tal
cost.

Chapter 5

Groups

A
bi-
nary
op-
er-
a-
tion
on
a
set
is,
roughly
speak-
ing,
a
way
of
com-
bin-
ing
pairs
of
el-
e-
ments
from
a
set
to
pro-
duce
a
new
el-
e-
ment.
We
see
many
ex-

CHAPTER

5.

GROUPS

am-
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of
bi-
nary
op-
er-
a-
tions
in
math-
e-
mat-
ics
in
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top-
ics
we
have
stud-
ied
up
to
now.
We
see
num-
bers,
of
var-
i-
ous
sorts,
and
the
arith-
metic
op-
er-
a-
tions
of
ad-
di-
tion,
mul-
ti-
pli-
ca-
tion
etc
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can
per-

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on
them.
We
have
met
ma-
tri-
ces
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ca-
tion.
We
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a-
tions
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di-
tion,
mul-
ti-
pli-
ca-
tion
and
com-
po-
si-
tion.
In
these
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i-

CHAPTER

5.

GROUPS

ous

sys-
tems

we
see

a
lot

of
sim-

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lar-

i-
ties

in
how

the
op-

er-
a-

tions
work.

These
com-

mon
prop-

er-
ties

of
op-

er-
a-

tions
on

sets
are

what
forms

the
sub-

ject
mat-

ter
of

ab-
stract

al-
ge-

bra.
And

one
of

the
pri-

mary
types

of

5.1.
GROUP
AXIOMS
ob-
ject
stud-
ied
in
ab-
stract
al-
ge-
bra
is
a
group.

5.1 Group ax- ioms

Definition

5.1.
A
group
 $(G, *)$
is
a
pair
con-
sist-
ing
of
a
non-
empty
set
 G
to-
gether
with
a
bi-
nary
op-
er-
a-
tion
 $*$
de-
fined
on
 G
such
that
the
fol-
low-

CHAPTER

5.

GROUPS

ing

ax-

ioms

hold,

1.

(*Closure*)

$\forall a, b \in$

$G \quad a*$

$b \in$

$G,$

2.

(*Associativity*)

$\forall a, b, c \in$

$G \quad a*$

$(b*$

$c) =$

$(a*$

$b)*$

$c,$

3.

(*Existence*

of

iden-

tity)

$\exists e \in$

$G \quad \forall a \in$

$G \quad e*$

$a =$

$a*$

$e =$

$a,$

4.

(*Existence*

of

in-

verses)

$\forall a \in$

$G \quad \exists a^{-1} \in$

$G \quad a*$

$a^{-1} =$

$a^{-1}*$

$a =$

$e.$

Note

that

the

group

is

the

pair

con-

sist-

ing

5.1.
GROUP
AXIOMS

of
the
set
and
the
op-
er-
a-
tion
that
to-
gether
sat-
isfy
the
four
group
ax-
ioms.
This
is
an
im-
por-
tant
point
as
there
may
be
more
than
one
op-
er-
a-
tion
that
one
can
de-
fine
on
a
given
set
 G
such
that
it
forms
a
group
with
re-
spect

CHAPTER

5.

GROUPS

to
that
op-
er-
a-
tion.
How-
ever
in
prac-
tice
we
may
re-
lax
this
for-
mal-
ism
and
sim-
ply
re-
fer
to
'the
group
 G ',
as
long
as
what
the
op-
er-
a-
tion
is
is
un-
der-
stood
and
no
con-
fu-
sion
will
arise.

Definition

5.2.

A
group
 $(G, *)$
is

5.1.
GROUP
AXIOMS
Abelian

if
and
only
if
the
op-
er-
a-
tion
*
is
com-
mu-
ta-
tive
on
 G ,
i.e.

$$\forall a, b \in G \quad a*b = b*a.$$

The
com-
mu-
ta-
tive
prop-
erty
is
not
in-
cluded
as
a
group
ax-
iom
be-
cause
there
are
many
im-
por-
tant
ex-
am-
ples
of
sys-
tems
that
are
not
com-
mu-

CHAPTER

5.

GROUPS

ta-

tive,

for

ex-

am-

ple

any

sys-

tems

of

func-

tions

with

the

com-

po-

si-

tion

op-

er-

a-

tion,

or

a

sys-

tem

of

ma-

tri-

ces

with

the

mul-

ti-

pli-

ca-

tion

op-

er-

a-

tion.

In-

deed,

be-

ing

Abelian

is

con-

sid-

ered

quite

a

spe-

cial

prop-

erty

5.1.
GROUP
AXIOMS
 of
 a
 group,
 and
 it
 sim-
 pli-
 fies
 much
 of
 the
 the-
 ory.

Definition

5.3
 (Group
 or-
 der).
 The
*or-
 der*
 of
 a
 group
 G
 is
 the
 num-
 ber
 of
 el-
 e-
 ments
 con-
 tained
 in
 G .

CHAPTER

5.

GROUPS

5.2 Some

im-
me-
di-
ate
con-
se-
quences
of
the
group
ax-
ioms

Before

any

fur-

ther

in-

ves-

ti-

ga-

tion

into

the

var-

i-

ous

ex-

am-

ples

of

groups

that

ex-

ist

in

math-

e-

mat-

ics

we

can

im-

me-

di-

ately

draw

some

con-

se-

quences

from

the

5.2.

SOME

IM-

GROUP

THE

DI-

IONS

THE

THEOREM

SH.

QUENCES

OF

THE

GROUP

AXIOMS

ments

of

a

group

is

unique.

Proof.

Suppose

that

e

and

f

are

two

iden-

tity

el-

e-

ments

of

a

group

$(G, *)$.

Con-

sider

the

group

el-

e-

ment

$e*$

f .

Since

e

is

an

iden-

tity

we

know

that

$e*f = f$.

However

f

CHAPTER

5.

GROUPS

is

also

an

iden-

tity

and

so

$$e * f = e.$$

Immediately

then

we

can

draw

the

con-

clu-

sion

that

$$e = f.$$

And

so

the

iden-

tity

el-

e-

ment

in

a

group

is

unique.

□

Theorem

5.2.

The

in-

verse

of

a

group

el-

e-

ment

is

unique.

Proof.

Let

$(G, *)$

be

a

group

5.2.
 SOME
 IM-
 ME-
 DI-
 ATE
 CON-
 SE-
 QUENCES
 OF
 THE
 GROUP
 AXIOMS
 91
 ments
 $f, h \in$
 G
 are
 both
 in-
 verses
 of
 g ,
 i.e.

$$g*f = f*g = e \text{ and } g*h = h*g = e,$$

where
 e
 is
 the
 iden-
 tity
 in
 $(G, *)$.

To
 show
 that
 $f =$
 h
 we
 shall
 con-
 sider
 the
 el-
 e-
 ment

$f*$
 $g*$
 h ,
 mak-
 ing
 use
 of
 the
 as-
 so-
 cia-
 tiv-
 ity

CHAPTER

5.

GROUPS

of

*,

Note

that

$$f = f * e = f * (g * h) = (f * g) * h = e * h = h,$$

as

re-

quired.

□

Theorem

5.3.

The

can-

cel-

la-

tion

law

holds,

i.e.

$$\forall a, b, c \in G \quad a * c = b * c \Rightarrow a = b,$$

and

$$\forall a, b, c \in G \quad c * a = c * b \Rightarrow a = b.$$

Proof.

This

is

proved

im-

me-

di-

ately

by

mul-

ti-

ply-

ing

both

sides

of

the

given

equa-

tion

by

the

el-

e-

ment

c^{-1} ,

on

the

right

5.2.
SOME
IM-
ME-
left
DI-
as
ATE
at
CON-
pre-
SE-
QUENCES
of
OF
THE
GROUP
The
AXIOMS
next

re-
sult
should
re-
mind
you
of
sim-
i-
lar
re-
sults
in
the
con-
text
of
ma-
tri-
ces
and
func-
tions.
It
is
a
gen-
er-
al-
i-
sa-
tion
of
those
re-
sults
to
any
group.

Theorem

5.4.
The
in-
verse
of

CHAPTER

5.

GROUPS

a

prod-

uct

is

the

prod-

uct

of

the

in-

verses,

but

in

the

op-

po-

site

or-

der,

i.e.

$$\forall a, b \in G \quad (a * b)^{-1} = b^{-1} * a^{-1}.$$

Proof.

This

is

proved

us-

ing

the

as-

so-

cia-

tiv-

ity

prop-

erty

and

the

def-

i-

ni-

tion

of

in-

verses,

$$\begin{aligned}(a * b) * (b^{-1} * a^{-1}) &= a * (b * b^{-1}) * a^{-1}, \text{ by associativity} \\ &= a * e * a^{-1} \\ &= a * a^{-1} \\ &= e.\end{aligned}$$

And

sim-

i-

larly

5.2.

SOME

IM-

QUE-

ME-

can

DI-

show

THE

that

CON-

SE-

QUENCES

OF

THE

THE

GROUP

AXIOMS

(A1).

(A2).

(A3).

(A4).

(A5).

(A6).

(A7).

(A8).

(A9).

(A10).

(A11).

(A12).

(A13).

(A14).

(A15).

(A16).

(A17).

(A18).

(A19).

(A20).

(A21).

(A22).

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(A26).

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(A28).

(A29).

(A30).

(A31).

(A32).

(A33).

(A34).

(A35).

(A36).

(A37).

(A38).

(A39).

(A40).

(A41).

(A42).

(A43).

(A44).

(A45).

(A46).

(A47).

(A48).

(A49).

$$(a^{-1} * a^{-1}) * (a * b) = e.$$

Therefore

THE

GROUP

AXIOMS

(A1).

(A2).

(A3).

(A4).

(A5).

(A6).

(A7).

(A8).

(A9).

(A10).

(A11).

(A12).

(A13).

(A14).

(A15).

(A16).

(A17).

(A18).

(A19).

(A20).

(A21).

(A22).

(A23).

(A24).

(A25).

(A26).

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(A29).

(A30).

(A31).

(A32).

(A33).

(A34).

(A35).

(A36).

(A37).

(A38).

(A39).

(A40).

(A41).

(A42).

(A43).

(A44).

(A45).

(A46).

(A47).

(A48).

(A49).

(A50).

(A51).

(A52).

(A53).

(A54).

Theorem

5.5

(Gen-

eral

as-

so-

cia-

tiv-

ity).

In

any

group

$(G, *)$

gen-

eral

as-

so-

cia-

tiv-

ity

holds.

Gen-

eral

as-

so-

cia-

tiv-

ity

says

that

in

any

prod-

uct

of

n

el-

e-

ments

$a_1, a_2, \dots, a_n$

the

prod-

uct

is

CHAPTER

5.

GROUPS

well-

defined

no

mat-

ter

how

paren-

the-

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are

in-

serted

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prod-

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to

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di-

cate

how

it

should

be

eval-

u-

ated

by

form-

ing

pair-

wise

prod-

ucts,

e.g.

$$(5.1) \quad ((a_1 * (a_2 * a_3)) * a_4) = (a_1 * (a_2 * (a_3 * a_4))) = etc$$

As

we

have

said

above

there

are

many

ex-

am-

ples

of

groups

in

the

math-

e-

5.3.
NUM-
BER
GROUPS
ics
that
we
have
stud-
ied
so
far.
We
now
de-
scribe
some
im-
por-
tant
classes
of
groups,
some
us-
ing
quite
fa-
mil-
iar
con-
cepts
and
some
that
may
be
new
to
you.

5.3 Number groups

We
can
form
many
groups
out
of
the
fa-
mil-
iar
num-
ber
sys-

CHAPTER

5.

GROUPS

terms
and
op-
er-
a-
tions.

Re-
mem-
ber
that

$\mathbb{Z}$
de-
notes

the
set
of
in-
te-
gers,

$\mathbb{Q}$
de-
notes

the
set
of
ra-
tio-
nal
num-
bers
and

$\mathbb{R}$
de-
notes
the
set
of
real
num-
bers.

Example

5.1.

The
fol-
low-
ing
pairs
all
form
groups

$(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{Q} - \{0\}, \times)$, $(\mathbb{R} - \{0\}, \times)$.

For
the
ad-

5.3. NUM- BER GROUPS

tion
op-
er-
a-
tion
the
iden-
tity
el-
e-
ment
if
al-
ways
the
num-
ber
0
and
the
in-
verse
of
an
el-
e-
ment
 x
is
more
tra-
di-
tion-
ally
writ-
ten
as
 $-x$.
For
mul-
ti-
pli-
ca-
tion
the
iden-
tity
is
the
num-
ber
1
and
the
in-

CHAPTER

5.

GROUPS

verse

of

x

will

be

the

re-

cip-

ro-

cal

$\frac{1}{x}$.

Note

that

it

is

nec-

es-

sary

to

ex-

clude

0

when

us-

ing

the

op-

er-

a-

tion

of

mul-

ti-

plic-

ca-

tion.

This

is

be-

cause

zero

has

no

mul-

ti-

plica-

tive

in-

verse.

However

not

all

such

pair-

ings

5.3.
NUM-
BER
all
GROUPS
form
a
group.

Example

5.2.
Why
does
the
pair
 $(\mathbb{Z}, \times)$
not
form
a
group
?

These
ex-
am-
ples
of
course
are
all
in-
fi-
nite
groups
-
the
un-
der-
ly-
ing
sets
in
each
case
are
in-
fi-
nite.
But
there
also
ex-
ist
fam-
i-
lies
of
fi-
nite
groups
closely

CHAPTER

5.

GROUPS

re-
lated
to
the
in-
te-
gers
and
the
op-
er-
a-
tions
of
ad-
di-
tion
and
mul-
ti-
pli-
ca-
tion.

5.4 Modular arith- metic groups

Let
 $n \in$
 $\mathbb{Z}$
such
that
 $n >$
1.
We
can
de-
fine
op-
er-
a-
tions
on
 $\mathbb{Z}$
called
*ad-
di-
tion*
*mod-
ulo*
 n
and
mul-

5.4.
 MOD-
 U-
~~U-~~
 LAR
~~U-~~
 ARITH-
~~U-~~
 METIC
~~U-~~
 GROUPS
~~U-~~
 mod-
 ulo
 n
 as
 fol-
 lows.
 For
 in-
 te-
 gers
 a, b
 their
 sum
 mod-
 ulo
 n
 is
 de-
 fined
 to
 be
 the
 in-
 te-
 ger
 c ,
 which
 is
 the
 non-
 negative
 re-
 main-
 der
 left
 upon
 di-
 vi-
 sion
 of
 the
 in-
 te-
 ger
 $a +$
 b
 by
 the
 mod-
 u-
 lus
 n .

CHAPTER

5.

GROUPS

So

we

take

the

in-

te-

gers

a

and

b

and

add

them

in

the

usual

way,

then

we

di-

vide

this

sum

by

the

in-

te-

ger

n

and

take

the

non-

negative

re-

main-

der

c

$(0 \leq$

$c <$

$n)$.

This

num-

ber

c

is

the

sum

of

a

and

b

mod-

ulo

n

and

5.4.
 MOD-
 U-
 VAR
 while
 ARITH-
 METIC
 $a+b=c \pmod n$.
 GROUPS
 Modular
 mul-
 ti-
 pli-
 ca-
 tion
 is
 de-
 fined
 in
 a
 sim-
 i-
 lar
 way.
 For
 in-
 te-
 gers
 a, b
 their
 prod-
 uct
 mod-
 ulo
 n
 is
 de-
 fined
 to
 be
 the
 in-
 te-
 ger
 c ,
 which
 is
 the
 non-
 negative
 re-
 main-
 der
 left
 upon
 di-
 vi-
 sion
 of
 the
 in-

CHAPTER

5.

GROUPS

te-

ger

ab

by

the

mod-

u-

lus

n .

So

we

take

the

in-

te-

gers

a

and

b

and

mul-

ti-

ply

them

in

the

usual

way,

then

we

di-

vide

this

prod-

uct

by

the

in-

te-

ger

n

and

take

the

non-

negative

re-

main-

der

c

$(0 \leq$

$c <$

$n)$.

This

num-

ber

5.4.
MOD-
U-
LAR
is
ARITH-
METIC
GROUPS

uct
of
a
and
b
mod-
ulo
n
and
we
write

$$ab = c \pmod n.$$

Definition

5.4.
The
group
 $(\mathbb{Z}_n, +)$
is
the
group
formed
by
the
in-
te-
gers

$$0, 1, 2, 3 \dots n-1$$

under
the
op-
er-
a-
tion
of
ad-
di-
tion
mod-
ulo
n.

It
is
called
the
ad-
di-
tive
group
mod-

CHAPTER

5.

GROUPS

ulo

n.

The

def-

i-

ni-

tion

of

a

group

us-

ing

mul-

ti-

plica-

tion

mod-

ulo

n

is

not

so

straight-

for-

ward.

The

is-

sue

is

the

ex-

is-

tence

of

mul-

ti-

plica-

tive

in-

verses

mod-

ulo

n.

An

in-

te-

ger

a

will

only

have

an

in-

verse

5.4.
MOD-
U-
for
LAR
the
ARITH-
OP-
METIC
GROUPS
a-
tion
of
mul-
ti-
pli-
ca-
tion
mod-
ulo
 n
if
and
only
if
it
is
co-
prime
to
 n ,
i.e.
if
 a
and
 n
share
no
com-
mon
fac-
tors
apart
from
1.

Example 5.3. 1.

The
mul-
ti-
plica-
tive
in-
verse
of
2
mod-
ulo
5
is
3

CHAPTER

5.

GROUPS

as

$$2 \times 3 = 6 \equiv 1 \pmod{5}.$$

2.

The
mul-
ti-
plica-
tive
in-
verse
of
3
mod-
ulo
8
is
it-
self
as

$$3 \times 3 = 9 \equiv 1 \pmod{8}.$$

3.

The
el-
e-
ment
2
does
not
have
a
mul-
ti-
plica-
tive
in-
verse
mod-
ulo
8,
as
2
is
not
co-
prime
to
8.

4.

The
mul-
ti-
plica-
tive

5.4.
MOD-
U-
LAR in-
ARITH-verse
METIC
GROUPS
mod-
ulo
11
is
9
as

$$5 \times 9 = 45 = 4 \times 11 + 1 = 1 \pmod{11}.$$

Definition

5.5.

We
de-
fine
the
set
 $\mathbb{Z}_n^\times$
to
con-
sist
of
those
in-
te-
gers
be-
tween
1
and
 n
that
are
co-
prime
to
 n ,
i.e.

$$\mathbb{Z}_n^\times = \{a \in \mathbb{Z} : 1 \leq a \leq n, \text{ } a \text{ is co-prime to } n\}.$$

Example

5.4.

De-
ter-
mine
the
el-
e-
ments
con-
tained
in
the

CHAPTER

5.

GROUPS

fol-
low-
ing

1.

$$\mathbb{Z}_4^\times = \{\dots\}$$

2.

$$\mathbb{Z}_5^\times = \{\dots\}$$

3.

$$\mathbb{Z}_6^\times = \{\dots\}$$

4.

$$\mathbb{Z}_7^\times = \{\dots\}$$

5.

$$\mathbb{Z}_8^\times = \{\dots\}$$

From
these
ex-
am-
ples
can
you
de-
scribe
the
el-
e-
ments
con-
tained
in
 $\mathbb{Z}_p^\times$,
where
 p
is
a
prime
num-
ber,
i.e.
a
num-
ber
with
no
pos-
i-
tive

5.5.
CAY-
LEY
TABLES
in
te-
ger
fac-
tors
other
than
it-
self
and
1.

Definition

5.6.

The
pair
 $(\mathbb{Z}_n^\times, \times)$
where
the
op-
er-
a-
tion
 $\times$
is
mul-
ti-
pli-
ca-
tion
mod-
ulo
 n
forms
a
group.
It
is
called
the
*mul-
ti-
plica-
tive
group
mod-
ulo
 n .*

5.5 Cayley ta- bles

Finite
groups,
such

CHAPTER

5.

GROUPS

as

these

mod-

u-

lar

arith-

metic

groups,

can

be

stud-

ies

by

way

of

their

Cay-

ley

ta-

ble.

Definition

5.7

(*Cay-*

ley

ta-

ble).

A

Cay-

ley

ta-

ble

for

a

fi-

nite

group

$(G, *)$

is

a

two-

dimensional

ta-

ble,

whose

rows

and

columns

are

in-

dexed

by

the

el-

e-

ments

5.5.
CAY-
LEY
TABLES
95
G.

The
en-
tries
in
the
body
of
the
ta-
ble
show
the
re-
sults
of
com-
bin-
ing
the
row
and
col-
umn
in-
dex
el-
e-
ments
with
the
op-
er-
a-
tion

*.
See
the
generic
Cay-
ley
ta-
ble
in
fig-
ure

[5.1.](#)
Take
note
of
the
con-
ven-
tion
re-

CHAPTER

5.

GROUPS

gard-

ing

the

or-

der-

ing

of

the

fac-

tors

in

the

prod-

uct

g_i^*

g_j ,

the

row

in-

dex

el-

e-

ment

is

on

the

left

and

the

col-

umn

in-

dex

el-

e-

ment

is

on

the

right.

It

is

im-

por-

tant

to

have

this

con-

ven-

tion

and

to

stick

to

it

5.5.
 CAY-
 LEY
 TABLES
 are
 not
 all
 groups
 are
 Abelian.

*	...	g_j	...
⋮			
g_i	...	$g_i * g_j$	...
⋮			

Figure 5.1:
 A
 generic
 Cay-
 ley
 ta-
 ble

Example
5.5.
 We
 show
 some
 Cay-
 ley
 ta-
 ble
 for
 some
 mod-
 u-
 lar
 arith-
 met-
 ic
 groups.

1. The
 ta-
 ble
 for
 the
 group
 $(\mathbb{Z}_4, +)$
 is

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

CHAPTER
5.
~~GROUPS~~
($\mathbb{Z}_4, +$)

2.
The
ta-
ble
for
the
group
($\mathbb{Z}_5^\times, \times$)
is

$\times$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

($\mathbb{Z}_5^\times, \times$)

3.
The
ta-
ble
for
the
group
($\mathbb{Z}_8^\times, \times$)
is

$\times$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

($\mathbb{Z}_8^\times, \times$)

In
the
ex-
er-
cises
you
are
asked
to
pro-
vide
the
Cay-
ley
ta-
bles
for
some

5.5.
 CAY-
 LEY
 the
 1930s
 mod-
 u-
 lar
 arith-
 metic
 groups.
 Note
 that,
 be-
 cause
 they
 are
 based
 on
 the
 usual
 arith-
 metic
 op-
 er-
 a-
 tions
 of
 ad-
 di-
 tion
 and
 mul-
 ti-
 pli-
 ca-
 tion,
 all
 of
 the
 mod-
 u-
 lar
 arith-
 metic
 groups
 are
 Abelian,
 i.e.
 both
 types
 of
 op-
 er-
 a-
 tion
 are
 al-
 ways
 com-

CHAPTER

5.

GROUPS

mu-
ta-
tive.

The
study
of
mod-
u-
lar
arith-
metic
is
an
im-
por-
tant
as-
pect
of
num-
ber
the-
ory,
and
this
topic
and
some
of
its
group-
theoretic
as-
pects
will
be
ex-
plored
fur-
ther
in
the
Level
5
unit
Num-
ber
The-
ory
and
Cryp-
tog-
ra-
phy.

5.6.
CYCLIC
GROUPS
5.6 Cyclic
AND
GENERATORS
and
gen-
er-
a-
tors

Before
 con-
 tin-
 u-
 ing
 with
 ex-
 am-
 ples
 of
 groups
 we
 give
 some
 more
 def-
 i-
 ni-
 tions
 and
 in-
 tro-
 duce
 some
 more
 group
 the-
 ory.

Definition

5.8
 (Mul-
 ti-
 plica-
 tive
 no-
 ta-
 tion).
 Let
 g
 and
 h
 be
 el-
 e-
 ments
 of

CHAPTER

5.

GROUPS

a

group

$(G, *)$.

The

group

prod-

uct

$g*$

h

can

also

be

de-

noted

gh .

In

this

way

we

of-

ten

omit

the

use

of

a

seper-

ate

sym-

bol

for

the

group

op-

er-

a-

tion

and

sim-

ply

use

this

mul-

ti-

plica-

tive

no-

ta-

tion

for

the

group

prod-

ucts

as

long

5.6.
CYCLIC
GROUPS
 AND
 GENERATORS

as
 the
 ac-
 tual
 op-
 er-
 a-
 tion
 be-
 ing
 used
 is
 clearly
 un-
 der-
 stood
 from
 the
 con-
 text.
 Sim-
 i-
 larly
 the
 no-
 ta-
 tion
 g^n ,
 where
 n
 is
 a
 pos-
 i-
 tive
 in-
 te-
 ger
 de-
 notes
 the
 prod-
 uct

$$g^n = g * g * \cdots * g,$$

where
 there
 are
 n
 fac-
 tors
 g
 in
 the
 prod-
 uct.

CHAPTER

5.

GROUPS

When

the

power

is

neg-

a-

tive

it

de-

notes

prod-

ucts

of

the

in-

verse

el-

e-

ment

of

g ,

i.e.

if

$m >$

0

then

$$g^{(-m)} = g^{-1} * g^{-1} * \cdots * g^{-1},$$

where

there

are

m

fac-

tors

g^{-1}

in

the

prod-

uct.

Lastly,

the

no-

ta-

tion

g^0

de-

notes

the

iden-

tity

el-

e-

ment

of

the

group.

5.6.
CYCLIC
GROUPS
Definition
AND
5.9
GENERATORS
(Cyclic
group).

A
group
 G
is
cyclic
if
there
is
an
el-
e-
ment
 $g \in$
 G
such
that

$$G = \{g^n : n \in \mathbb{Z}\}.$$

The
el-
e-
ment
 g
is
called
a
gen-
er-
a-
tor
of
 G
and
we
use
the
no-
ta-
tion

$$G = \langle g \rangle.$$

A
cyclic
group
may
have
more
than
one
gen-
er-

CHAPTER

5.

GROUPS

a-
tor.

In
the

ex-
er-
cises

you
are
asked

to
ex-
plore
mod-

u-
lar
arith-
metic
groups

and
search
for

ex-
am-
ples
of

cyclic
groups
amongst
them.

You
should
find
there
that

gen-
er-
a-
tors

are
not
unique,
i.e.

a
cyclic
group

G
can
have
more
than
one
gen-
er-
a-
tor.

5.6.
CYCLIC
GROUPS
 In
 AND
 1991
 GENERATORS

if
 g
 is
 a
 gen-
 er-
 a-
 tor
 of
 G
 then
 so
 will
 g^{-1}
 be
 (al-
 though
 g^{-1}
 need
 not
 nec-
 es-
 sar-
 ily
 be
 a
 dif-
 fer-
 ent
 el-
 e-
 ment
 to
 g .

Definition

5.10
 (Cyclic
 sub-
 group,
 sub-
 group).
 If
 we
 take
 any
 el-
 e-
 ment
 g
 of
 a
 group
 G ,

CHAPTER

5.

GROUPS

then

the

set

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\},$$

will

form

a

group

un-

der

the

same

op-

er-

a-

tion

used

in

G .

This

is

called

a

cyclic

sub-

group

of

G

gen-

er-

ated

by

g .

More

gen-

er-

ally

a

sub-

group

of

a

group

G

is

a

group

H

un-

der

the

same

op-

er-

a-

5.6.
CYCLIC
GROUPS
 tion
 AND
 389
 GENERATORS
 that
 used
 in
 G
 such
 that
 $H \subset$
 G .

Definition

5.11

(Or-
 der
 of
 an
 el-
 e-
 ment).

Let
 g
 be
 an
 el-
 e-
 ment
 of
 a
 group
 G .

The
 or-
 der
 of
 the
 cyclic
 sub-
 group
 $\langle g \rangle$

is
 also
 re-
 ferred
 to
 as
 the
*or-
 der
 of
 the
 el-
 e-
 ment*
 g .

The

CHAPTER

5.

GROUPS

or-

der

of

a

group

el-

e-

ment

g

can

be

char-

ac-

terised

as

the

small-

est

pos-

i-

tive

in-

te-

ger

n

such

that

$g^n =$

e ,

the

iden-

tity

el-

e-

ment

of

the

group.

5.7 Matrix groups

We

can

form

ma-

trix

groups

us-

ing

ei-

ther

the

op-

er-

5.7.
MA-
TRIX
GROUPS
tion
of
ma-
trix
ad-
di-
tion
or
ma-
trix
mul-
ti-
pli-
ca-
tion.
We
need
to
take
care
to
dis-
tin-
guish
the
dif-
fer-
ent
uses
of
the
words
iden-
tity
and
in-
verses.
These
might
be
used
in
their
matrix-
theoretic
sense
or
their
group-
theoretic
sense.
Soem-
times
the
mean-

CHAPTER

5.

GROUPS

ings

will

co-

in-

cide

and

some-

times

not.

It

should

be

clear

from

the

con-

text

what

is

meant

in

each

case.

Definition

5.12.

The

group

$(M(n, \mathbb{R}), +)$

is

the

group

formed

by

all

$n \times$

n

ma-

tri-

ces

with

real

en-

tries,

un-

der

the

op-

er-

a-

tion

of

ma-

trix

ad-

di-

5.7. MA- TRIX GROUPS

The
iden-
tity
in
this
group
is
the
 $n \times$
 n
zero
ma-
trix.
The
in-
verse
of
an
el-
e-
ment
 A
is
the
ma-
trix
 $-A$.

Similar
groups
can
also
be
formed
by
re-
strict-
ing
the
ma-
trix
en-
tries
to
come
from
the
other
num-
ber
sys-
tems
 $\mathbb{Z}, \mathbb{Q}, \mathbb{C}$.
Or
even

CHAPTER

5.

GROUPS

more

gen-

er-

ally,

we

could

choose

any

group

as

the

source

of

the

en-

tries.

For

the

op-

er-

a-

tion

of

ma-

trix

mul-

ti-

plic-

ation

the

iden-

tity

el-

e-

ment

will

al-

ways

be

the

ap-

pro-

pri-

ately

sized

iden-

tity

ma-

trix

(in

the

lin-

ear

al-

ge-

5.7.
MA-
TRIX
GROUPS
(in
sense)
since
mul-
ti-
pli-
ca-
tion
by
an
in-
den-
tity
ma-
trix
leaves
the
other
ma-
trix
un-
changed.
The
in-
verse
el-
e-
ments
(in
the
group
the-
o-
retic.
sense)
will
be
the
in-
verse
ma-
tri-
ces
(in
the
lin-
ear
al-
ge-
bra
sense).
This
leads
us
to
the

CHAPTER

5.

GROUPS

fol-
low-
ing
def-
i-
ni-
tion

Definition

5.13

(Gen-
eral
lin-
ear
group).

The
*Gen-
eral
Lin-
ear
Group*
 $GL(n, \mathbb{R})$

is
the
group
formed
by
the
set
of
non-
singular
ma-
tri-
ces
(i.e.
with
non-
zero
de-
ter-
mi-
nant)
of
size
 $n \times$
 n
un-
der
the
op-
er-
a-
tion
of
ma-

5.7.
MA-
TRIX
GROUPS
The
mul-
ti-
pli-
ca-
tion.
The
iden-
tity
el-
e-
ment
of
this
group
is
the
 $n \times$
 n
iden-
tity
ma-
trix
 I_n
and
the
in-
verse
group
el-
e-
ment
of
a
ma-
trix
 A
is
the
in-
verse
ma-
trix
 A^{-1} .

Recall
the
mul-
ti-
plica-
tive
prop-
erty
of
the
de-

CHAPTER

5.

GROUPS

ter-

mi-

nant,

namely

$$\det(AB) = \det(A)\det(B),$$

for

$n \times$

n

ma-

tri-

ces

A

and

B .

This

prop-

erty

en-

sures

that

the

clo-

sure

ax-

iom

is

sat-

is-

fied

for

the

fol-

low-

ing

ma-

trix

group.

Definition

5.14

(Spe-

cial

lin-

ear

group).

The

Spe-

cial

Lin-

ear

Group

$\mathrm{SL}(n, \mathbb{R})$

is

the

group

5.7.
 MA-
 TRIX
 GROUPS
 formed
 by
 the
 set
 of
 ma-
 tri-
 ces
 with
 de-
 ter-
 mi-
 nant
 equal
 to
 1,
 of
 size
 $n \times$
 n
 un-
 der
 the
 op-
 er-
 a-
 tion
 of
 ma-
 tric
 mul-
 ti-
 pli-
 ca-
 tion.
 The
 iden-
 tity
 el-
 e-
 ment
 of
 this
 group
 is
 the
 $n \times$
 n
 iden-
 tity
 ma-
 trix
 I_n
 and
 the
 in-

CHAPTER

5.

GROUPS

verse

group

el-

e-

ment

of

a

ma-

trix

A

is

the

in-

verse

ma-

trix

A^{-1} .

Example

5.6.

Prove

that

$\mathrm{SL}(n, \mathbb{R})$

forms

a

group.

See

the

ex-

er-

cises

for

ex-

am-

ples

of

other

more

spe-

cialised

ma-

trix

groups.

5.8 Permutation groups

Definition

5.15

(Per-

mu-

ta-

tion).

A

per-

5.8.
~~PER-~~
~~MU-~~
~~TA-~~
~~TI-~~
~~ON~~
~~US-~~
~~GROUPS~~
of
the
 n
ob-
jects

$1, 2, 4, \dots, n$

is
a
bi-
jec-
tive
func-
tion
with
do-
main
and
co-
domain
both
equal
to
 $\{1, 2, 3, \dots, n\}$.

Definition

5.16

(Sym-
met-
ric
group).

The
sym-
met-
ric

group

S_n

is

the

group

formed

by

all

the

per-

mu-

ta-

tions

of

$1, 2, \dots, n$

un-

der

the

CHAPTER

5.

GROUPS

op-
er-
a-
tion
of
com-
po-
si-
tion.

Theorem

5.6.

The
group
 S_n
has
or-
der
 $n!$.

Notation

for
per-
mu-
ta-
tions

There
are
two
main
types
of
no-
ta-
tion
used
for
per-
mu-
ta-
tions:
the
two-
line
no-
ta-
tion
and
the
cy-
cle
no-
ta-
tion.

ments,

CHAPTER

5.

GROUPS

i.e.

$$\sigma(1) = 2, \sigma(2) = 4, \sigma(3) = 1, \sigma(4) = 3.$$

Example

5.7.

Re-
mem-
ber-
ing
that
func-
tion
com-
po-
si-
tions
are
read
from
the
right,
find
the
two-
line
no-
ta-
tion
for
the
fol-
low-
ing
prod-
ucts
of
per-
mu-
ta-
tions.
Also
the
con-
ven-
tion
is
that,
al-
though
these
prod-
ucts
are
ex-
actly
com-

5.8.
PER-
MU-
TA-
PER-
MUTATION
GROUPS
 of

func-
 tions
 (the
 per-
 mu-
 ta-
 tions),
 we
 usu-
 ally
 omit
 the
 func-
 tion
 com-
 po-
 si-
 tion
 sym-
 bol
 o
 and
 sim-
 ply
 con-
 cate-
 nate
 the
 two-
 line
 no-
 ta-
 tions
 to
 de-
 note
 the
 prod-
 uct.

1.

In
 S_3 ,

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

2.

In
 S_8 ,

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 5 & 1 & 6 & 8 & 2 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 4 & 6 & 1 & 3 & 5 & 7 & 2 \end{pmatrix}.$$

5.18

(Cy-
cle
no-
ta-
tion).

A
*cy-
cle*
is
a
per-
mu-
a-
tion
that
*cycli-
cally*
per-
mutes

a
se-
lec-
tion
of
the
ob-
jects
 $1, 2, \dots, n$.

For
ex-
am-
ple
the
cy-
cle
 $\sigma \in S_8$
de-
noted
by
the
fol-
low-
ing
*cy-
cle*
*no-
ta-
tion*

$$\sigma = (1\,3\,5\,2\,4),$$

is
the
per-

5.8.
 PER-
 MU-
 PL-
 TA-
 TION
 is-
 sat-
 is-
 fy-
 ing

$$\sigma(1) = 3, \sigma(3) = 5, \sigma(5) = 2, \sigma(2) = 4, \sigma(4) = 1,$$

and
 where
 the
 ob-
 jects
 not
 men-
 tioned
 in
 the
 cy-
 cle
 no-
 ta-
 tion
 are
 left
 fixed
 by
 the
 per-
 mu-
 ta-
 tion,
 i.e.

$$\sigma(6) = 6, \sigma(7) = 7, \sigma(8) = 8.$$

A
 nice
 way
 of
 rep-
 re-
 sent-
 ing
 per-
 mu-
 a-
 tions
 is
 to
 ex-
 press
 them
 as
 prod-
 ucts

CHAPTER

5.

GROUPS

of

dis-

joint

cy-

cles.

Definition

5.19

(Dis-

joint

cy-

cles).

Two

cy-

cles

$\sigma, \tau \in$

S_n

are

dis-

joint

if

the

ob-

jects

that

are

not

left

fixed

by

σ

are

all

dif-

fer-

ent

to

the

ob-

jects

not

left

fixed

by

τ .

Example

5.8.

Find

ex-

pres-

sions

for

the

per-

mu-

ta-

5.8.
PER-
MU-
TI-
ON
of
the
GROUPS
 previous
 ous
 ex-
 am-
 ple
 as
 prod-
 ucts
 of
 dis-
 joint
 cy-
 cles.

Example

5.9.
 Find
 dis-
 joint
 cy-
 cle
 prod-
 uct
 ex-
 pres-
 sions
 for
 the
 fol-
 low-
 ing
 prod-
 ucts
 of
 cy-
 cles,

1.
 $(13)(245)(235)$
2.
 $(123)(13)(45)$
3.
 $(246)(12)$

Another
 nice
 way
 of
 rep-
 re-
 sent-
 ing

CHAPTER

5.

GROUPS

per-

mu-

a-

tions

is

to

ex-

press

them

sa

prod-

ucts

of

cy-

cles

of

length

two.

Such

cy-

cles

are

called

trans-

po-

si-

tions.

Definition

5.20

(Trans-

po-

si-

tion).

A

trans-

po-

si-

tion

$\sigma \in$

S_n

is

a

cy-

cle

of

length

2.

Theorem

5.7.

Any

per-

mu-

ta-

tion

$\sigma \in$

5.8.
 PER-
 MU-
 TA-
 TION
 GROUPS
 ex-
 pressed
 as
 a
 prod-
 uct
 of
 trans-
 pos-
 tions.

Proof.
 First
 find
 an
 ex-
 pres-
 sion
 for
 σ
 as
 s
 prod-
 uct
 of
 dis-
 joint
 cy-
 cles.
 Then
 ex-
 press
 each
 cy-
 cle
 as
 a
 prod-
 uct
 of
 trans-
 pos-
 tions
 us-
 ing
 the
 fol-
 low-
 ing
 re-
 sult,

$$(a_1 a_2 a_3 \dots a_m) = (a_1 a_2)(a_2 a_3)(a_3 a_4) \dots (a_{m-2} a_{m-1})(a_{m-1} a_m).$$

□

Example
5.10.

Express
the
fol-
low-
ing
per-
mu-
a-
tions
from
 S_5
as
prod-
ucts
of
trans-
po-
si-
tions,

1.

$$\left(\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 2 & 1 \end{array} \right)$$

2.

$$\left(\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{array} \right)$$

The
sig-
nif-
i-
cance
of
trans-
pos-
tions
is
given
by
the
fol-
low-
ing
re-
sult.

Theorem
5.8.
When

5.8.
 PER-
 MU-
 TA-
 TION
 GROUPS
~~PER-~~
~~MU-~~
~~TION~~
~~GROUPS~~
mu-
a-
tion
 σ
is
ex-
pressed
as
a
prod-
uct
of
trans-
pos-
tions,
the
num-
ber
of
such
trans-
pos-
tions
will
al-
ways
be
even,
or
al-
ways
be
odd.

This
 is
 called
 the
par-
ity
 of
 the
 per-
 mu-
 ta-
 tion
 σ .
 Any
 per-
 mu-
 a-
 tion
 has

CHAPTER

5.

GROUPS

ei-
ther
even
or
odd
par-
ity.

Theorem

5.9.

*The
set
of
all
even
pe-
mu-
ta-
tions
from
the
group
 S_n
is
de-
noted
 A_n ,
and
it
forms
a
group
of
or-
der
 $\frac{n!}{2}$.*

Definition

5.21

(Al-
ter-
nat-
ing
group).
The
group
 A_n
is
called
the
Al-
ter-
nat-
ing
group
of
de-

5.8.
PER-
MU-
TA-
TION
~~63~~*GROUPS*

CHAPTER

5.

GROUPS Exercises

5.1.

These
prob-
lems
re-
quire
the
ap-
pli-
ca-
tion
of
var-
i-
ous
con-
cepts
and
re-
sults
from
this
chap-
ter.
We
shall
use
these
as
a
ba-
sis
for
dis-
cus-
sion
in
the
tu-
to-
ri-
als.

1.

Construct
Cay-
ley
ta-
bles
for
a
se-
lec-
tion
of

5.8.

PER-

MU-

TA- the
TION mod-

GROUPS
GROUPS

lar
arithmetic
groups
 $(\mathbb{Z}_n, +)$
and
 $(\mathbb{Z}_n^\times, \times)$,
for
 $1 \leq$
 $n \leq$
20.

2.

For
the
groups
you
have
ex-
am-
ined
in
the
pre-
vi-
ous
ques-
tion,
iden-
tify
which
ones
are
cyclic
and
which
ones
re-
quire
more
than
one
gen-
er-
a-
tor.

3.

Write
down
in
two-
line
no-
ta-

CHAPTER

5.

~~638~~ GROUPS

tion
all
the
per-
mu-
ta-
tions
from
the
group
 S_3 .

4.

Write
down
in
two-
line
no-
ta-
tion
all
the
per-
mu-
ta-
tions
from
the
group
 S_4 .

5.

In
the
group
 S_8
de-
ter-
mine
the
two-
line
no-
ta-
tion
form
of
the
fol-
low-
ing
prod-
ucts
or
per-
mu-

5.8.
 PER-
 MU-
 TA-
 TION^{ta-}
^{tions,}
 GROUPS_{a)}

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 7 & 5 & 6 & 8 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 6 & 7 & 8 & 1 \end{pmatrix}$$

b)

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 4 & 6 & 8 & 1 & 3 & 5 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 4 & 6 & 1 & 3 & 5 & 7 & 2 \end{pmatrix}$$

6.
 Now
 ex-
 press
 the
 prod-
 ucts
 you
 found
 in
 the
 pre-
 vi-
 ous
 ques-
 tion
 as
 prod-
 ucts
 of
 dis-
 joint
 cy-
 cles.

7.
 Express
 the
 fol-
 low-
 ing
 per-
 mu-
 ta-
 tion
 prod-
 ucts
 as
 prod-
 ucts
 of
 dis-
 joint
 cy-
 cles,

CHAPTER

5.

~~5.1~~ GROUPS

a)

$$(1\ 3\ 5)(2\ 4)(3\ 6),$$

b)

$$(2\ 4\ 6\ 7)(3\ 5\ 4)(6\ 2),$$

c)

$$(1\ 3)(2\ 5)(2\ 3)(1\ 4\ 7)(3\ 6\ 5).$$

8.

Express
the
fol-
low-
ing
per-
mu-
ta-
tions
as
prod-
ucts
of
trans-
po-
si-
tions
and
de-
ter-
mine
the
par-
ity
of
each
one,

a)

$$\sigma = (1\ 3\ 5\ 7)(2\ 4),$$

b)

$$\tau = (2\ 5\ 7\ 3)(6\ 4),$$

c)

$$\pi = (1\ 5\ 7\ 8\ 3\ 6\ 9)$$

9.

In
the
group
 S_5 ,
find
an
el-
e-
ment

5.8.

PER-

MU-

TA- τ

TION such

that

~~GROUPS~~

$\sigma\tau =$

$(1\ 5)(2\ 4),$

where

$\sigma =$

$(1\ 3)(4\ 5).$

10.

In

the

group

S_9

find

the

per-

mu-

ta-

tions

σ

and

τ

that

solve

the

fol-

low-

ing

equa-

tions,

a)

$\sigma(1\ 3\ 7\ 9) =$

$(1\ 2)(4\ 5),$

b)

$(1\ 3)(5\ 7)\tau(1\ 3\ 4) =$

$(1\ 3\ 6\ 9).$

11.

Are

any

of

the

sym-

met-

ric

groups

S_n

cyclic?

De-

ter-

mine

those

that

are

and

CHAPTER

5.

~~GROUPS~~
those
that
are
not.

12.

Are
any
of
the
sym-
met-
ric
groups
 S_n
Abelian?
De-
ter-
mine
those
that
are
and
those
that
are
not.

5.9 Going fur- ther: Sub- groups and La- grange's the- o- rem

Group
the-
ory
is
a
huge
sub-
ject
in
pure
math-
e-
mat-
ics.

5.9.
GO-
ING
Aside
FUR-
THER:
SUB-
GROUPS
AND
LA-
GRANGE'S
THEOREM
ar-
eas
of
math-
e-
mat-
ics,
one
of
the
main
con-
cerns
of
group
the-
ory
is
clas-
si-
fy-
ing
and
enu-
mer-
at-
ing
all
the
ex-
ist-
ing
groups,
both
fi-
nite
and
in-
fi-
nite.

To
give
a
small
taste
of
this
sub-

CHAPTER

5.

GROUPS

ject

we

shall

state

and

prove

La-

grange's

the-

o-

rem,

which

places

re-

stric-

tions

on

the

sizes

of

sub-

groups

that

might

ex-

ist

within

a

larger

group.

Definition

5.22

(Sub-
group).

Let

$(G, *)$

be

a

group

and

H

a

sub-

set

of

G .

We

say

that

H

forms

a

sub-

group

of

5.9.
 GO-
 ING
 (FUR-
 THER:
 if
 the
 SUB-
 GROUPS
 (H*)
 AND
 IS
 EA-
 GRANGE'S
 THEOREM
 Note
 that
 it
 is
 the
 same
 op-
 er-
 a-
 tion
 *
 in-
 volved
 in
 the
 con-
 tain-
 ing
 group
 G
 and
 the
 sub-
 group
 H .
 This
 is
 im-
 por-
 tant
 as
 in
 many
 con-
 texts
 there
 is
 more
 than
 one
 group
 like
 op-
 er-
 a-
 tion.

Theorem

CHAPTER

5.

GROUPS

5.10

(La-
grange's
the-

o-
rem).

*Sup-
pose
that*

*H
forms*

*a
sub-
group
of*

*a
fi-
nite
group*

G.

*Then
the*

*or-
der
of*

*H
di-
vides*

*the
or-
der*

*of
G.*

Appendices

Appendix A

A
(very)
brief
look
at
num-
ber
sys-
tems

A
pre-
cise
and
rig-
or-
ous
de-
scrip-
tion
and
con-
struc-
tion
of
the
var-
i-
ous
num-
ber
sys-
tems

APPENDIX

A.

A
would
(VERY)

is
BRIEF
quite

LOOK
AT
a

whole
NUM-
BER
course

is
SYSTEMS
it-

self.

We

shall

not

be

ex-

am-

in-

ing

these

tech-

ni-

cal

is-

sues

in

de-

tail

in

these

notes

but

we

may

come

across

some

of

them

from

time

to

time.

For

the

time

be-

ing

we

shall

in-

tro-

duce

and

agree

on

some

ba-

sic
 ter-
 mi-
 nol-
 ogy
 for
 the
 num-
 ber
 sys-
 tems
 we
 shall
 use,
 and
 in-
 di-
 cate
 some
 of
 the
 tech-
 ni-
 cal
 de-
 tail
 that
 dis-
 tin-
 guishes
 the
 real
 num-
 bers
 from
 the
 ra-
 tio-
 nal
 num-
 bers.

- The
Nat-
u-
ral
num-
bers,
 $\mathbb{N} =$
 $\{1, 2, 3, \dots\}$.
 These
 are
 the
 ba-
 sic

APPENDIX

A.

A

(VERY) count-

BRIEF ing

LOOK num-

AT bers¹.

NUM-

BER • The

SYSTEMS

te-

gers,

$\mathbb{Z} =$

$\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

These

con-

sist

of

the

nat-

u-

ral

num-

bers

and

their

neg-

a-

tives,

to-

gether

with

0.

•

The

Ra-

tio-

nal

num-

bers,

$\mathbb{Q} =$

$\{\frac{m}{n} : m, n \in \mathbb{Z}, m \neq 0\}$.

These

con-

sist

of

all

¹Some

au-

thors

re-

gard

0

as

a

nat-

u-

ral

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and
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•

The
Real
num-
bers,
 $\mathbb{R}$.
A
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num-
ber
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point
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(or
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uum).
They
can
be
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sented
in
many
ways,
for
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stance

APPENDIX

A.

A

(VERY^{as})

BRIEFⁱⁿ

LOOK^{fi}

AT^{nite}

NUM^{dec}

BERⁱ

SYSTEMS^{real}

554

ex-

pan-

sions

(or

in-

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nite

digit

ex-

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to

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base),

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The

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$\mathbb{C} =$

$\{a + bi : a, b \in \mathbb{R}\},$

where
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 $\sqrt{-1}.$

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APPENDIX

A.

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BRIEF

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$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \subset \dots$$

Examples

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$\sqrt{2}$,

$\sqrt{3}$,

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line.
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val
on
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ber
line,
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and
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bers.

Describing
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APPENDIX

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 set
 $A \subset$
 S
 which
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 above,
 there
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 ists
 in
 S
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least
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 A .

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 bers
 do
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 this
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 erty
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 ples
 such
 as

$$A = \{x \in \mathbb{Q} : x^2 < 2\}.$$

The
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 A
 is
 clearly
 bounded
 above
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 3
 for

APPENDIX

A.

A

in (VERY)

stage.

BRIEF

But

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SUBSYSTEMS

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